

Materials

Alloys - Materials that are composed of 2 or more metals or a metal with a non-metal



iron



alloy

Bonding in metals is nondirectional

• Attractive force between electrons and cation are in all directions equally

E.g. Bronze (copper + tin), Brass (copper + zinc),
Stainless Steel (iron + nickel + chromium)

Types of Alloy

1. Substitutional Alloy - Atoms of one metal are substituted by atoms of another metal

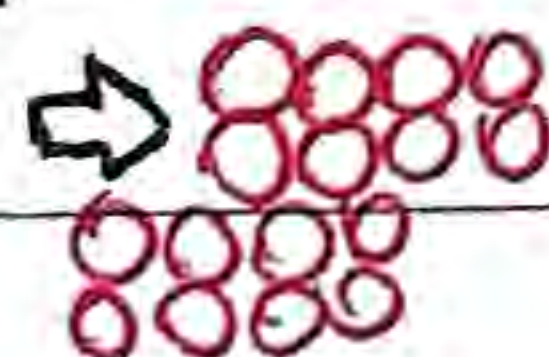


2. Interstitial Alloy - Different metal occupies interstitial spaces (gaps) in the lattice structure



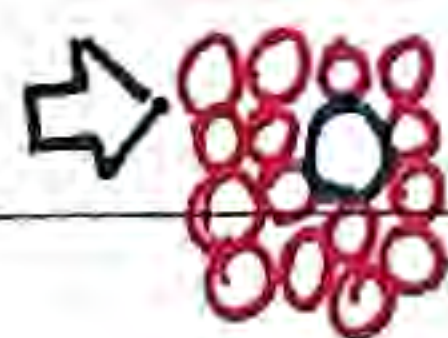
Alloys are generally stronger and harder (less malleable)

• The presence of multiple metal elements disrupts the uniform atomic structure of the pure metal (harder for layers to slide past each other)



PURE METAL

layers can slide



ALLOYS

layers cannot slide

E.g.

Alloy
Steel

Component metals

iron, carbon

Properties

high tensile strength
resistant to corrosion

Uses

construction
cooking implements

Stainless Steel

iron, nickel, chromium

Brass

Copper, zinc

Bronze

copper, tin

Pewter

copper, tin, antimony

Solder

lead, tin

Nichrome

nickel, chromium

low melting point

Pipes
coins, medals, tools
decorative ornaments
join metals in circuits
heating elements

Alkenes - Molecule with a Carbon-Carbon double bond
 • Shortest is Ethene since it requires at least two carbon atoms to form a double bond

Groups (replace a hydrogen atom)

Halogens: Bromo, Fluoro, Chloro, etc.

-Br -F -Cl

$\text{H}_3\text{C}-\text{Br}$ $\text{H}_3\text{C}-\text{F}$ $\text{H}_3\text{C}-\text{Cl}$

Methyl: Carbon atom bonded to 3 hydrogens
 -CH₃

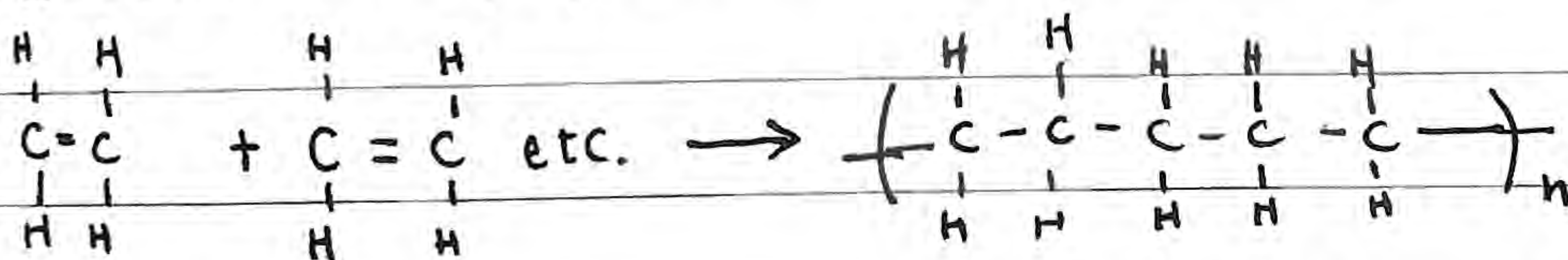
Phenyl: Benzene group 

Addition Polymerization

Turning monomers into a polymer
 (simple molecule)

• Monomers are unsaturated (alkenes or alkynes)

E.g. Ethene to polyethene



Polymers are drawn with a bracket outside and subscript n .

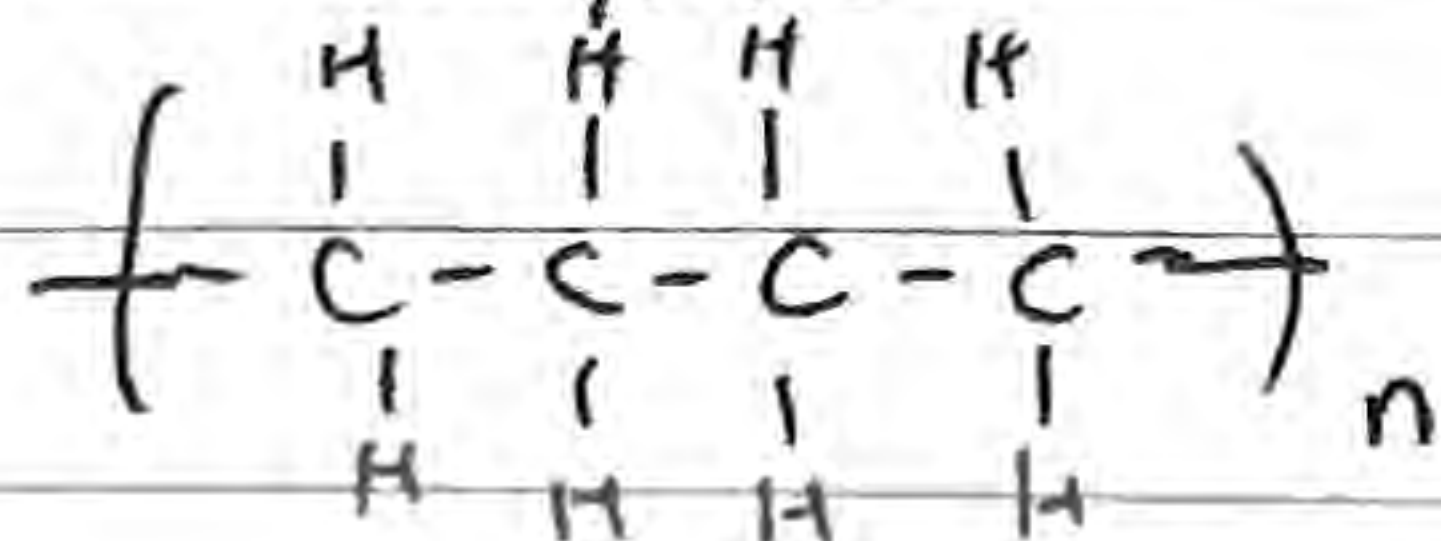
(segment of polymer) _{n}

even though
 alkenes
 become
 alkanes,
 do not
 change
 name
 to
 alkanes

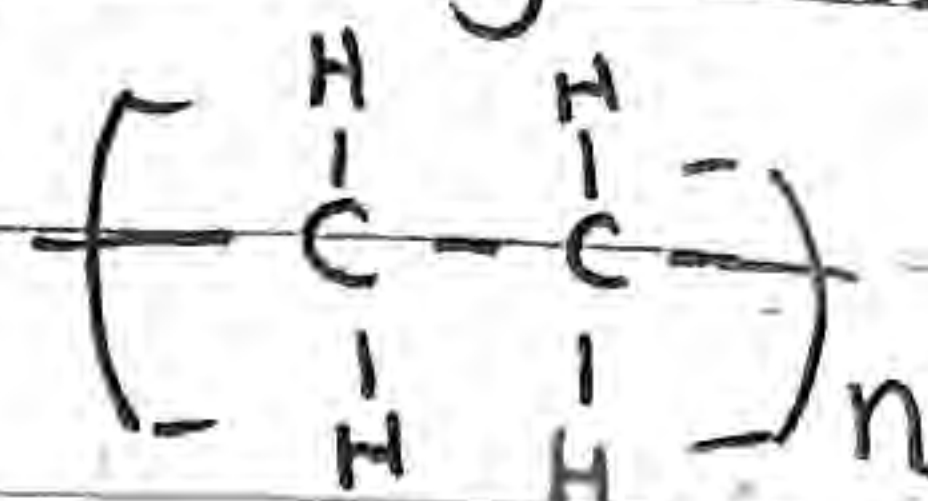
The name of the polymer is just poly + (name of monomer)

The repeating unit in polymers is just a saturated monomer (or the transformed monomer in bonds) ^{for alkenes}

E.g. Polymer



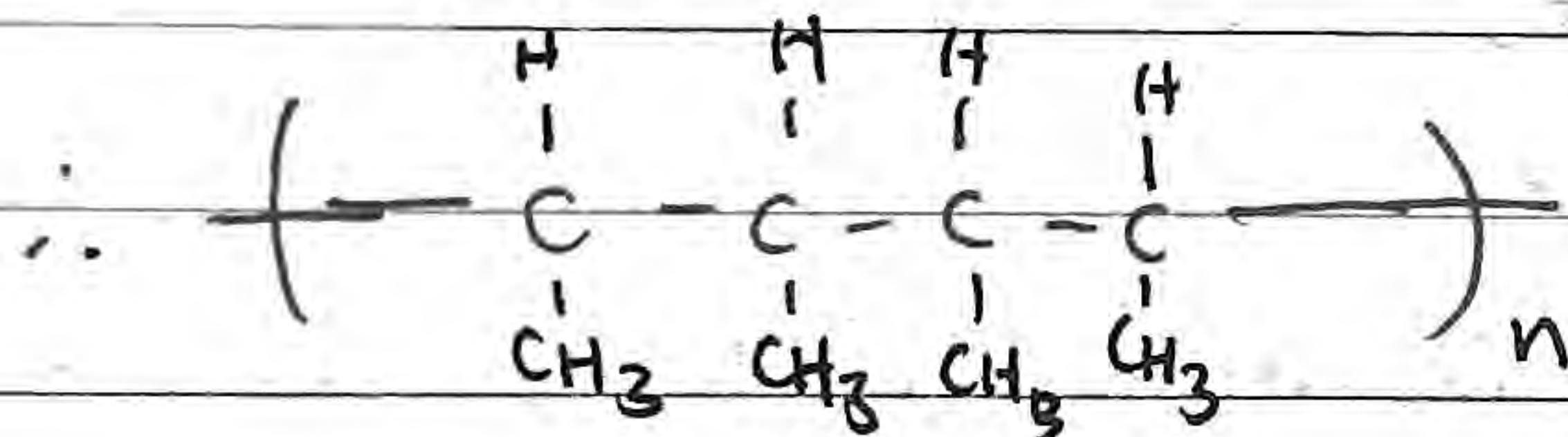
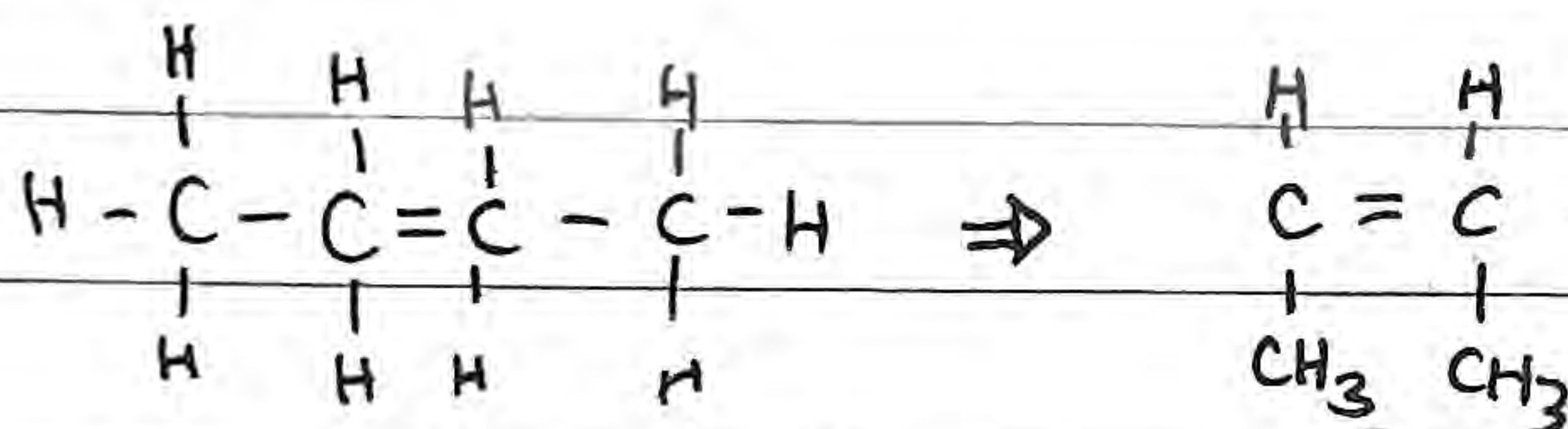
Repeating Units



How about more complicated monomers?

You can rearrange the molecule by "bending" it such that the double bonded carbons are on the extremes

E.g.



Polymers

Covalently bonded materials (mostly of Carbon and Hydrogen)

- Low thermal and electrical conductivity
- Low density

Macromolecules (very large)

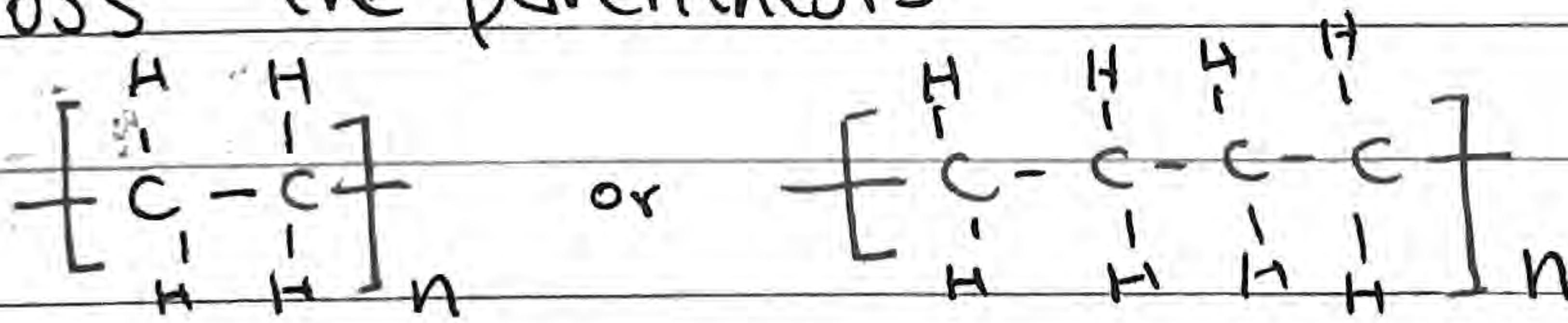
Synthetic polymers and natural polymers

Polymerization refers to joining of monomers into a polymer



Repeating Unit - Group of atoms that repeat along a polymer

When drawing polymers and repeating units, bond must cross the parenthesis



Properties of polymers

- Strong due to macromolecule nature

- Strength due to length, entanglement, IMFs,

Crosslinking

↑ branches of polymers "linking" together

branches entangle from different polymers

LDF

- No mobile charged particles

- Low thermal and electrical conductivity

Polymers

Polymers are molecular covalent but very high boiling and melting point (length and massive)

Properties depend on chain length, type of forces, type of monomers

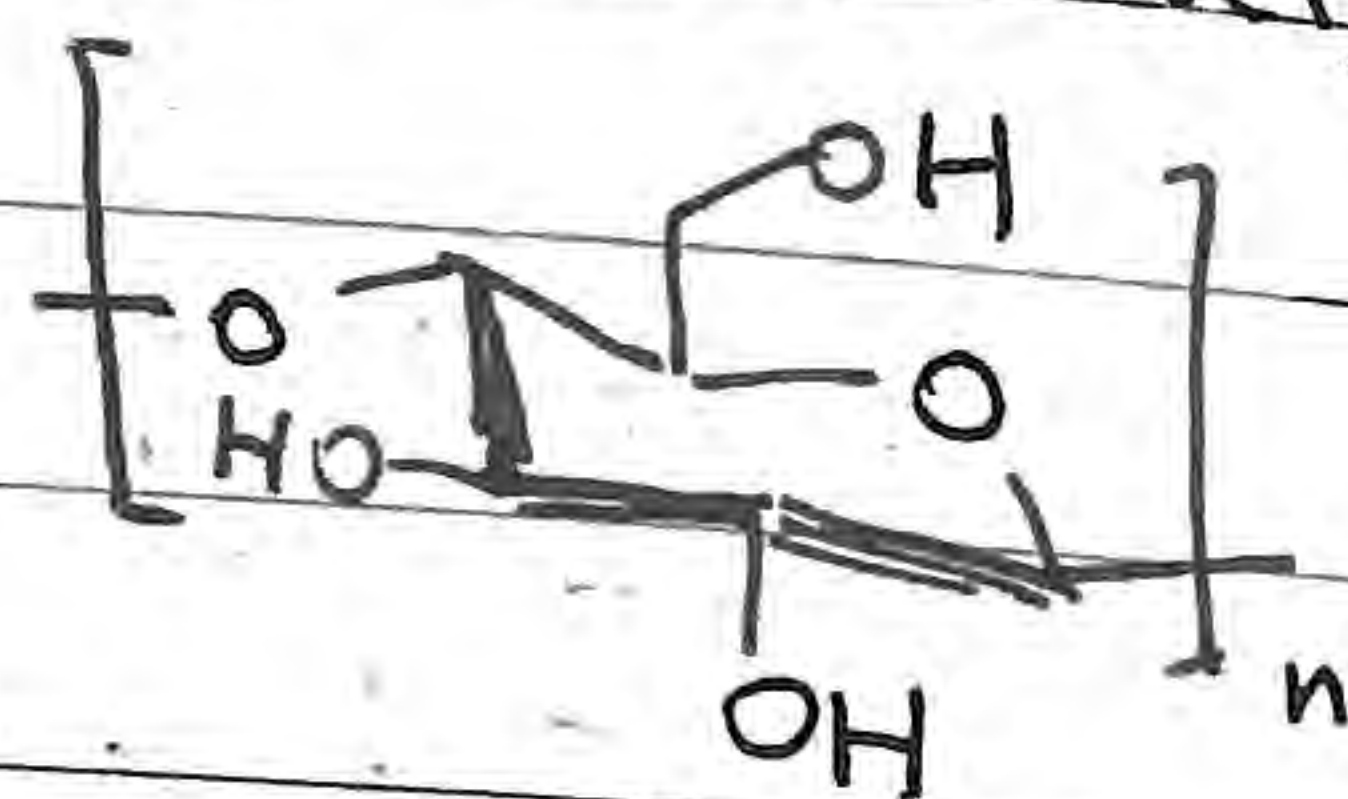
the [↑] branches attached

Starch (glucose polymer)

- Many energy store in plants
- Less strong (can be digested)
- Fewer hydrogen bonds (irregular branched structure)

Cellulose (carbohydrate)

- Natural linear polymer
- Strong (due to hydrogen bonding)
- Insoluble in water (prefer Hbond with self than water)



Synthetic Polymers

- Many are derived from crude oil
- E.g. PET bottles, nylon textiles, teflon

- Some are semi-synthetic

Renewable biological materials



Plastics

synthetic polymer (carbon and hydrogen)

• Unreactive

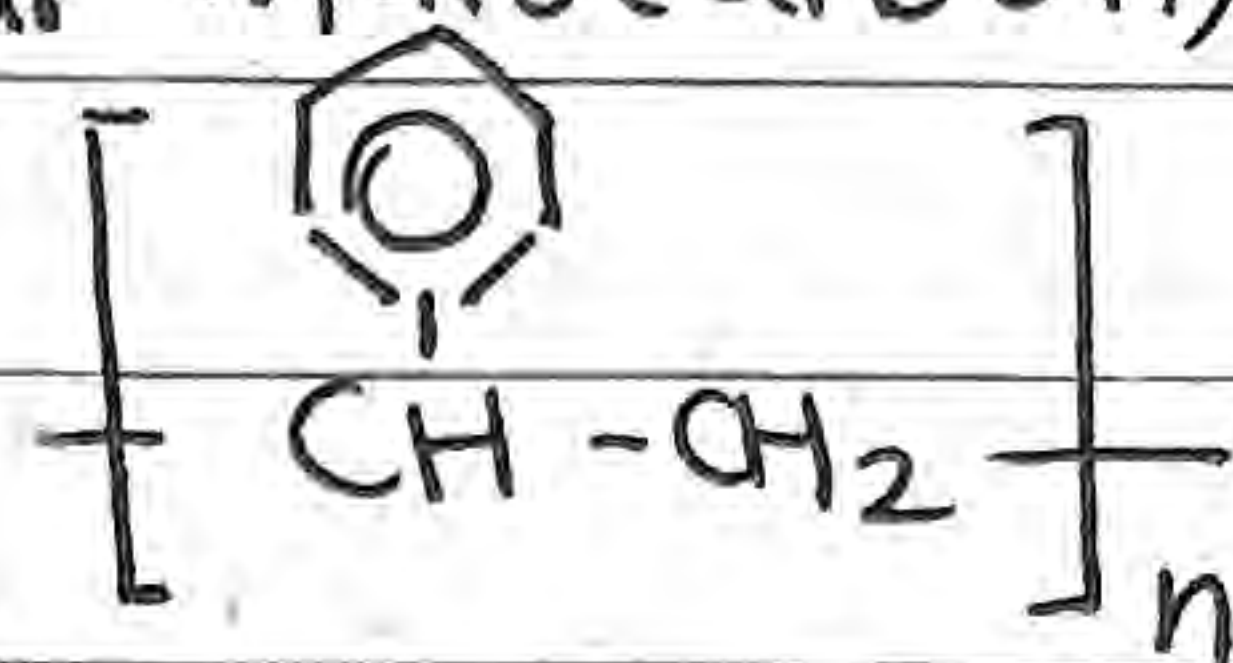
* Poly(phenylethene) or Polystyrene

Low solubility in water (nonpolar hydrocarbon)

Lower melting point than other polymers

Very brittle (large phenyl group)

Thermally insulating (non thermally conductive)



Plastics have strong covalent bonds

• Low reactivity means hard for bacteria to enter

Crosslinking - Polymer chains can covalently bond together

* How to make biodegradable?

• Facilitate bacterial action by having increased surface area, shorter polymer chains, and functional groups attractive to bacteria

• Add starch granules (absorb water and increases surface area).

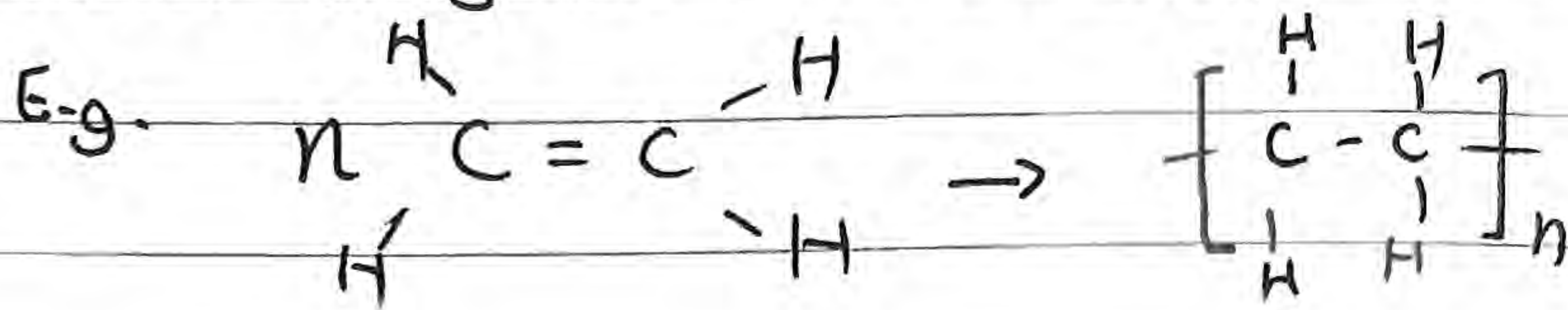
Microplastic (<5nm)

• Small fragments of plastic broken down over time

• Not biodegraded (unreacted)

Condensed structural formula

- Write diagram as one line



Condensation Polymers

E.g. Nylon, cellulose formed through condensation

Produced from monomers with reactive functional group on ends

- Unlike addition polymerization, condensation always has byproduct

Esterification

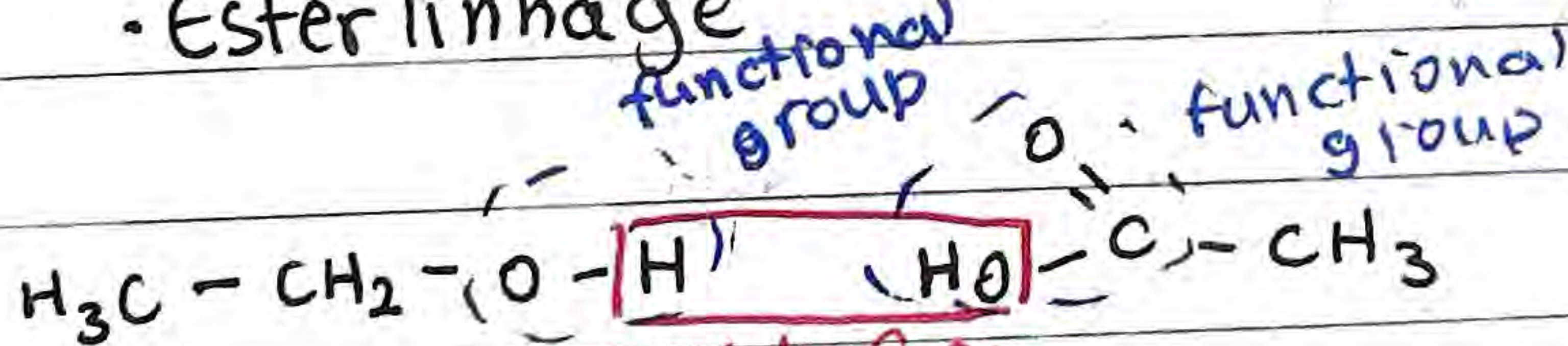
Esters formed in condensation reaction between alcohol and carboxylic acid

⇒ Hydroxyl (-OH) and Carboxyl (-COOH) groups react and release H_2O

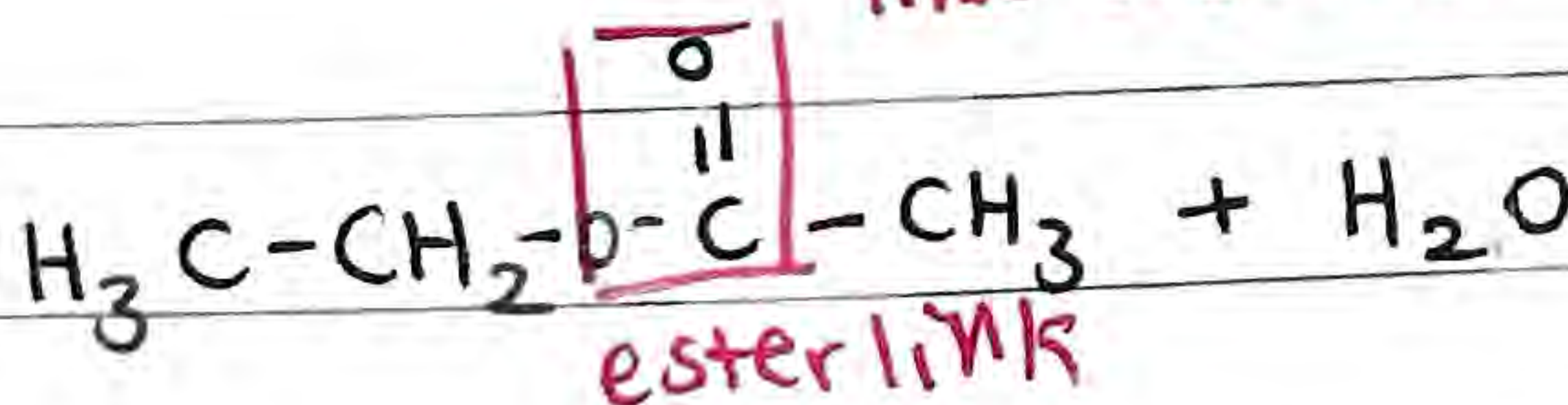
but can also be -COCl etc.

- Ester linkage

E.g.



react to form water and other parts link together

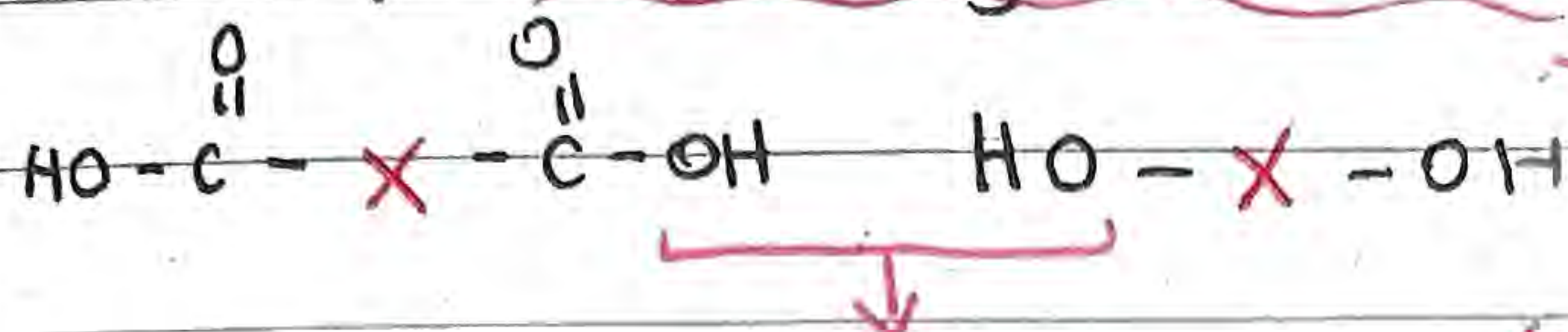


Ethyl Ethanoate

Condensation Polymers (naming through Poly- (link type))

- To form polymers, monomers must have reactive functional groups on both ends

(polymer of alternating monomer group)



Polyesters:



typically
-COOH but
can also be
others like
-COCl

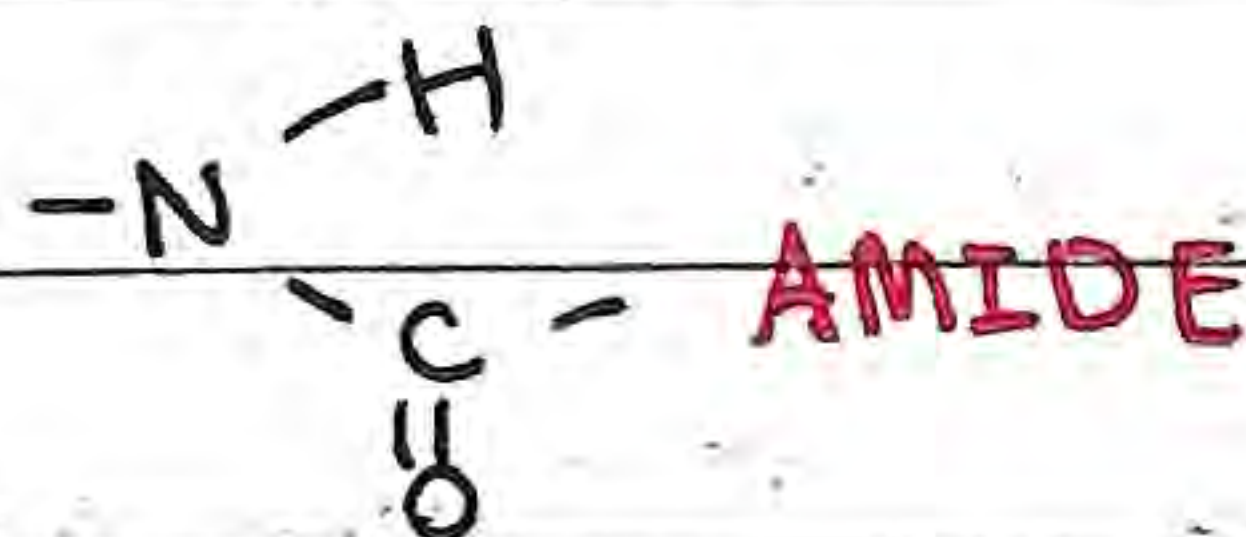
This is condensation reaction between 2 different monomers

Polyester link

Other types of links include...

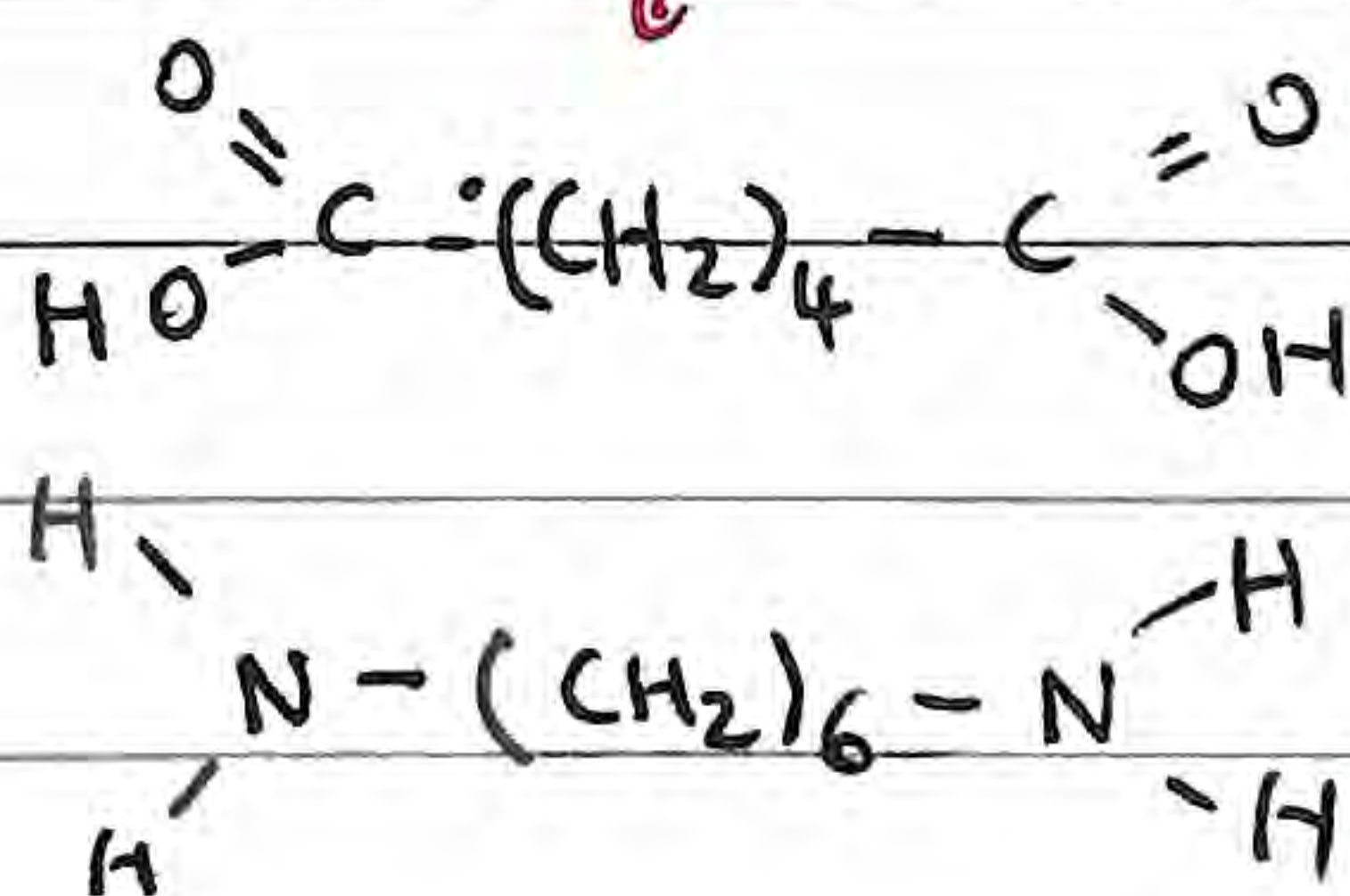
Polyamide link

Polyether (-OH and -OH)



formed between

can also
be with other
than -COOH
like -COCl



only AMIDE
when between
two (not one)

Condensation of same monomer

- Different functional groups on both ends

Eg. 3-hydroxy pentanoic acid with
(-OH) and (-COOH)

Polymers

Proteins are biological polymers
(metabolic processes)

polypeptide is single monomer
↓ (polyamide is multiple but both are NH_2, COOH)

Proteins are polypeptides (peptide link)

Polymers are made of monomers called 2-amino acid
(link between $-\text{NH}_2$ and $-\text{COOH}$)

Hydrolysis

all biological macromolecules form from condensation, break from hydrolysis

• Reverse of condensation (which formed a molecule too)

⇒ Taking in the molecule to break up the polymer

Monosaccharides are polymerized via condensation reaction

• Formed again by reverse hydrolysis (condensation)

• Two ends with hydroxyl group

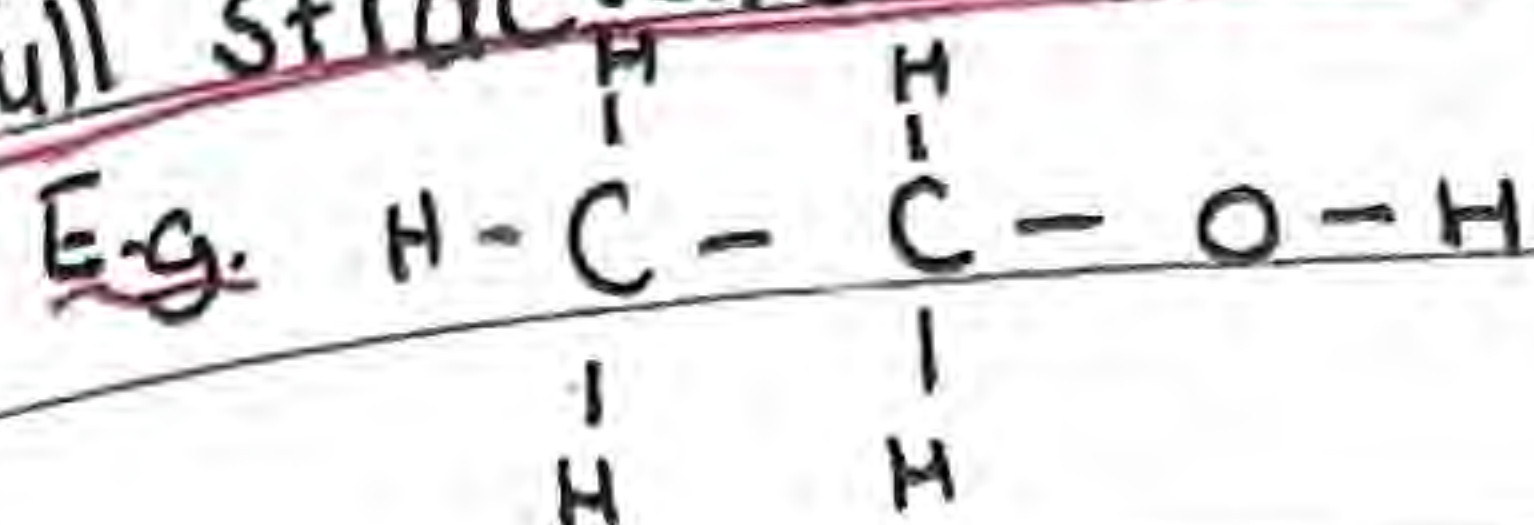
• Form glycosidic link

(polysaccharides = carbohydrates)

Organic Chemistry

Organic compounds can be represented in multiple ways.

Full structural formula: All bonds, all atoms



Condensed structural formula: Describes "main branch"



Skeletal formula: Ignore Hydrogens, carbons are "vertices"



Naming:

Meth - = 1

Eth - = 2

Prop - = 3

But - = 4

Pent - = 5

Hex - = 6

Hept - = 7

Oct - = 8

Non - = 9

Dec - = 10

Single bond = -ane (Alkane)

Double bond = -ene (Alkene)

Triple bond = -yne (Alkyne)

Homologous series - Series of organic compounds of the same trend (e.g. alkanes) but increasing number of carbons.

• Same ending, similar chemical properties, trend of physical properties, same general formula

Functional Groups - An atom or group of atoms that give an organic molecule its physical and chemical properties

Amides

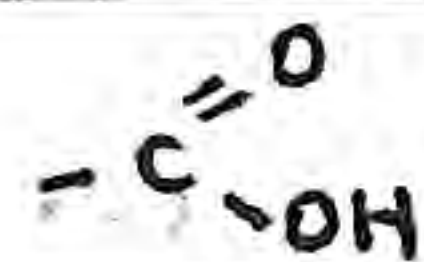
Hydroxyl

-OH

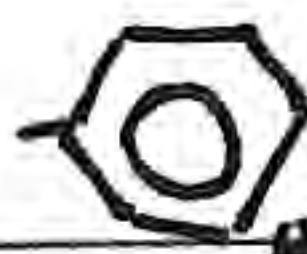
(forms alcohol)

Carboxyl

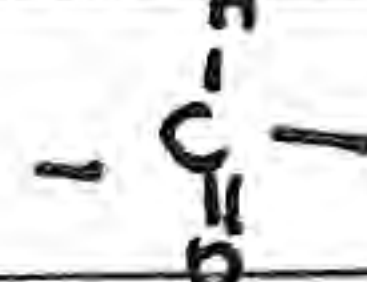
-COOH



Phenyl



Carbonyl



Halogeno

-Br, -F, -Cl, etc.

(homologous series)

Functional NameGroup FormulaSuffixClassExampleHalogeno $R-X$ Bromo-
chloro-
etc., prefix

halogenoalkanes

chloromethane

 H_3C-Cl Hydroxyl $R-OH$

-ol

alcohols

ethanol

 $H_3C-\underset{\substack{| \\ H}}{\overset{\substack{| \\ H}}{C}}-OH$ Carbonyl $R-\overset{\overset{O}{\parallel}}{C}-H$

-al

aldehydes

(position # always 1)

ethanal

 $H_3C-\overset{\overset{O}{\parallel}}{C}-H$ Carbonyl $R-\overset{\overset{O}{\parallel}}{C}-R'$

-one

ketones

must have at least one C on each side (or else it becomes an aldehyde)

propanone

 $H_3C-\overset{\overset{O}{\parallel}}{C}-CH_3$ Carboxyl $R-\overset{\overset{O}{\parallel}}{C}-OH$

-oic acid

carboxylic acid

ethanoic acid

 $H_3C-\overset{\overset{O}{\parallel}}{C}-OH$

Consists of alcohol + carbonyl

Alkoxy $R-O-R'$

-oxy-

ethers

methoxymethane

 $H_3C-O-CH_3$

The longest of the two is the main branch (the -ane or -ene, etc.)

Amino $R-NH_2$

-amine

primary amines

ethanamine

 $H_3C-\underset{\substack{| \\ H}}{\overset{\substack{| \\ H}}{C}}-NH_2$ Amido $R-\overset{\overset{O}{\parallel}}{C}-NH_2$

-amide

amides

ethanamide

 $H_3C-\overset{\overset{O}{\parallel}}{C}-NH_2$ Ester $R-\overset{\overset{O}{\parallel}}{C}-O-R'$

-oate

esters

methyl ethanoate

 $H_3C-\overset{\overset{O}{\parallel}}{C}-O-CH_3$

R is always -oate, R' is always -yl

phenyl $R-\text{C}_6\text{H}_5$

-yl/benzene aromatics

phenyl-

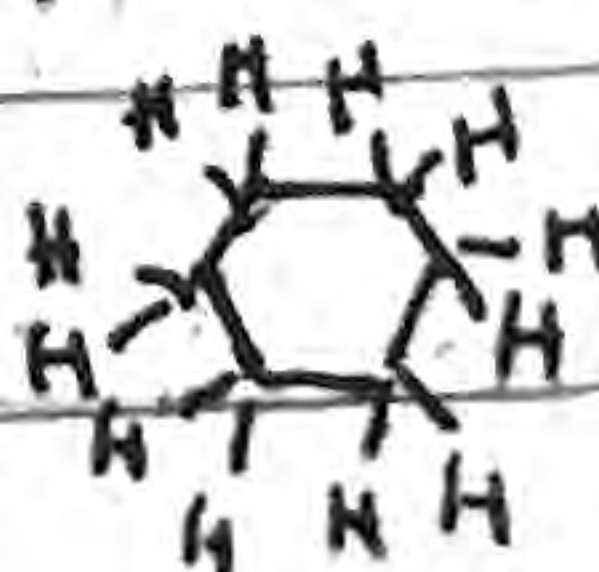
methylbenzene

 $H_3C-\text{C}_6\text{H}_5$ $R-NO_2$

Nitro

Cyclo-

cyclohexane



Naming Organic Compounds

It is usually necessary to add positioning before prefix/suffix when naming organic compounds.

- The position # should be minimized (in general)
- Position on the main (longest continuous carbon chain that contains all functional groups present)

- Position # should prioritize alphabetical order (chloro before fluoro) and specific functional groups

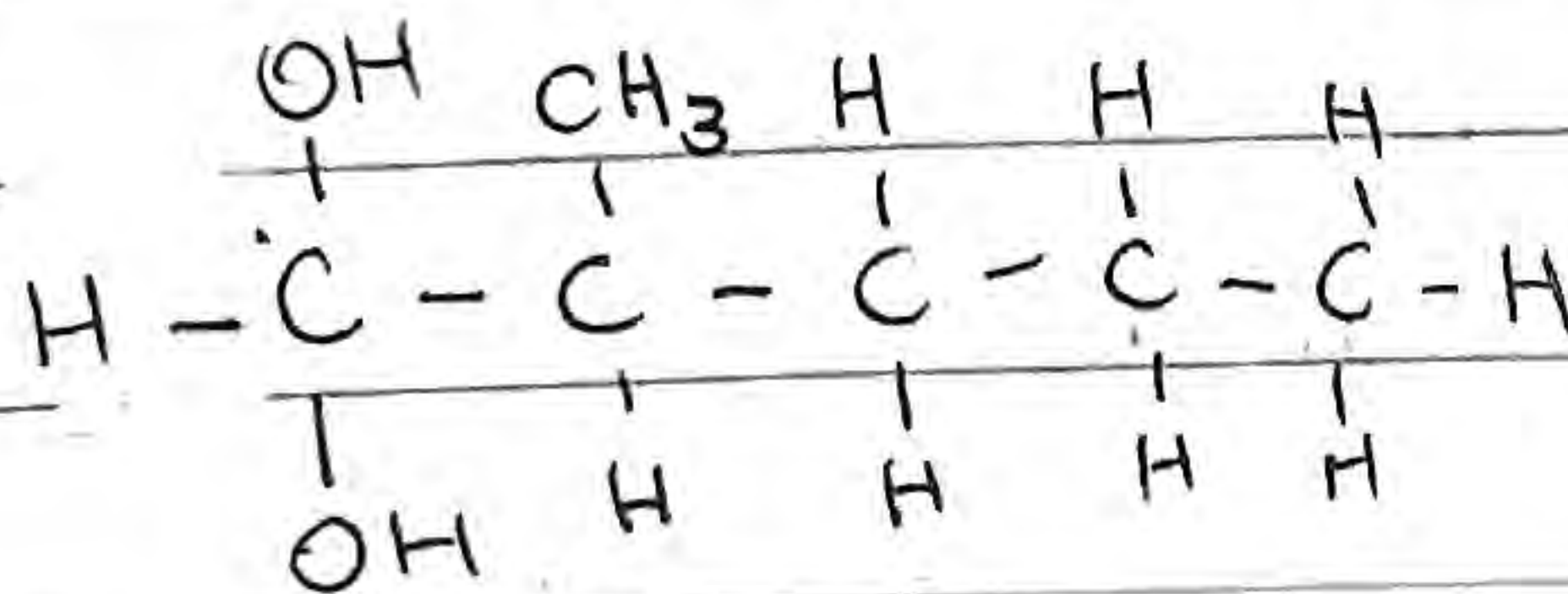
E.g. carbonyl should always be 1 in aldehydes

of the functional group, exclude "di", "tri", etc. (overall should be minimized first but if not possible same then go with functional groups)

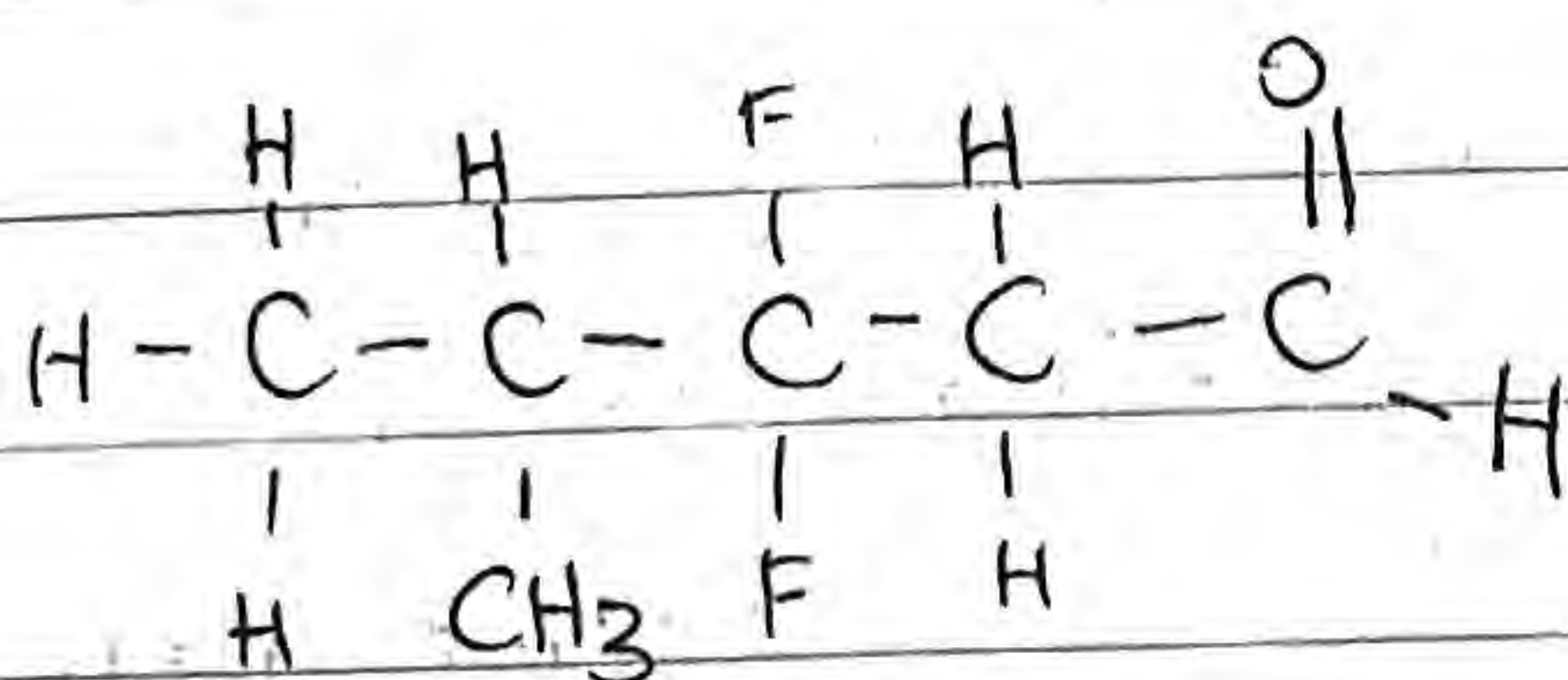
- When multiple of the same functional group is present, use prefixes (and make sure position of all of them are present)

di = 2
tri = 3
tetra = 4
penta = 5
etc.

E.g.



2-Methyl butan-1,1-diol



3,3-Difluoro-4-methylpentanal

Amides, carboxylic acids, and aldehydes do not need

#s in their name since they're always 1

when carbon branch is longer than benzene's # of carbons

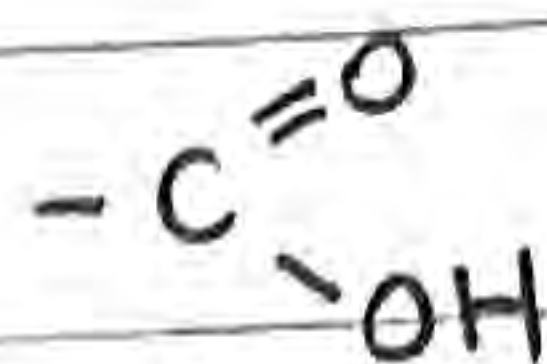
For phenyl groups, when the main branch is benzene, use

-benzene, if the main branch is hydrocarbon, then use phenyl-

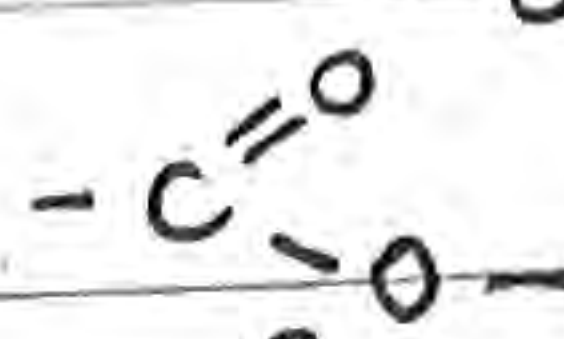
Functional group priority list

MOST PRIORITIZED

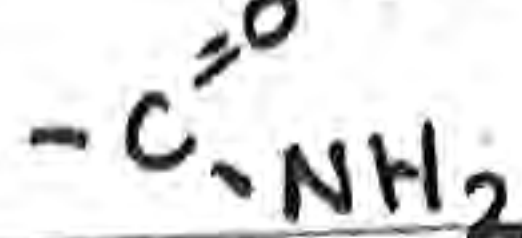
Carboxylic acid



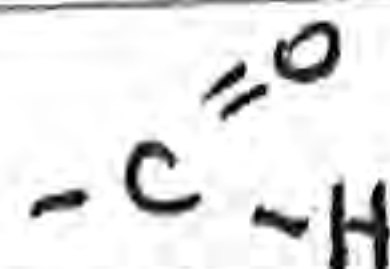
Ester



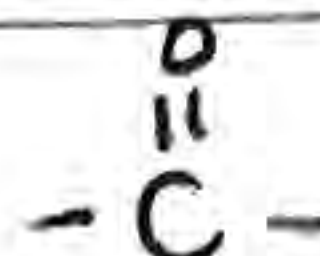
Amide



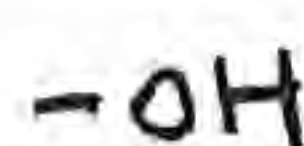
Aldehyde



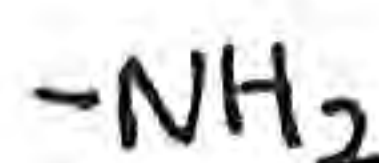
Ketone



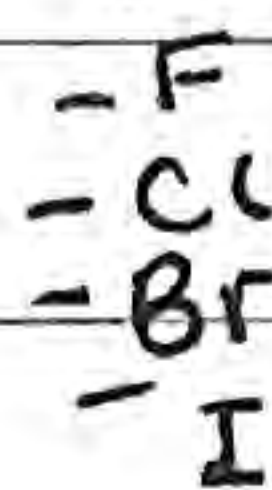
Alcohol



Amine



Halogeno



LEAST PRIORITIZED

↳ Alkyl $-\text{CH}_3$ etc.

Structural Isomers (constitutional isomers)

- Compounds with the same molecular formula but different structure (arrangement of atoms)
- Types: chain, positional, functional group
- Structural isomers should have a different structure (i.e. different IUPAC name)

E.g. Butane



2-Methylpropane



normal
cycloalkanes
are neither

Aliphatic - No aromatic rings and is not cyclic

Aromatic - Unsaturated and resonance

Homologous Series - Family of compounds that can be

same general
formula

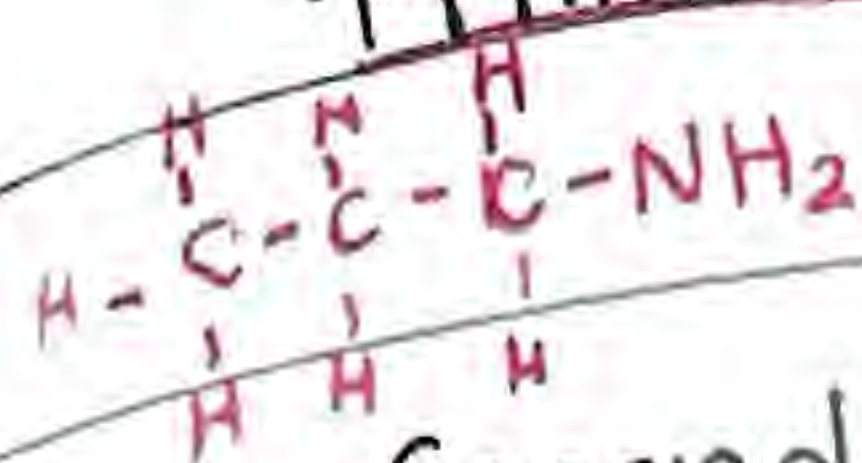
grouped based on similarities on structure/reactivity

Type of molecules

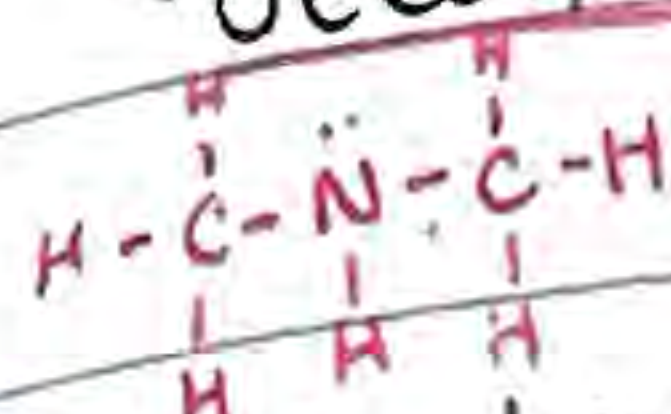
(applicable for amines, haloalkanes, alcohols)

sometimes it looks
at the carbon attached
to the FG
instead

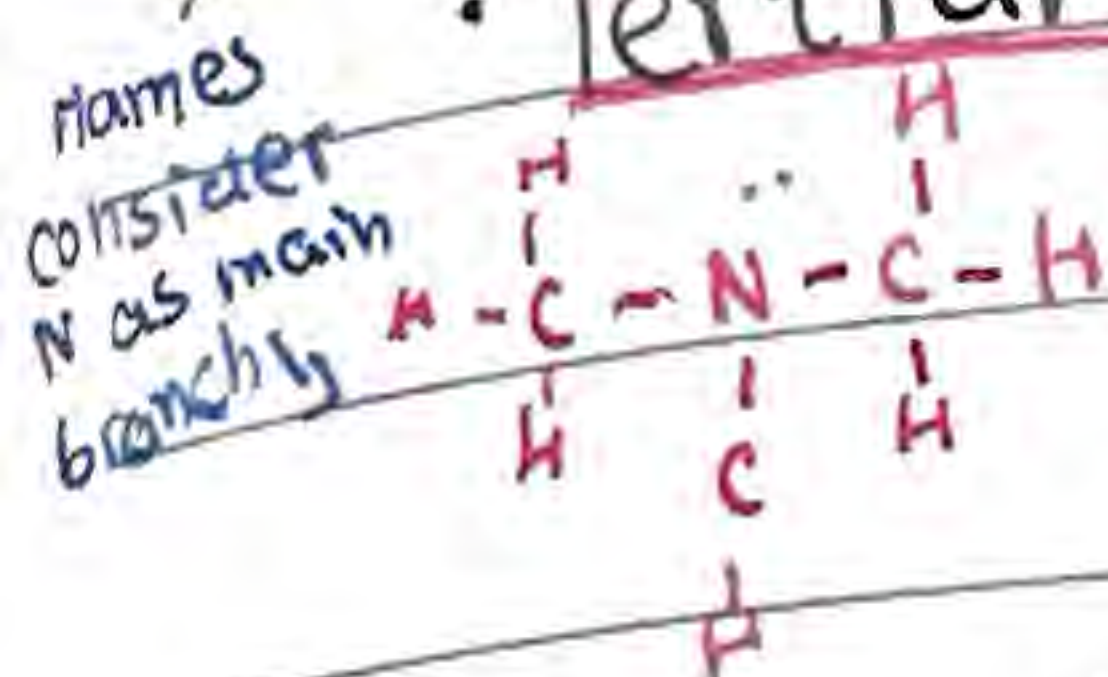
• Primary - The main atom of the functional group is attached to one carbon



• Secondary - The main atom of the functional group is attached to two carbons



• Tertiary - The main atom of the functional group is attached to three carbons



These differences can result in changes in reactivity

Physical Trends in Homologous Series

• Physical properties change gradually as chain gets longer (i.e. melting pt., boiling pt.)

⇒ Change in intermolecular forces

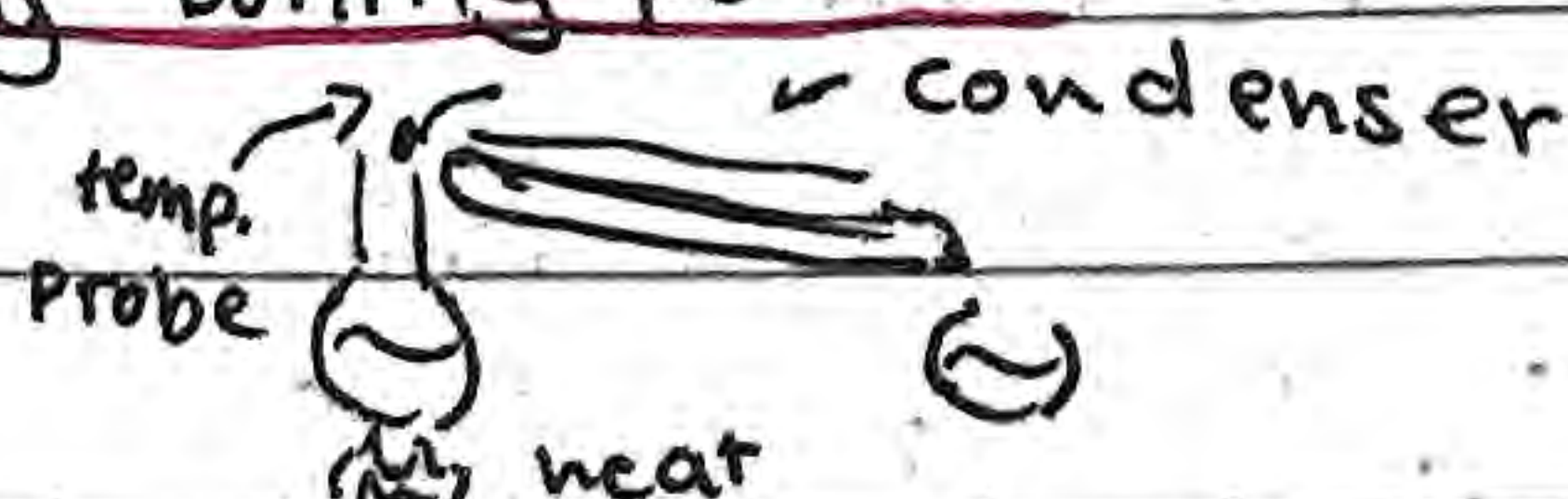
(London dispersion force increase as chain gets longer)

• Affects physical properties

E.g.	boiling pt.	melting pt.
Methane	-161°C	-183°C
Ethane	-89°C	-172°C
⋮	⋮	⋮
Pentane	36°C	-130°C
Hexane	69°C	-95°C

increasing
temperature

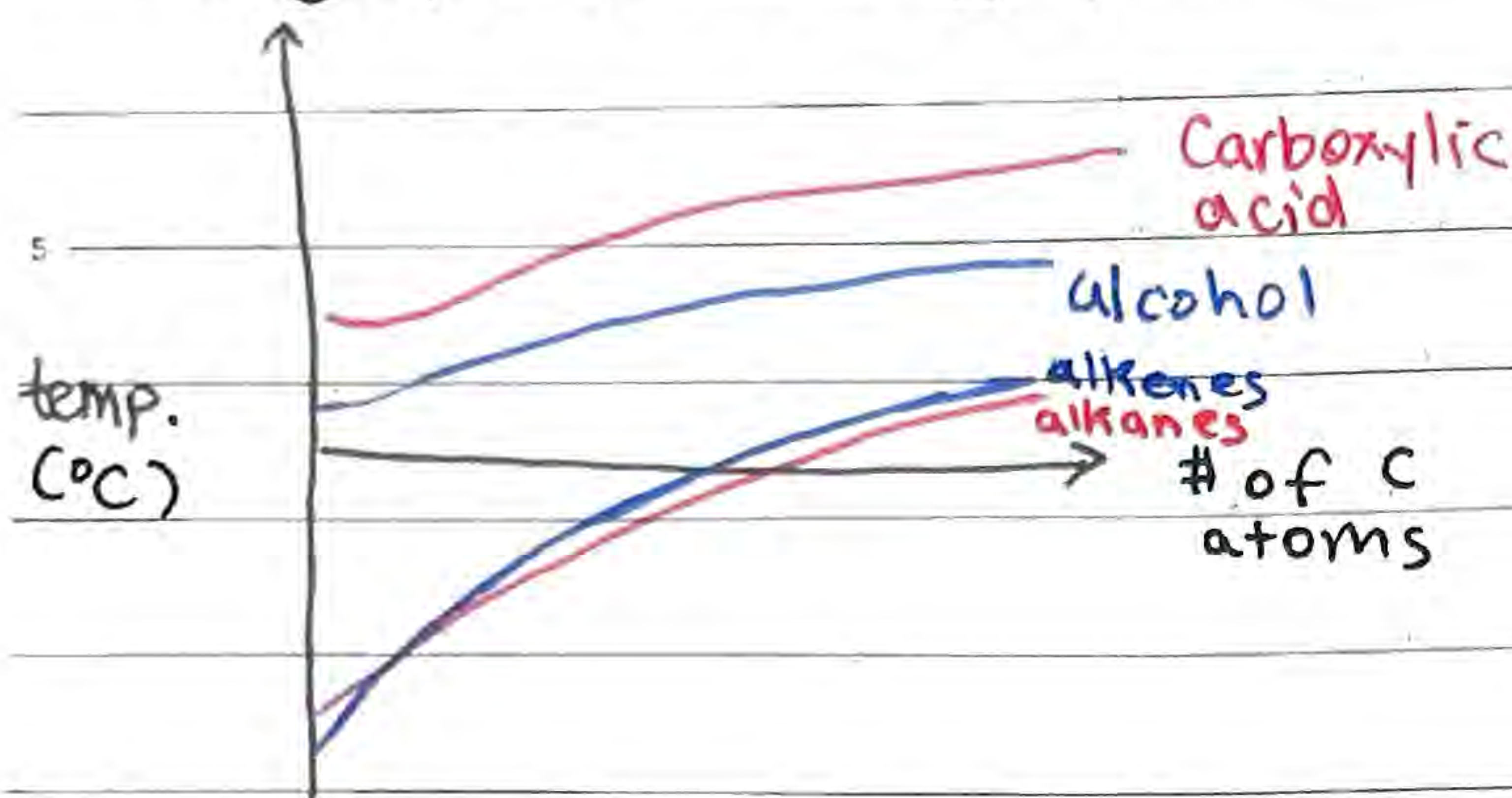
Measuring boiling points



* Density and viscosity
also increase as
chain length increases

Physical Trends

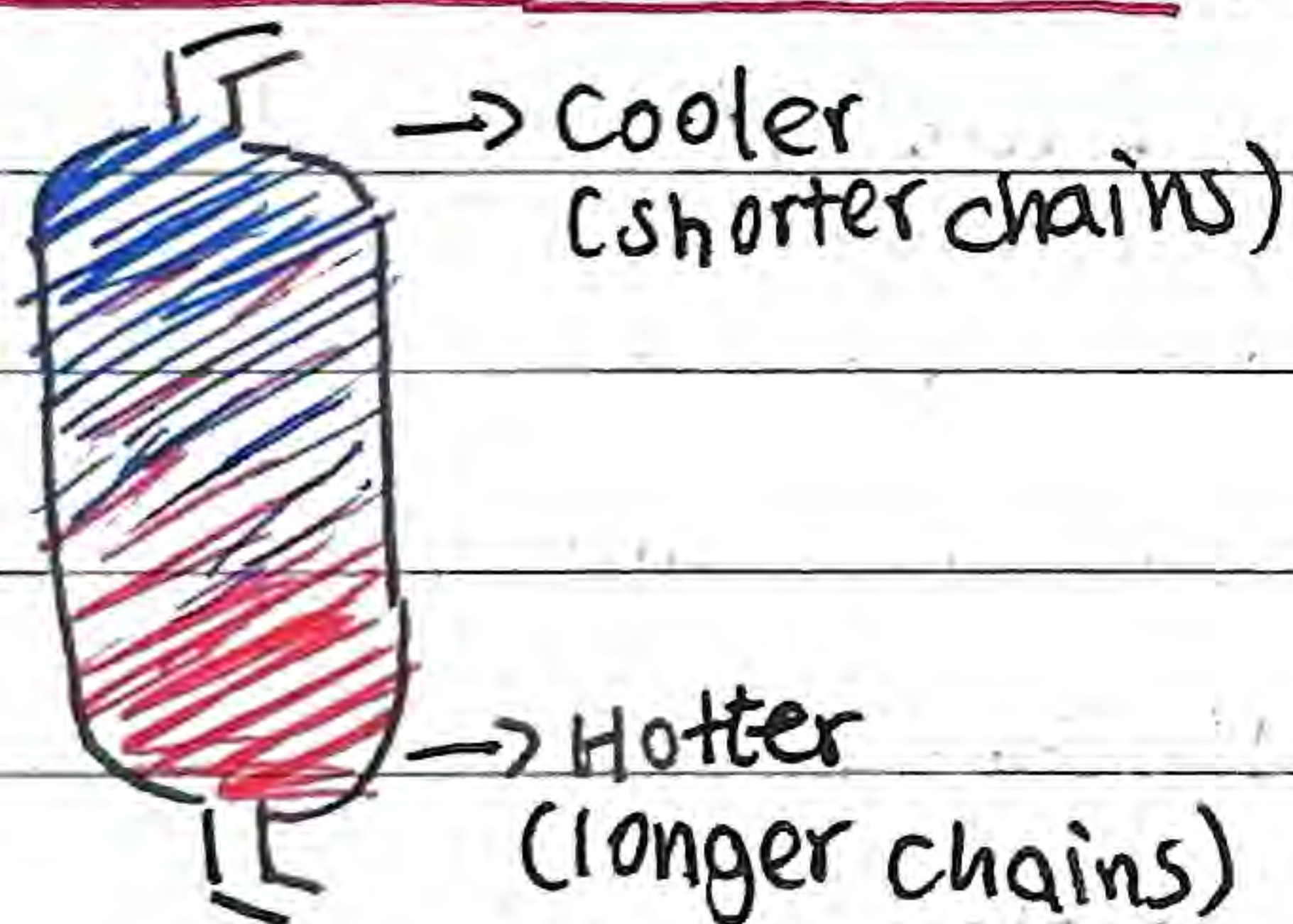
A graph for boiling points



Different boiling points in homologous series means condensation at different temperatures.

- Used to separate hydrocarbons in crude oil.

Fractional distillation is a physical separating process



Shorter hydrocarbon chains condense at the top

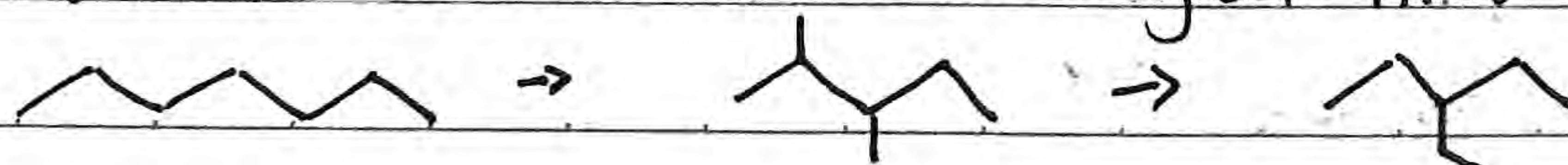
longer hydrocarbon chains condense at the bottom

Chain Isomerism - structural isomers where arrangement in carbon chain differs.

- Typically different chain lengths

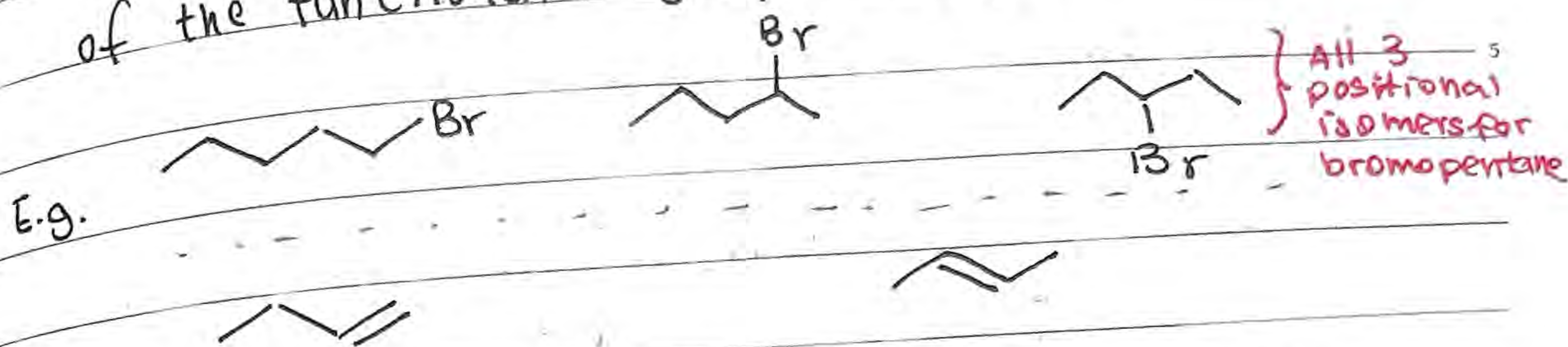
(some carbons are rearranged into branches)

E.g.



Positional Isomerism

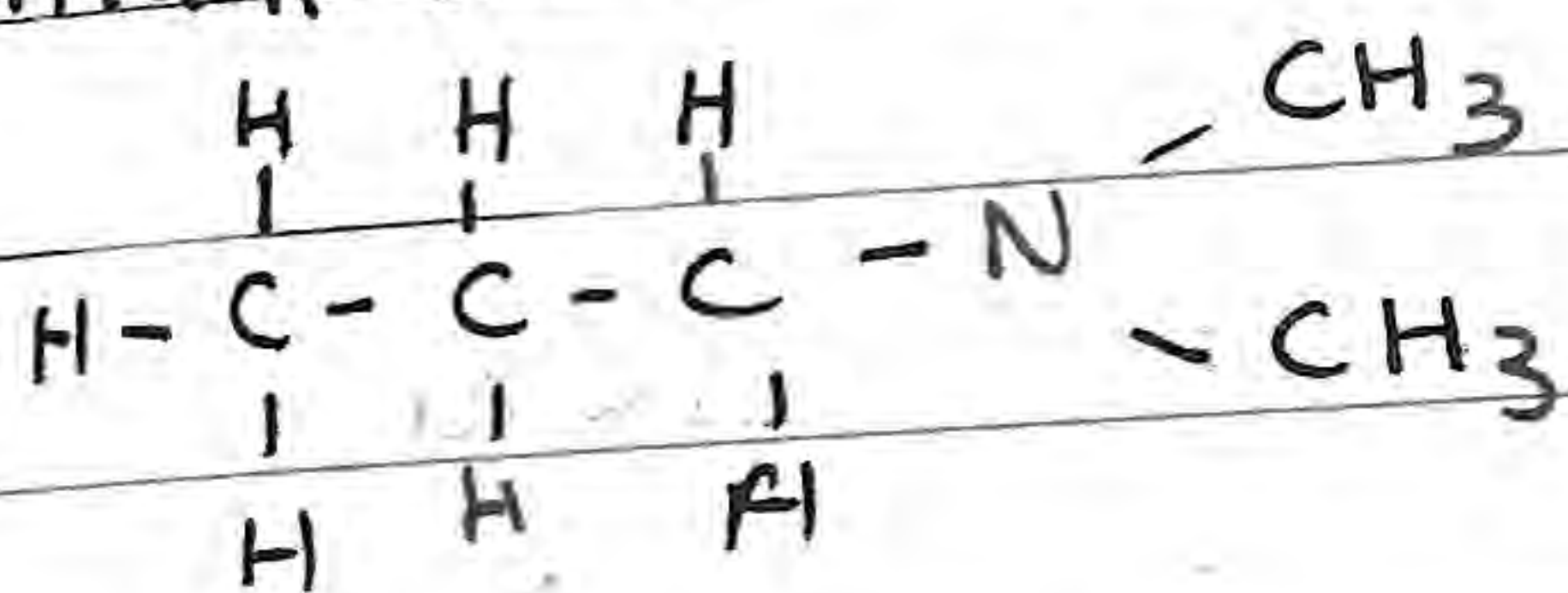
Refers to structural isomers where the position of the functional group changes



changing position of functional group affects chemical properties

* Sometimes, alkyl groups are directly bonded to a functional group

In that case, the position is the main atom of the functional group



N,N-dimethylpropanamide

Functional Group isomers

structural isomers with atoms arranged differently so they make different functional groups

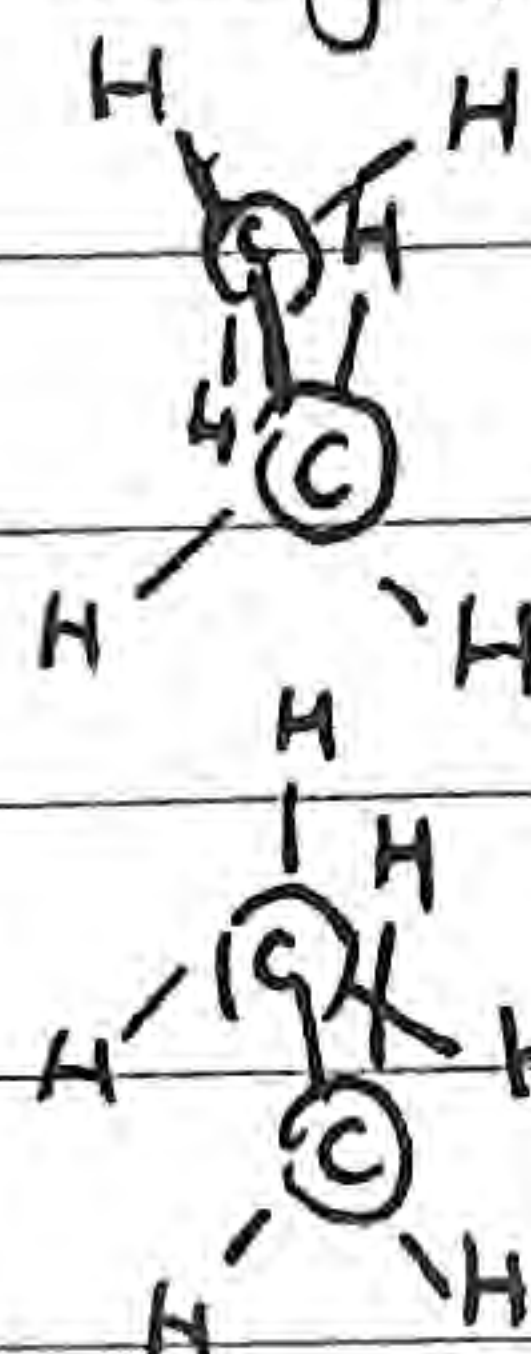
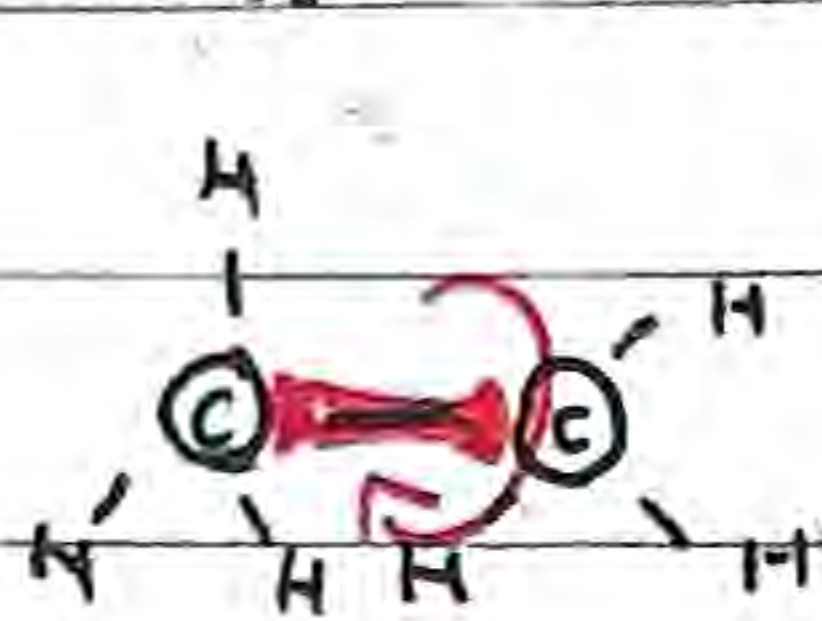
Typically...

- Alkoxy and hydroxy can form isomers
- Aldehyde and ketone can form isomers
- Carboxyl and ester can form isomers

Stereoisomers

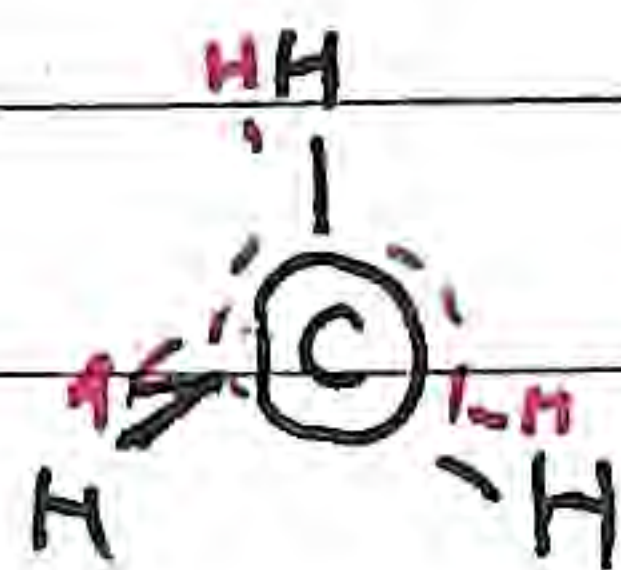
- Organic molecules of the same molecular formula but different spatial arrangement
- Conformational and configurational isomers

Notice that C-C bonds (single) can be rotated

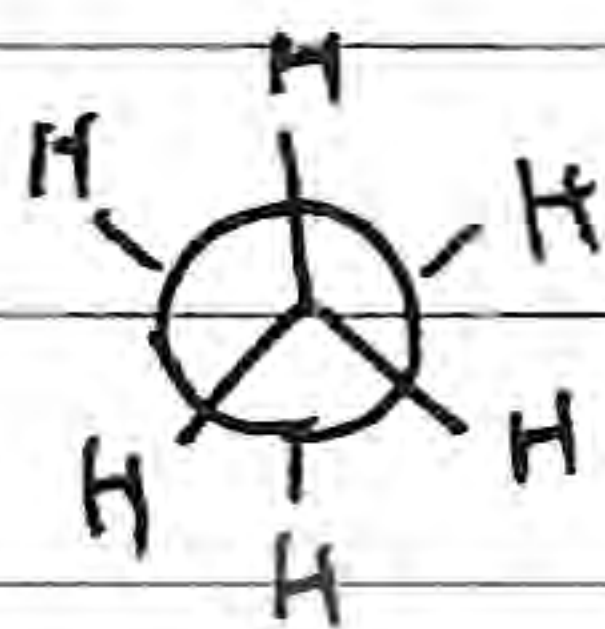
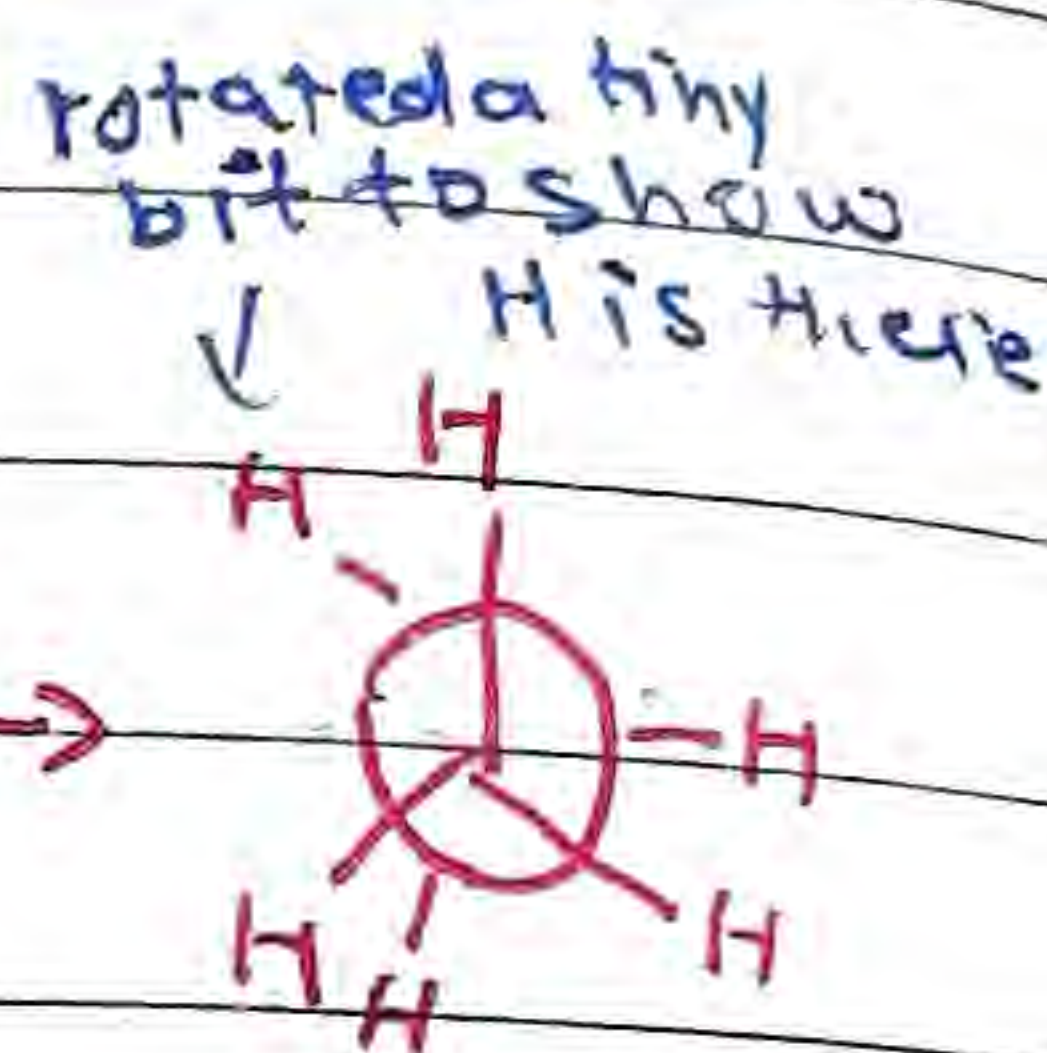


different rotations are also different isomers

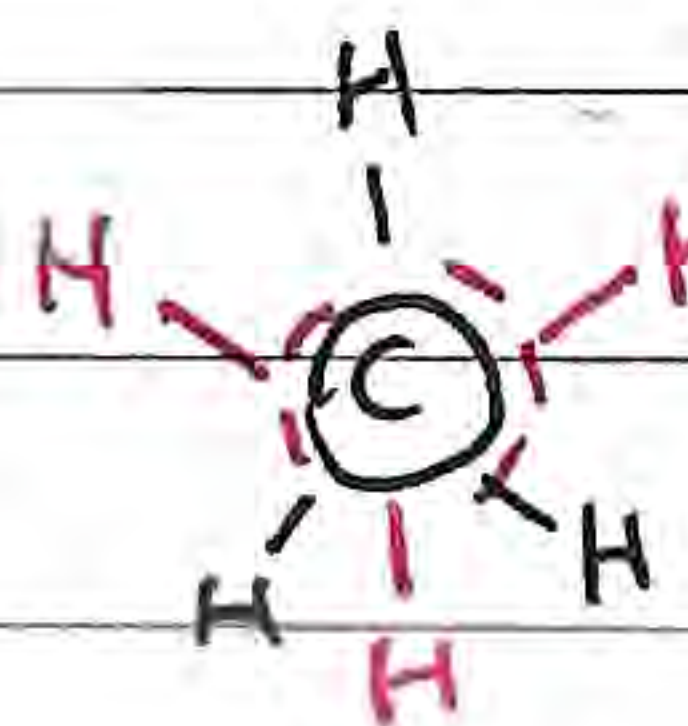
It is considered eclipsed when the "hydrogens" block the hydrogens behind it when viewed at an angle



Drawn as



Drawn as



It is considered staggered when the "hydrogens" are perfectly rotated such that none block other hydrogens.

This is known as conformational isomers (conformers)

- When 2 or more groups are bonding together by a single bond
- Rotation can occur to form these
- Does not involve breaking any bonds

Configurational isomerism

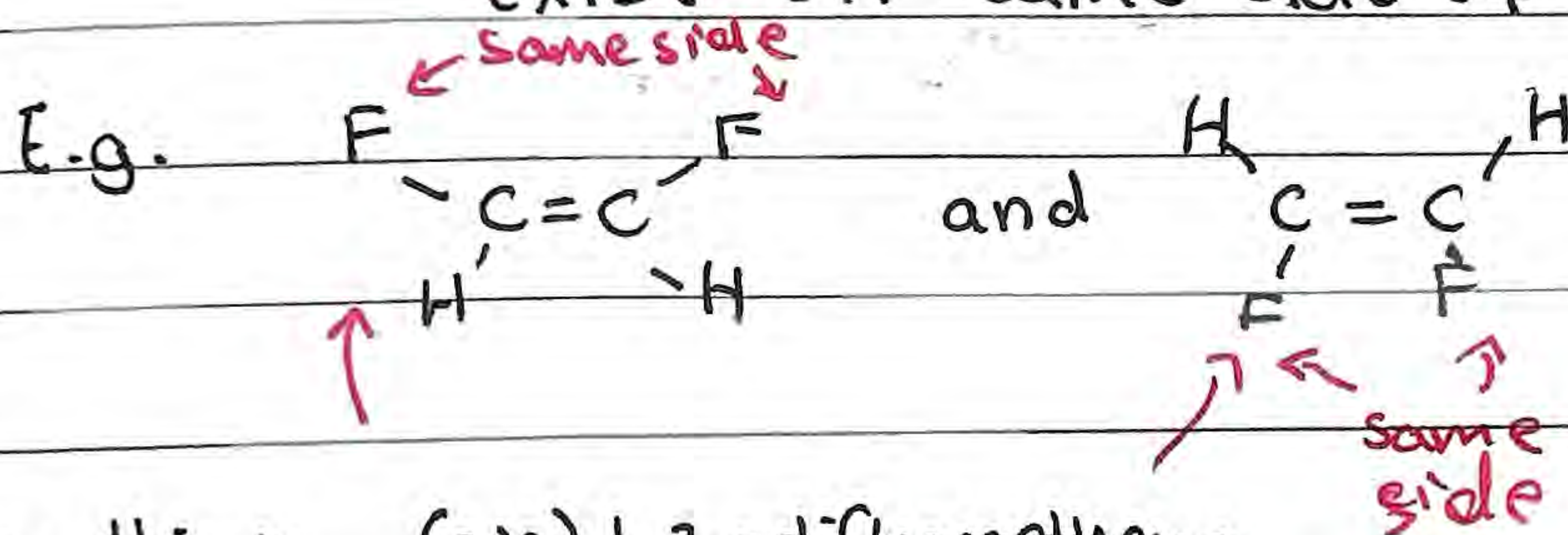
- created by breaking and remaking bonds
- Types: Cis/Trans, Optical isomerism

Cis-trans Isomers (only occurs in unsaturated)

- involves aliphatic (straight) alkenes

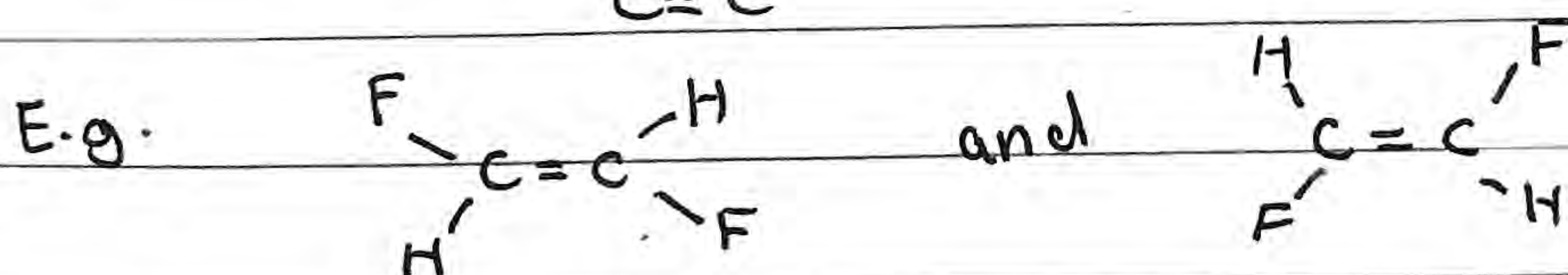
where groups attached to carbon double bond cannot rotate

Cis (same) - 2 identical substituents (referring to any group branched to a carbon) exist on same side of $C=C$



so this is (cis) 1,2-difluoroethene
(these two are the same since they're only physically rotated)

trans (different) - 2 identical substituents (functional groups) exist on different side of $C=C$

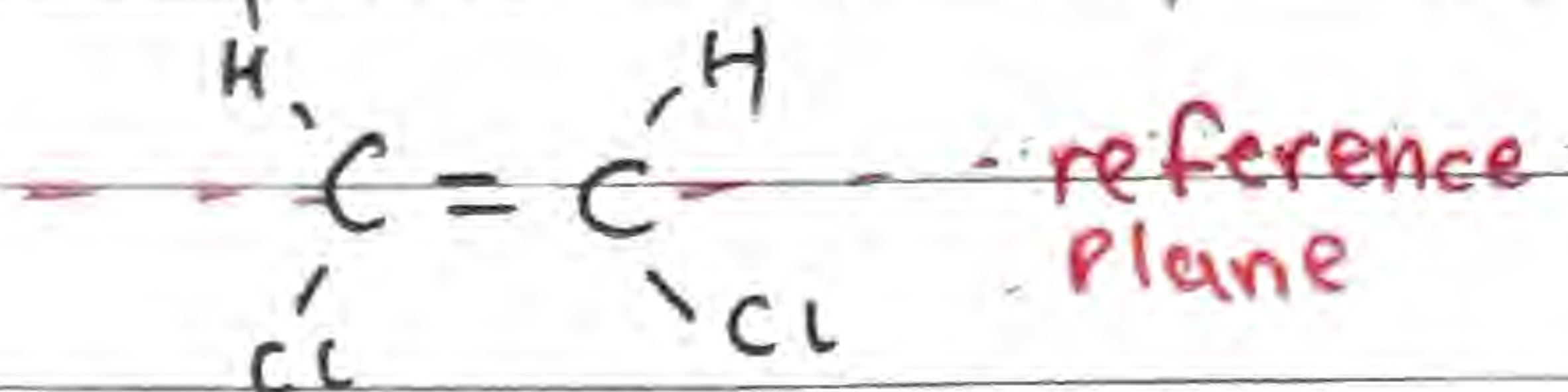


so this is (trans) 1,2-difluoroethene

- Basically, cis/trans are isomers where two organic molecules cannot change into each other through physical rotation

Cis/trans isomers

reference plane is usually along double bond

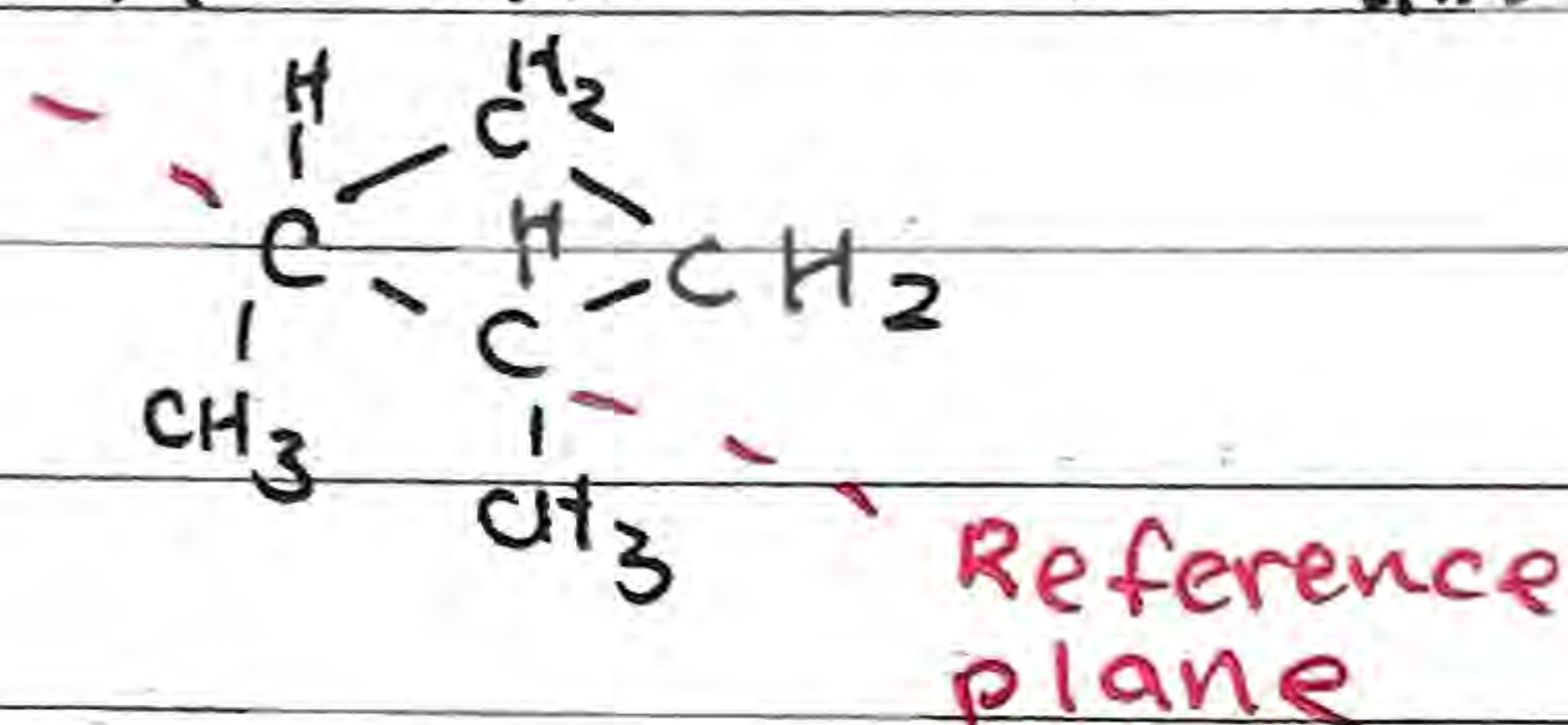


where substituents are attached on sides of reference plane

Cis/trans also occurs in cycloalkanes

(choose the bond with substituents on either carbons)

E.g.



Optical Isomers

• Chiral carbon is a carbon bonded to 4 different atoms/groups of atoms (tetrahedral) ↳ all has to be distinct

• Has a stereo center / asymmetric center CH₃
|
H-C-CH₃
|
H-CH₃ is not

• Optical isomerism occurs when molecules have one or more chiral carbons have configurational isomers.

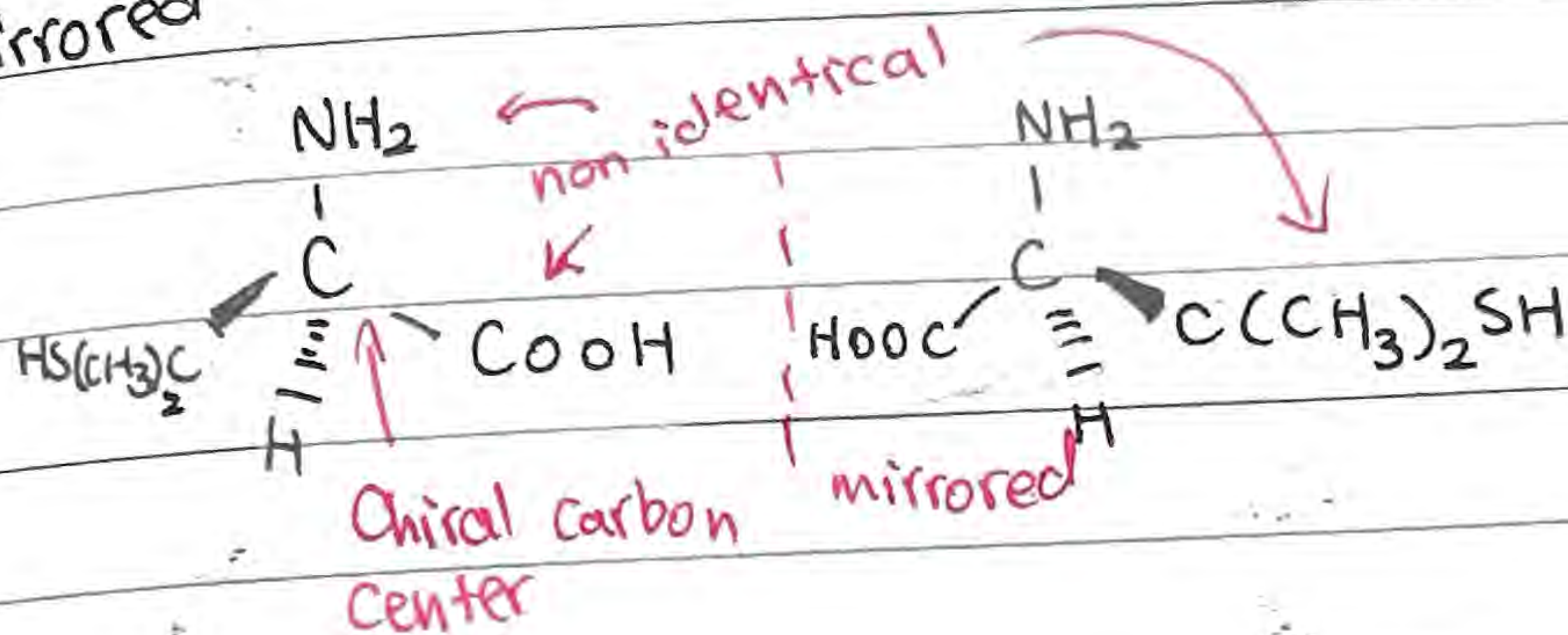
• Optical isomers are known as enantiomers

⇒ Mirror image (plane is the same so groups behind and in front of plane should not change).

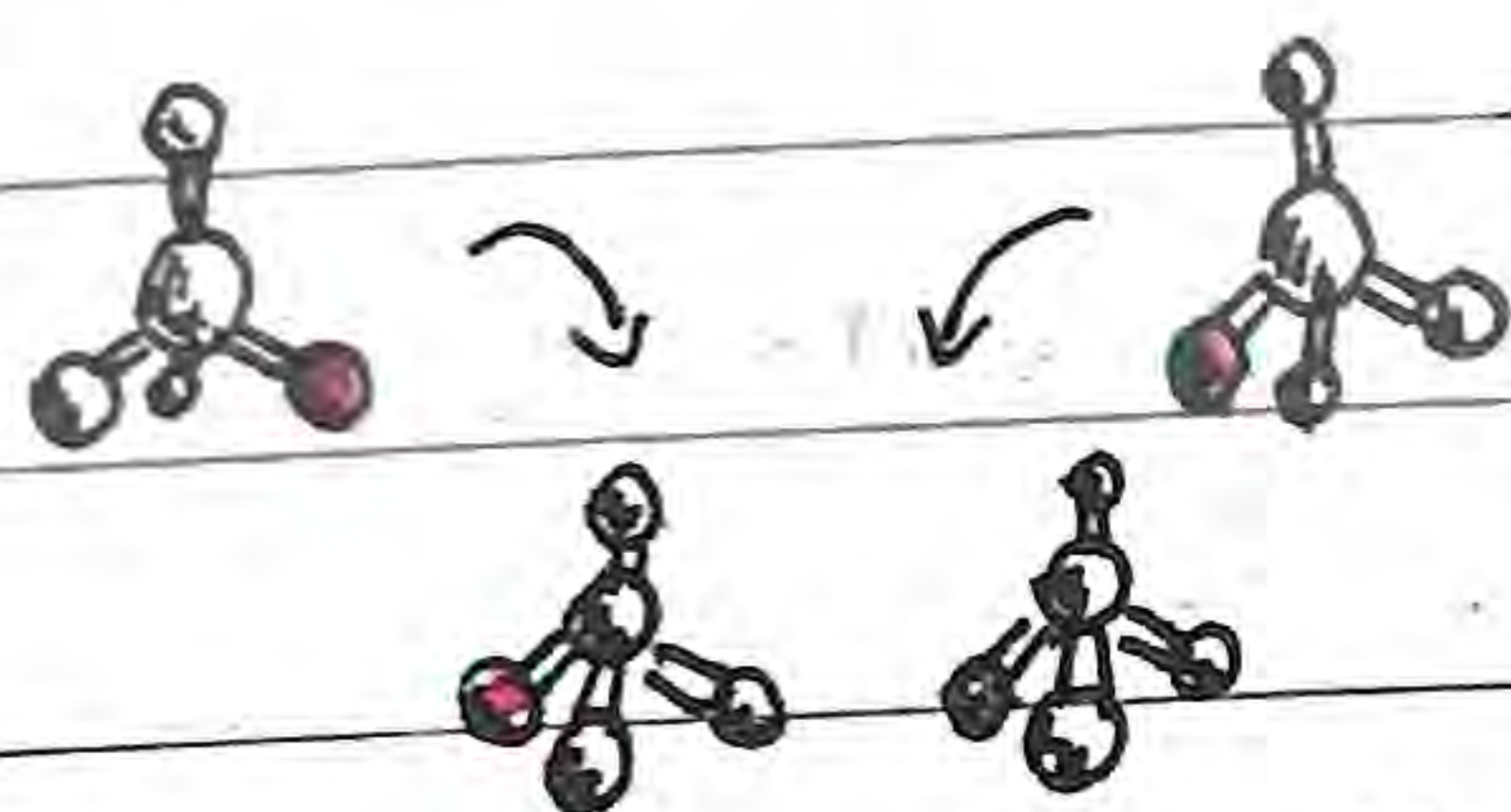
Optical isomers

when drawing enantiomers, use molecular geometry around the chiral carbon

- Drawing an enantiomer's mirrored version should be in the same plane and bonds are mirrored



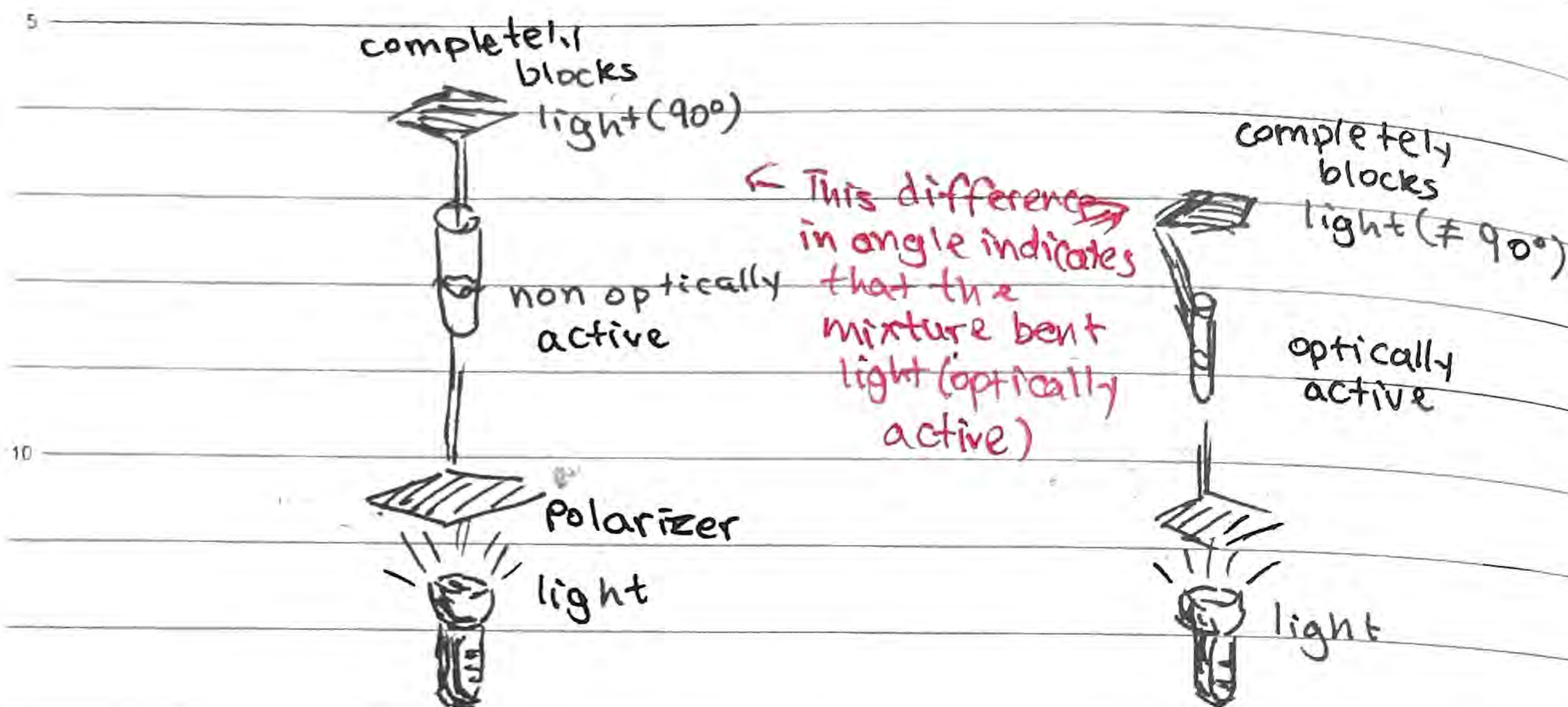
These molecules are non-superimposable (they are not rotationally congruent)



- Enantiomers are optically active
- Pair of enantiomers (under same conditions) will rotate plane polarized light by the same angle but in opposite directions
- One (+) enantiomer rotates light clockwise
- One (-) enantiomer rotates light anticlockwise
- Racemic Mixture - (50-50 composition)
- Does not rotate plane polarized light
- Non optically active mixtures also does not rotate plane polarized light

Optical Isomers

How do enantiomers rotate light?



Analytical techniques for organic chemistry

Mass spectrometry

- Previously used to calculate relative atomic mass
- Higher abundance isotopes are typically more stable (peak in mass spectrometry)

• When used on organic molecules, the molecules are broken into fragments (ion) $\Rightarrow$ shown on spectrometry

1. Samples vaporized
2. Hit with high energy electrons
3. Removes electron and creates positive ion
4. Ions accelerated from magnetic field
5. Deflected due to ion

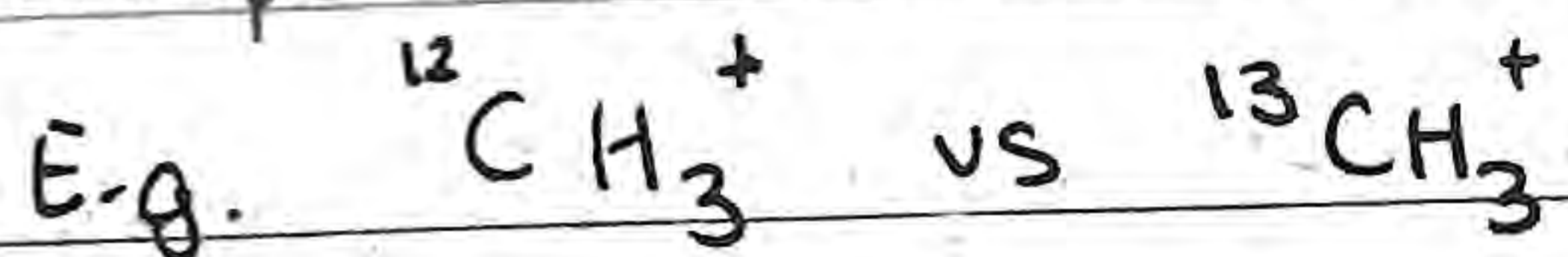
Mass Spectrometry

- Amount of ion deflected is inversely proportional to its mass/charge ratio (usually charge is $1+$ for simplicity)
- Mass spectrometry graph reflects the relative abundance for each m/z (type of fragment)

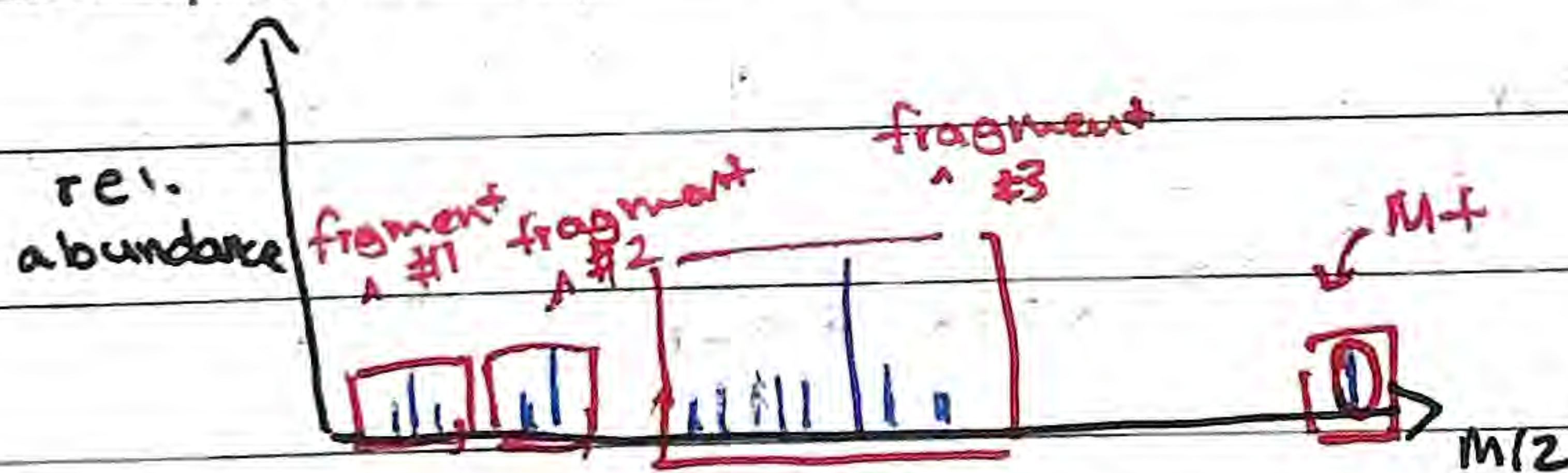
So how do you visualize data and thus see entire compound?

- Look at the peaks! (main fragments)

⇒ There are $\pm$ charges near peaks due to isotopes of the individual atoms in the fragment



⇒ You can spot peaks by looking at the maximum of each "bundle"



• The line with the highest mass to charge ratio is the original parent molecule (furthest right)

not necessarily greatest peak

⇒ This is known as the Molecular Ion Peak (M^+)

Every fragment is $1+$ since mass spectrometer ionizes the fragments

Infrared Spectroscopy

• Analytical technique that identifies types of bonds in organic compounds

⇒ As such can determine the functional group

• Releases IR radiation, absorbed by bonds and they vibrate ⇒ shown on graph

• IR spectroscopy is a type of vibrational spectroscopy

Types of vibration: stretch, bend (diatomics can't bend)

Frequency of vibration depends on bond enthalpy and the masses of atoms in the bond

• Higher bond enthalpy refers to stronger bond

⇒ restricts amount vibrated so shorter wavelength and higher frequency

• Larger masses means lower frequency of vibration

Absorption is described by wavenumber, $\bar{\nu}$

where $\bar{\nu} = \frac{1}{\lambda}$

i.e. H-Cl

H-Br

H-I

Bond enthalpy

frequency

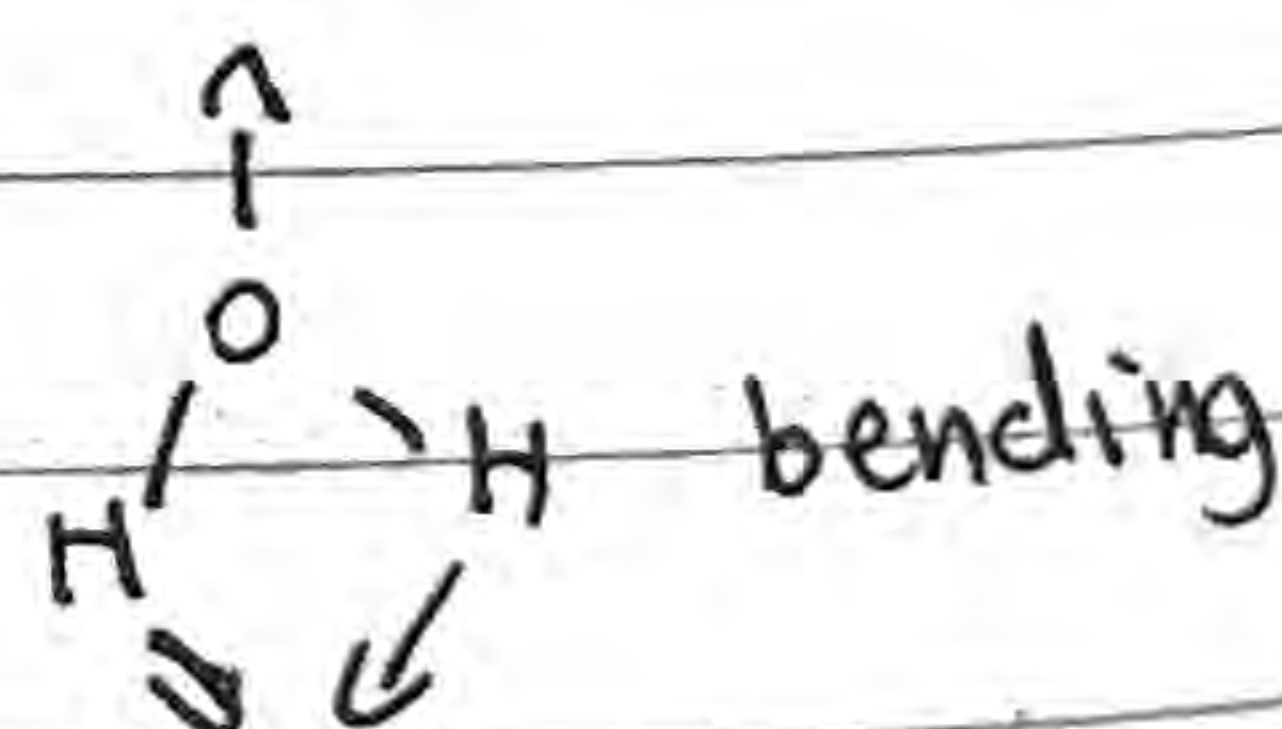
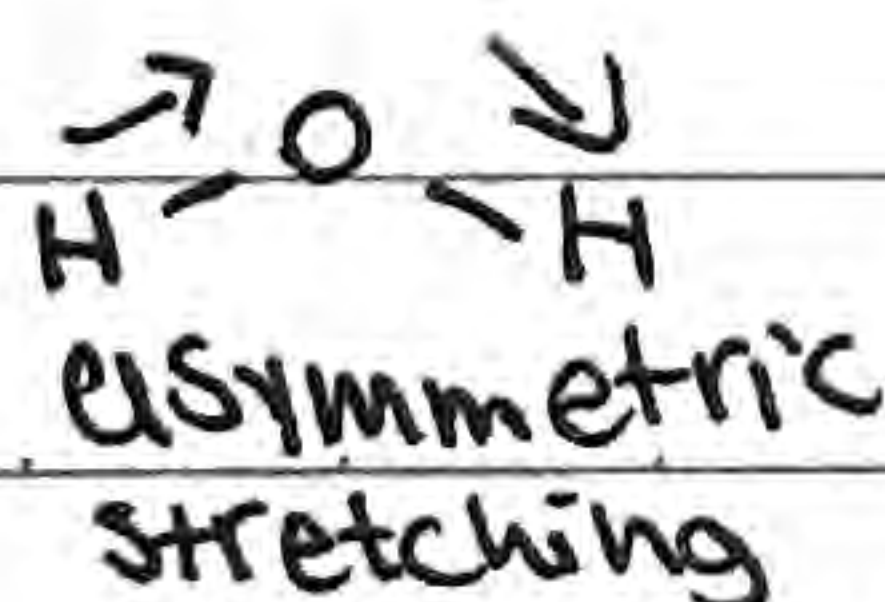
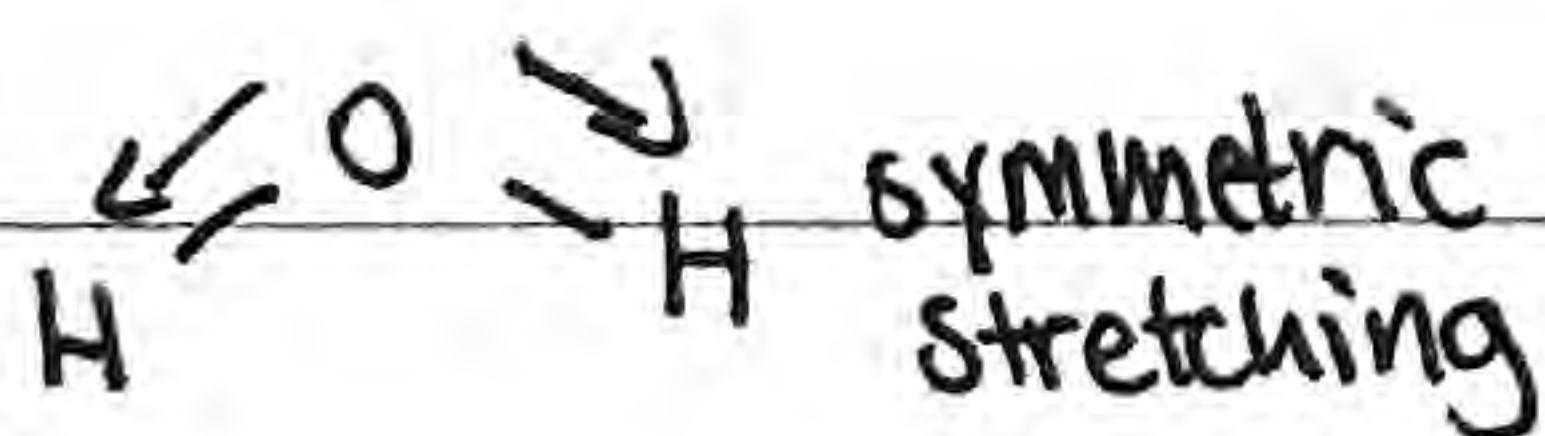
$\bar{\nu}$

decr.

decr.

decr.

Poly atomic molecules can bend or stretch (symmetric, asymmetric)
(bond angle changes)



A molecule is considered IR active if the molecule changes its dipole when vibrated (so diatomic won't work)

- Only IR active molecules work with IR spectroscopy
- So diatomic molecules are IR inactive (does not go through stretching/compressing)

• Certain wavenumbers correspond to specific functional groups (their bonds)

- You can only determine the presence of a specific bond if it matches the wave number and intensity specified (and broadness, etc.)

H¹NMR (Proton nuclear magnetic resonance)

- Chemical environment is determined by the surrounding atoms attached to an atom.
- H¹NMR in this case determines the # of different chemical environments of H atoms
- Also known as spin resonance spectroscopy

The nucleus of a hydrogen atom contains a single positive proton

- Has 2 possible spin orientations
- Nuclear spin quantum number ($m_F = +\frac{1}{2}, -\frac{1}{2}$)
- Subjecting a hydrogen nucleus to exactly the right energy

↑
energy is influenced by electrons (neighboring atoms may attract electrons more, reducing their effect, vice versa)

- Thus different energies required.

H¹NMR

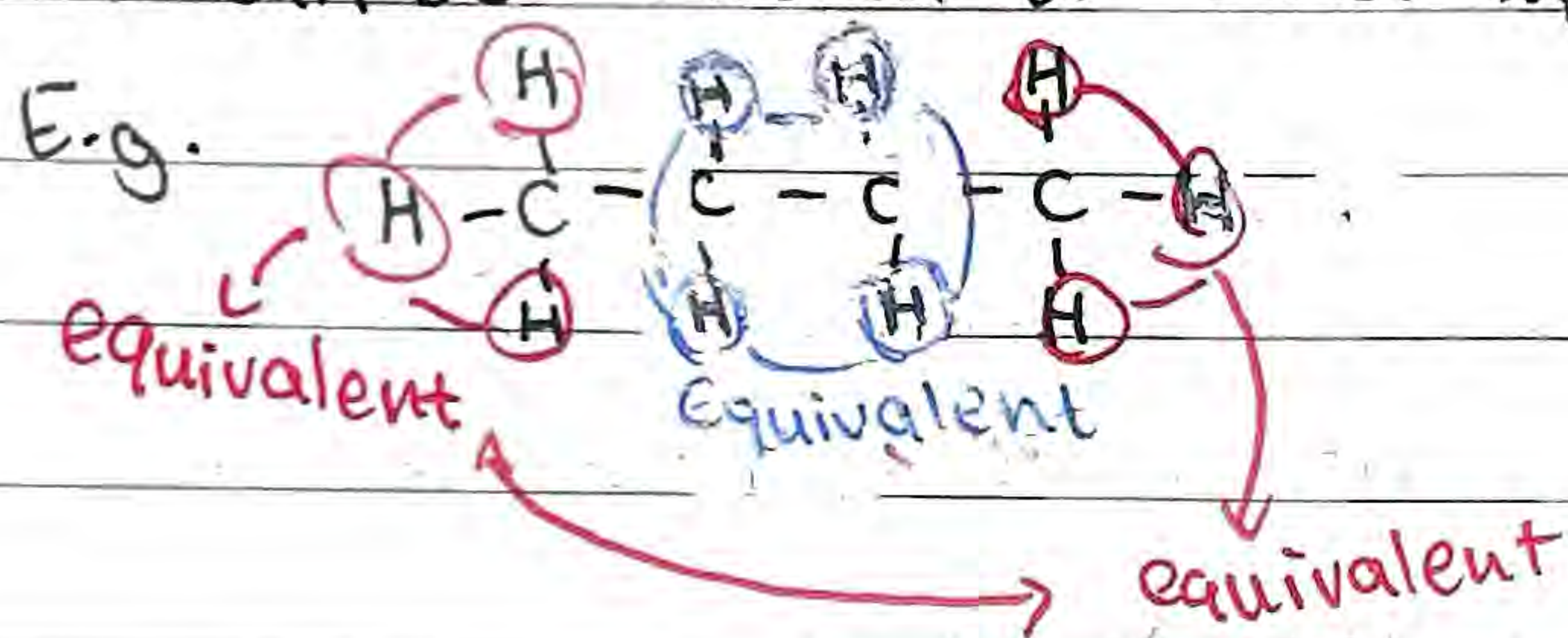
- H¹NMR basically passes a strong magnetic field $\Rightarrow$ Hydrogens react to specific wavelengths based on their environment

- H¹NMR is to verify other techniques (not used alone)

Hydrogens are equivalent when its surrounding environment is the same

- Equivalent when fragment left and right are same for hydrogens

- Or can be rotated to other hydrogen's places



Signals refer to a chemical environment

- Given in terms of their "chemical shift" (δ)

- in parts per million

- Chemical shift is measured using a relative

Scale $\Rightarrow$ Usually with respect to tetramethylsilane

- TMS has $\delta = 0$ ppm, used because unreactive

$\hookrightarrow$ right most

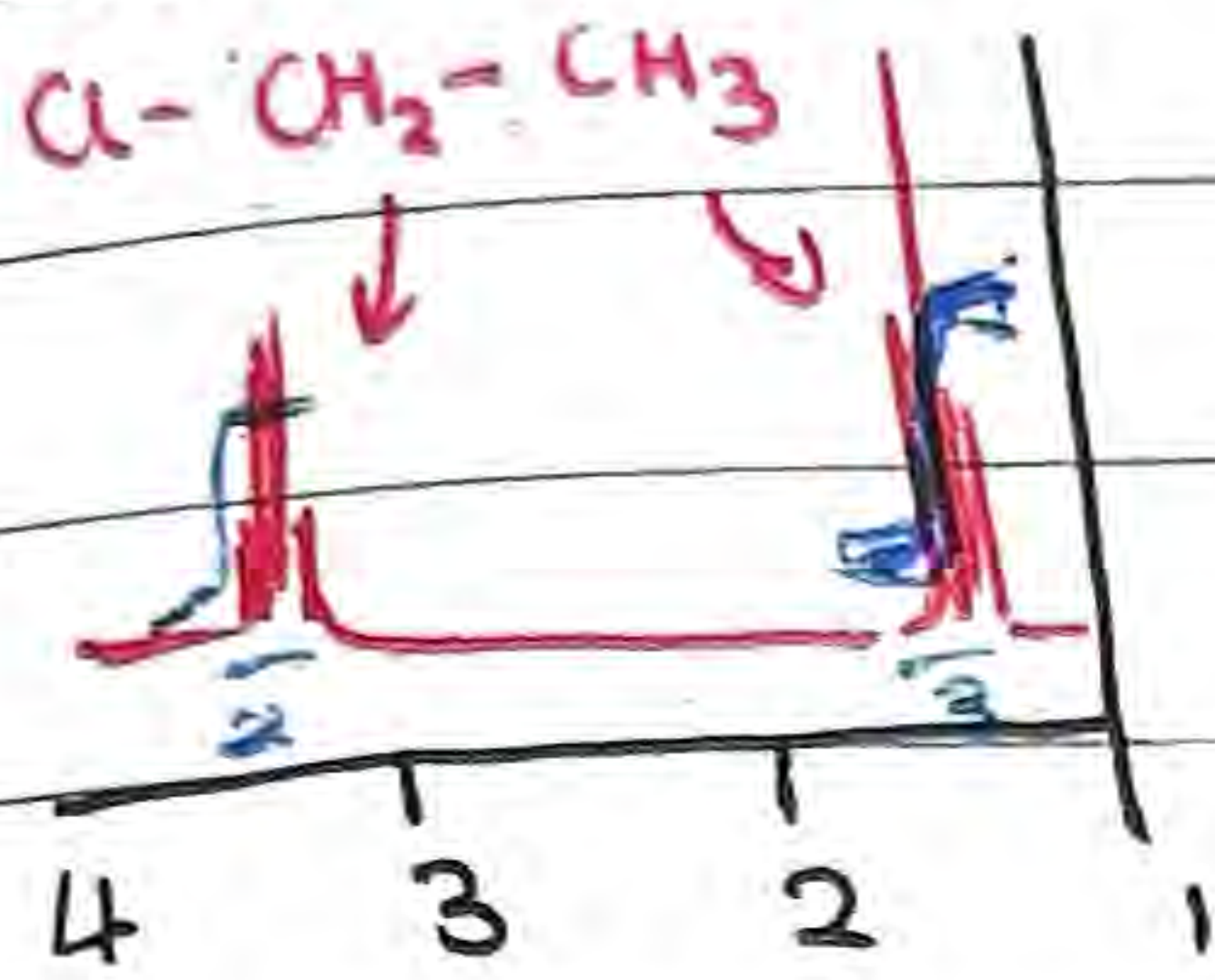
$\hookrightarrow$ exactly same environment for each H

Low resolution H¹NMR (does not identify splitting)

Area under each signal is directly proportional to the same number of H¹ nuclei generating the signal (# of hydrogens)

Low resolution ^1H NMR

- The area under a signal is small and hard to measure so integration trace is used



Integration trace

can sometimes be
a ratio (# of H in environment 1
: # of H in environment 2)
or directly the #

^1H NMRs have axis
reversed (1 on the right)

NOTE: When using ^1H NMR table, pay attention to the environment

E.g. $\text{H}_3\text{C}-\underline{\text{CH}_2}-\text{Cl}$ refers to $-\text{CH}_2-\text{Hal}$ not $-\text{CH}_2-\text{R}$

High resolution ^1H NMR

- splits the peaks of low resolution into clusters of peaks
- Tells us about # of hydrogens attached to neighboring (by 1) atoms in molecules

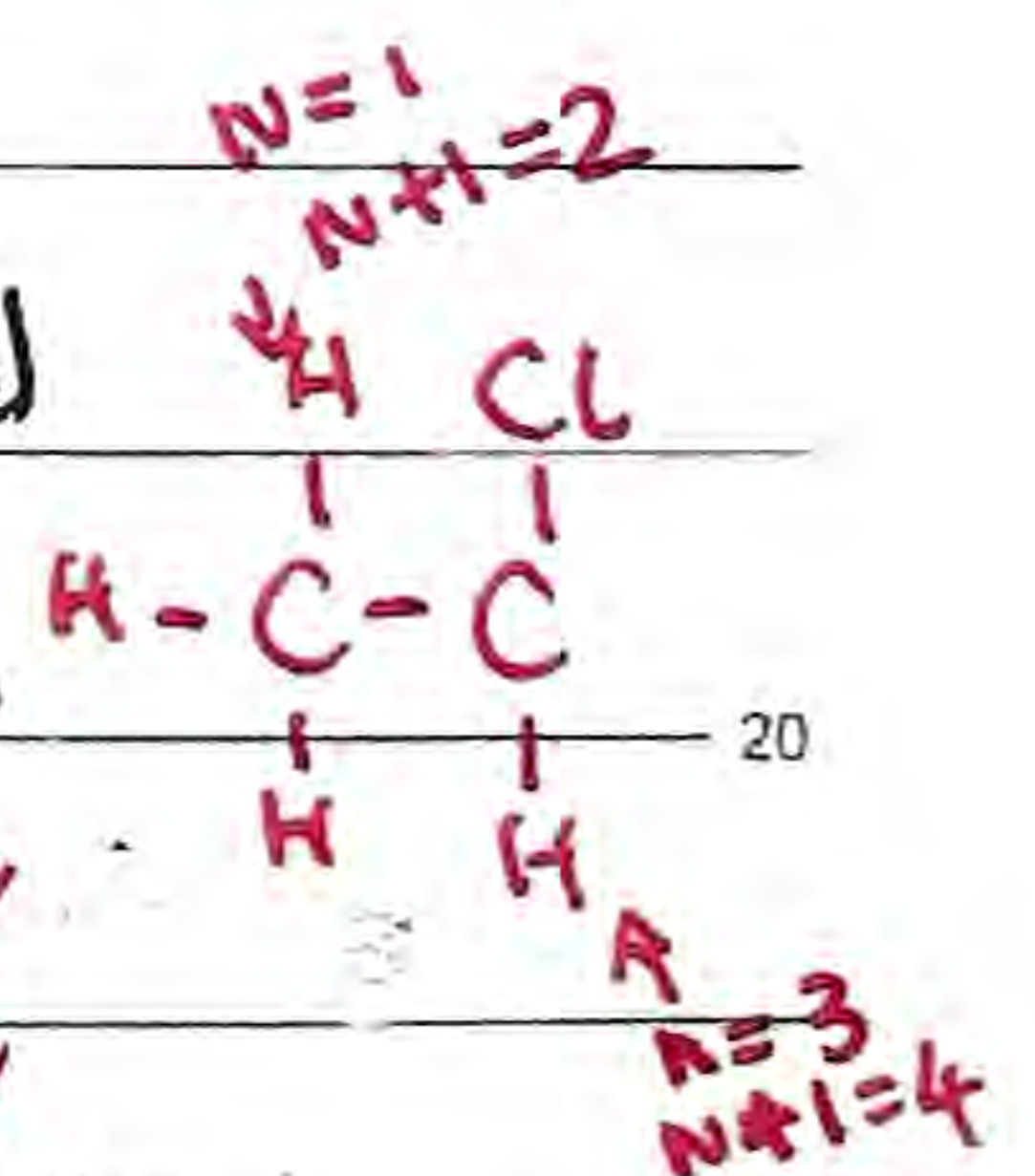
specifically, # of peaks = $N + 1$

where N = # of hydrogens attached to neighboring molecule

one peak is known as singlet, two is known as doublet, etc.

- The peak ratio is the pascal triangle

N	$N+1$		peak ratio
0	1	Singlet	1
1	2	doublet	1:1
2	3	triplet	1:2:1
3	4	quartet	1:3:3:1



High Res. ^1H NMR

singlet



doublet



triplet



quartet

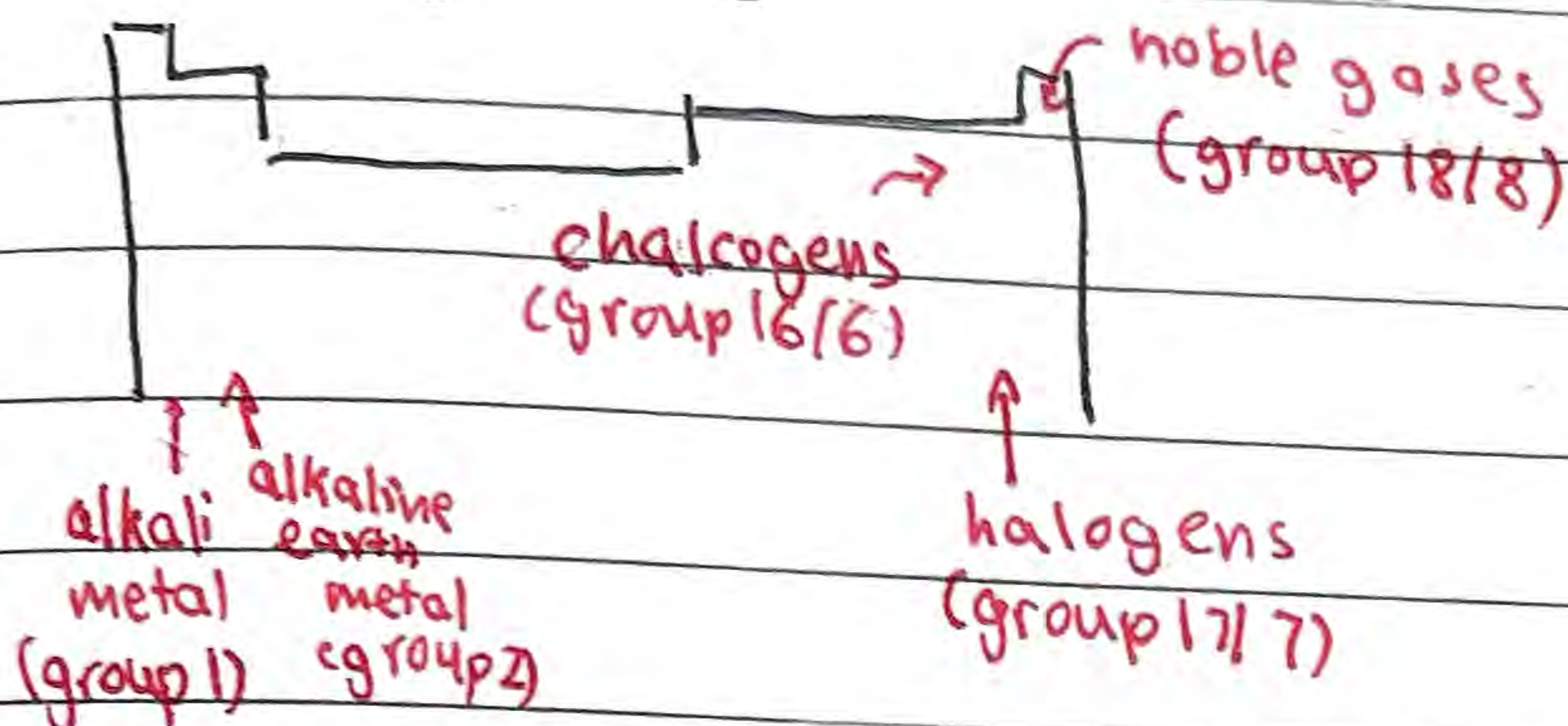


^1H NMR is typically used with combustion analysis (empirical formula), mass spectroscopy, IR spectroscopy, ^1H NMR spectroscopy

The periodic table of elements

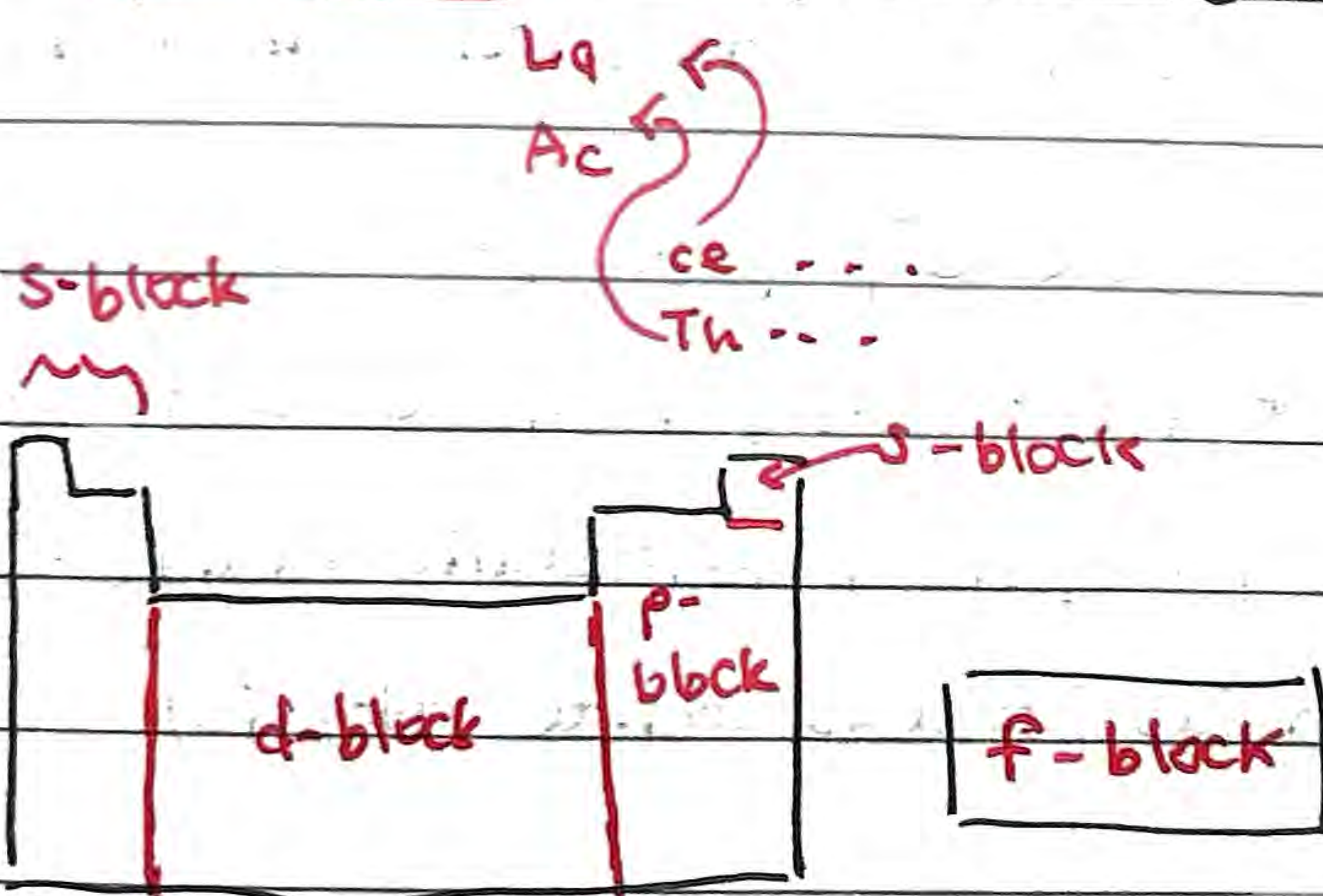
- Usefulness of the structure: Increasing periods, energy levels
 ↳ Quick way to obtain properties
- Increasing groups, charges

General grouping of elements



Periodic table is composed of periods, groups and blocks

- d shells - Transition Metals
- f shells - Lanthanide - Actinide

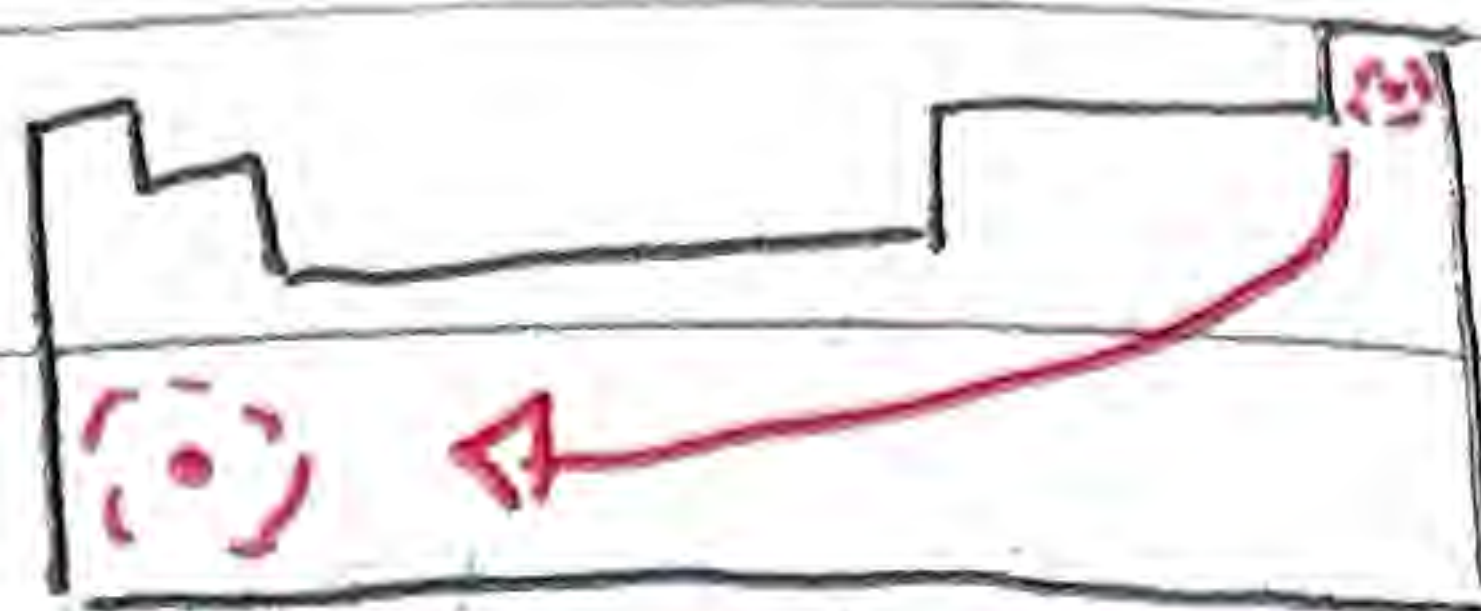


Metalloids - Under certain conditions behave as metal or non-metal

Periodicity

Atomic radius

increases
as period ↑



decreases
as group ↑

Increases down a group
(as period increases)

↳ More shells
+ electron shielding



atomic # →

Decreases across a period
(as group increases)

↳ More protons, greater attraction (due to charge e^{-} in nuc.)
between nucleons and electrons $\Rightarrow$ smaller radius

Ionic radius

Cations - Lost valence e^{-} , lost energy level,
ionic radius decreases (rel. to atomic)

Anions - Gains valence e^{-} , proton-electron ratio,
same # of protons per more electrons,
weaker per electron electrostatic attraction,
ionic radius increases (rel. to atomic)

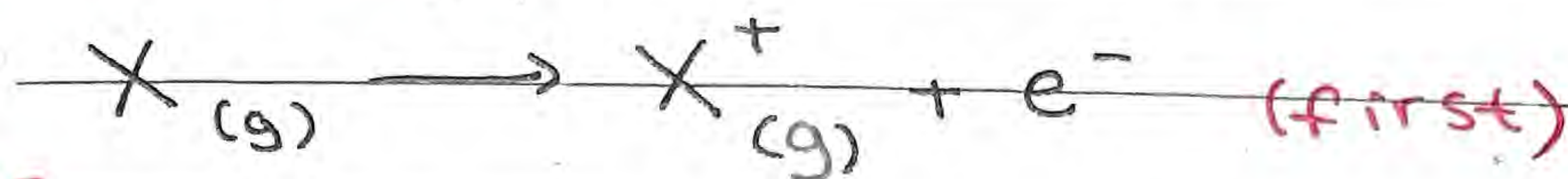
Ionization energy

• Min energy, remove 1 mol gas. e^{-} , from 1 mol gas atom,
form 1 mol $1+$ ion, ground state (first ion. energy)

↳ no excited electron

↳ electrons in base level

Ionization energy



make sure not diatomic (since ioniz. energy is specifically 1mol)

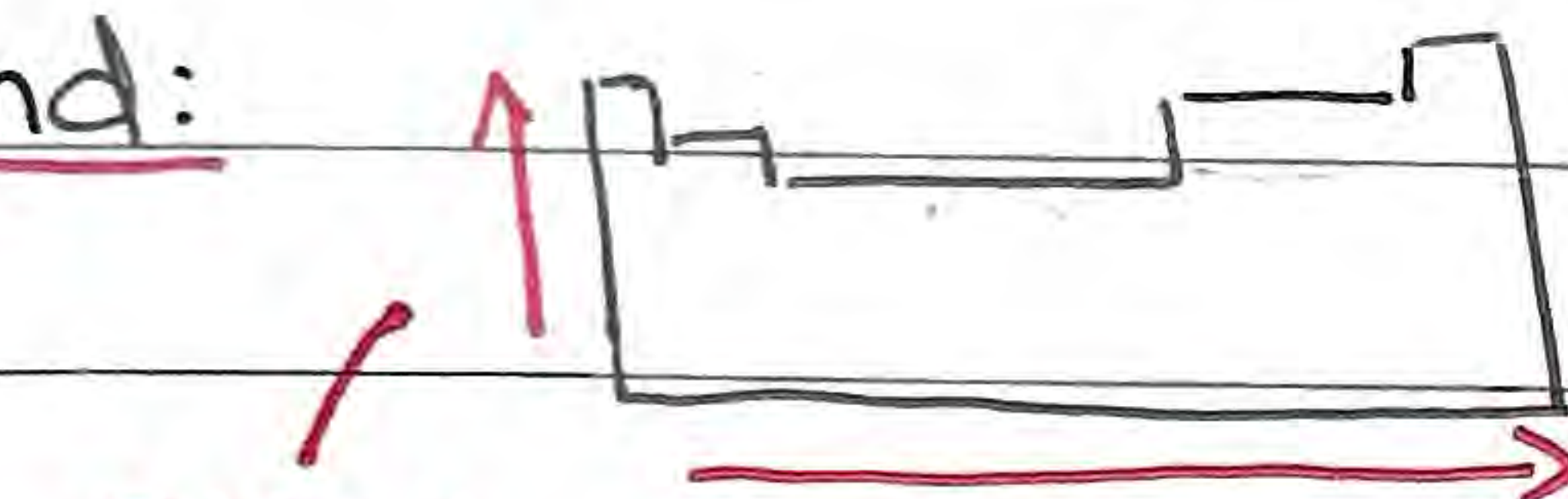
nth ionization energy



only taking one at a time

Removing valence electrons require energy (unfav.)

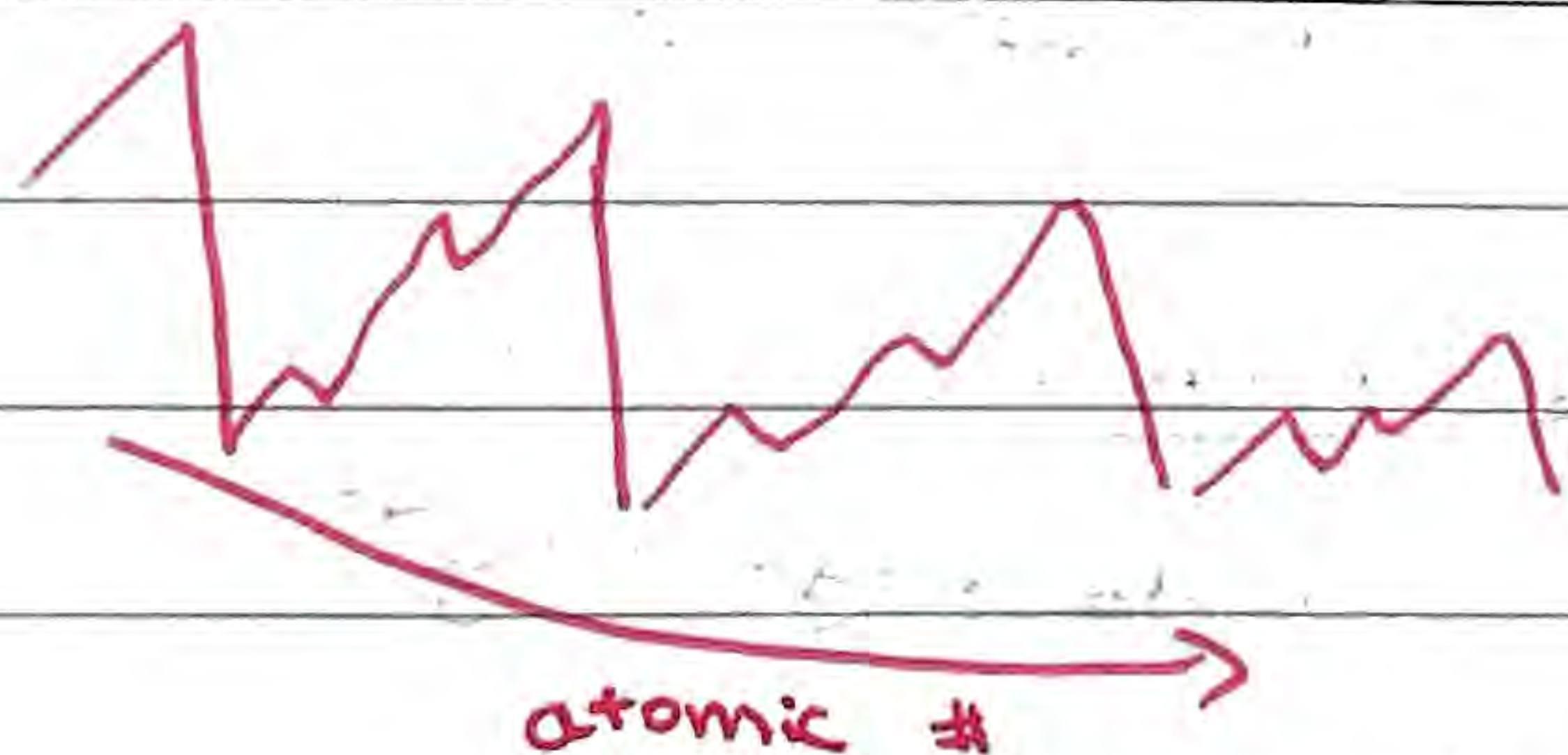
Trend:



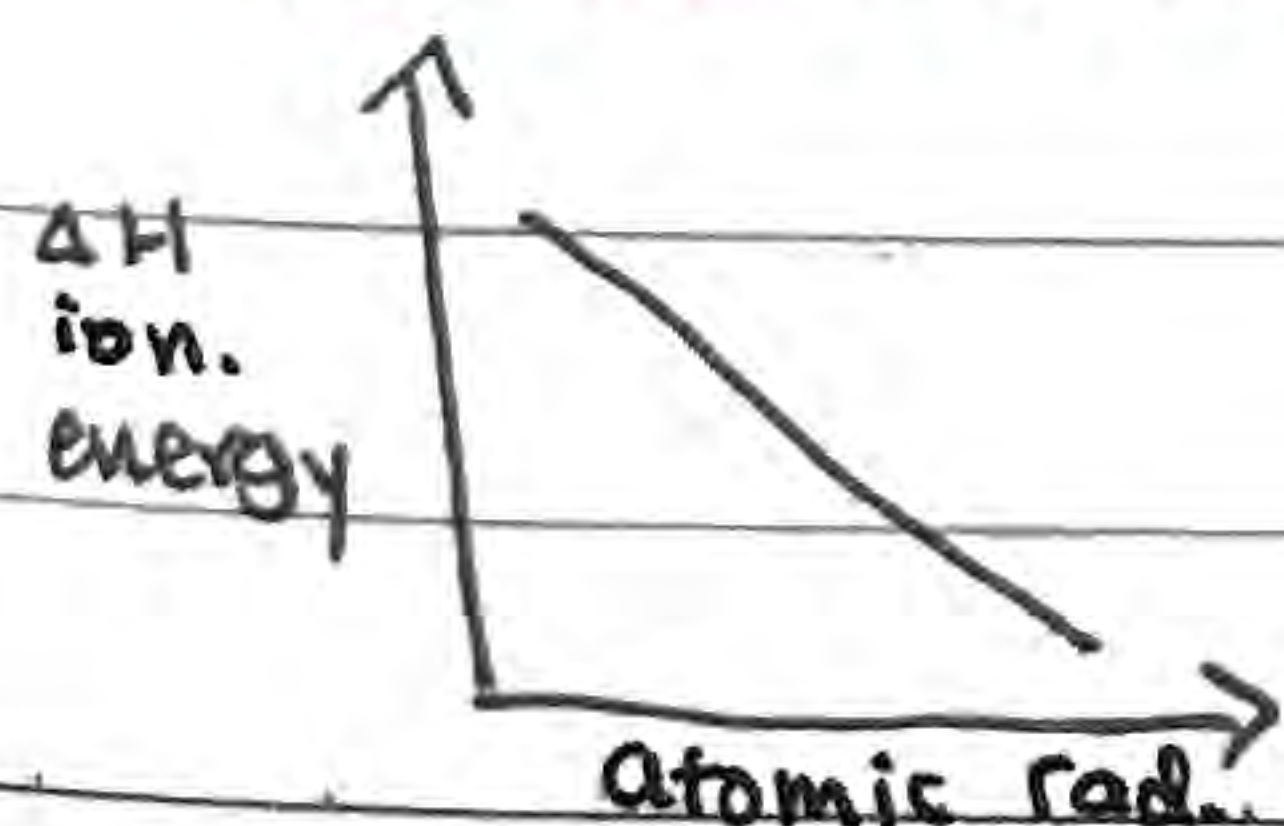
electron shielding from more shells
∴ easier to remove

more protons, outer e^- held with more attraction

$$\Delta H > 0$$



Atomic radius and ionization energy



negative relationship (linear)

$$\Delta H = kJ\ mol^{-1}$$

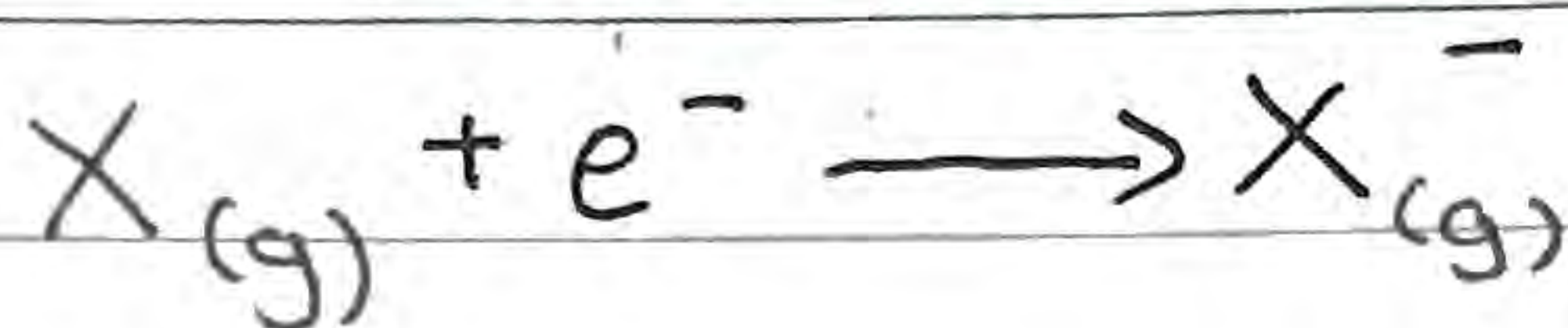
Electron affinity

↪ "like" electrons

"reverse" of ionization energy

minimum energy change required to add 1 mole of gaseous electrons to 1 mole of gaseous atoms in ground state.

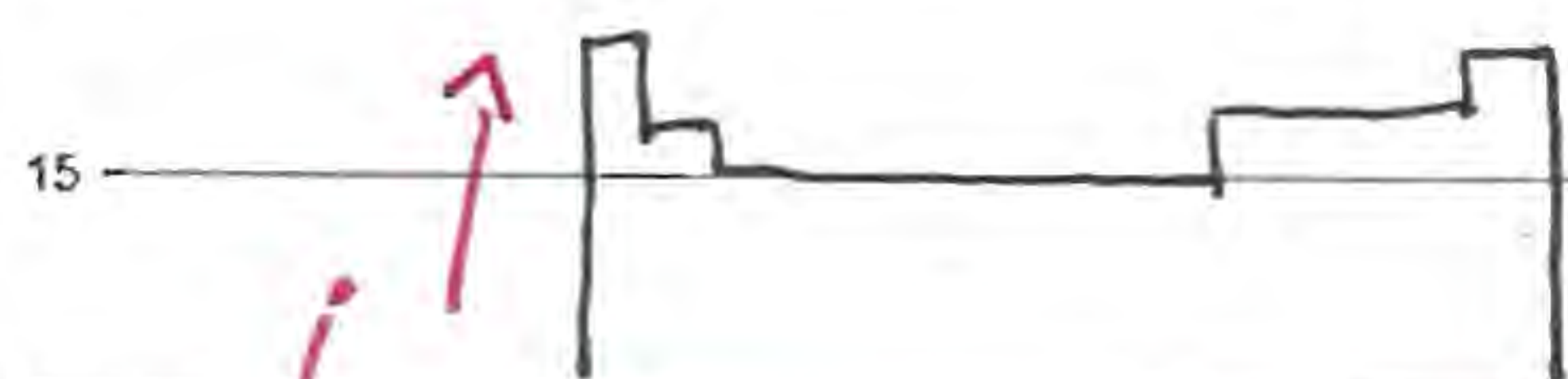
↪ where electrons are all unexcited, in "normal" position



position

↪ ignore diatomic (i.e. Cl not $\frac{1}{2}Cl_2$)

nth electron affinity



↪ harder as you go down (greater electron shielding)

↪ increase in magnitude

↪ greater protons, nuclear charge, thus attraction ↑

$$\Delta H < 0$$

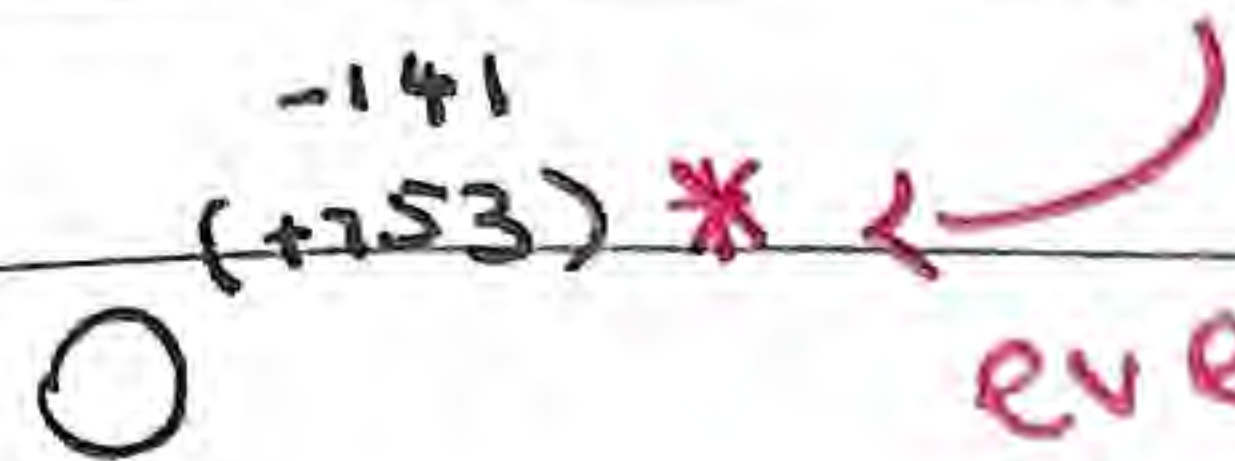
for iso-electronic elements

• Greater # of protons (e.g. Mg^{2+})

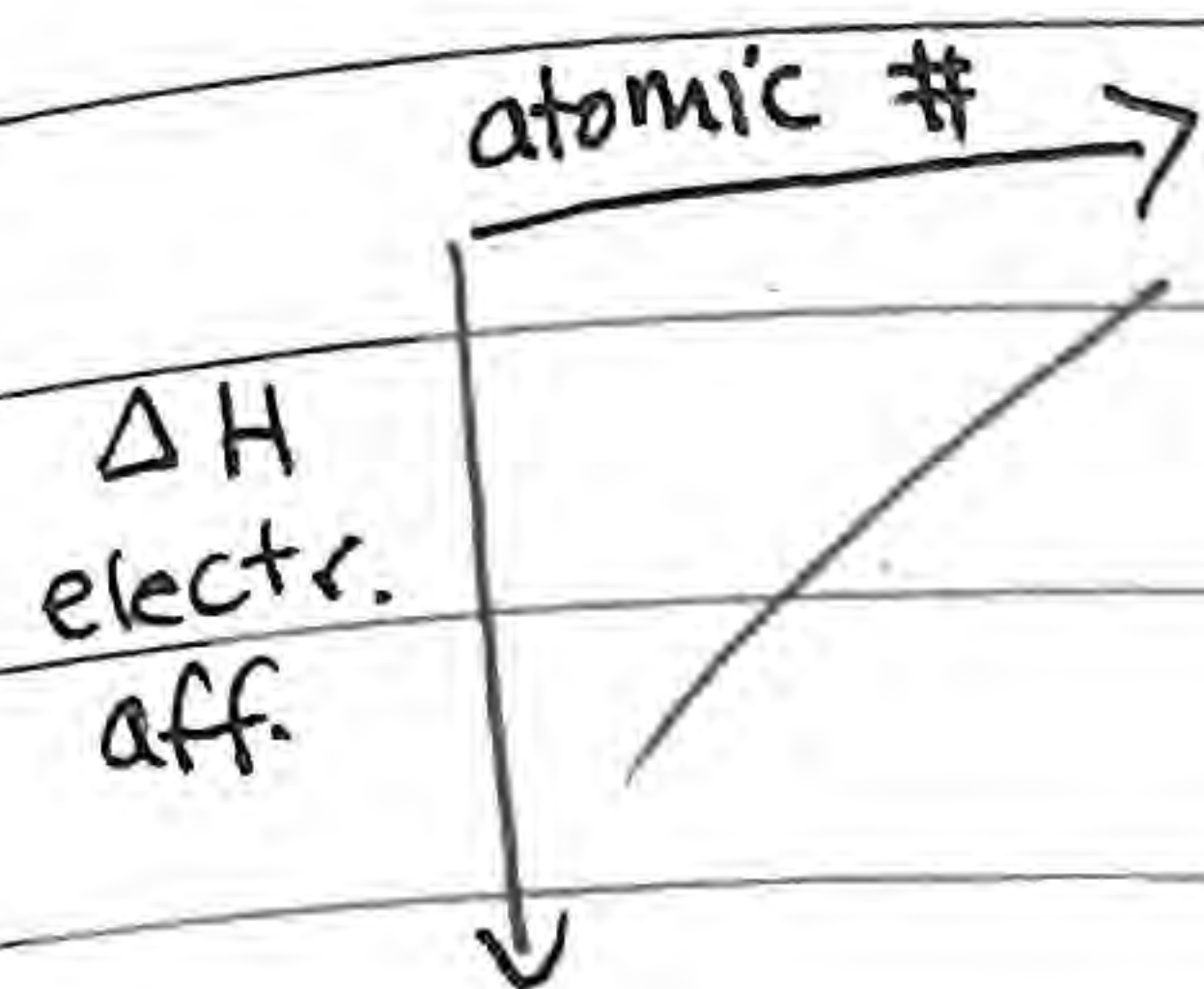
↪ Thus greater ioniz energy

2nd Electron Affinity

- Tends to require energy due to repulsion between anion and electron



even tho O wants to be O²⁻



positive relation since magnitude decreases

when actually bonding, lattice enthalpy way higher so O becomes O²⁻

Exothermic

Energy released

(-) values

Energetically favorable

• Electron Affinity (1st)

Endothermic

Energy required

(+) values

Energetically unfavorable

• First ioniz. energy

• Electron affinity (2nd)

Exceptions to 1st electron affinity

Group 2 (Be, Mg) - Less ⊖ than expected, e⁻ enters empty higher energy p-subshell

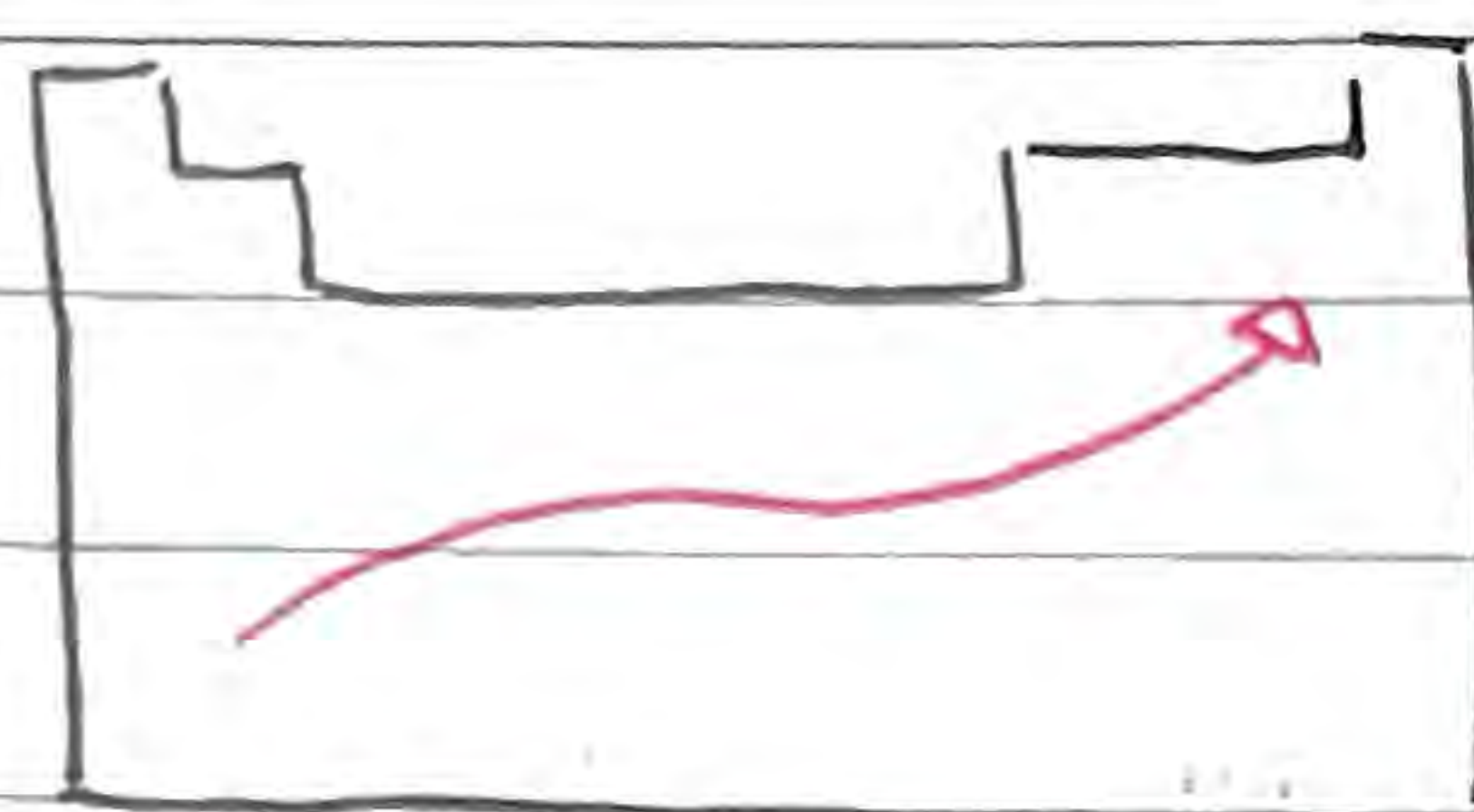
Group 15 (N, P) - Less ⊖ than expected, stable partially filled p orbital

Group 17 (F, Cl) - Fluorine very small size leads to e⁻e⁻ repulsion

Chlorine more favorable (bigger) for added e⁻

Electronegativity

A measure of an atom's ability to attract shared electrons in a covalent bonds



decrease
as you go
down a group

⇒ More shells, greater
electron shielding

increase as you go
across a period

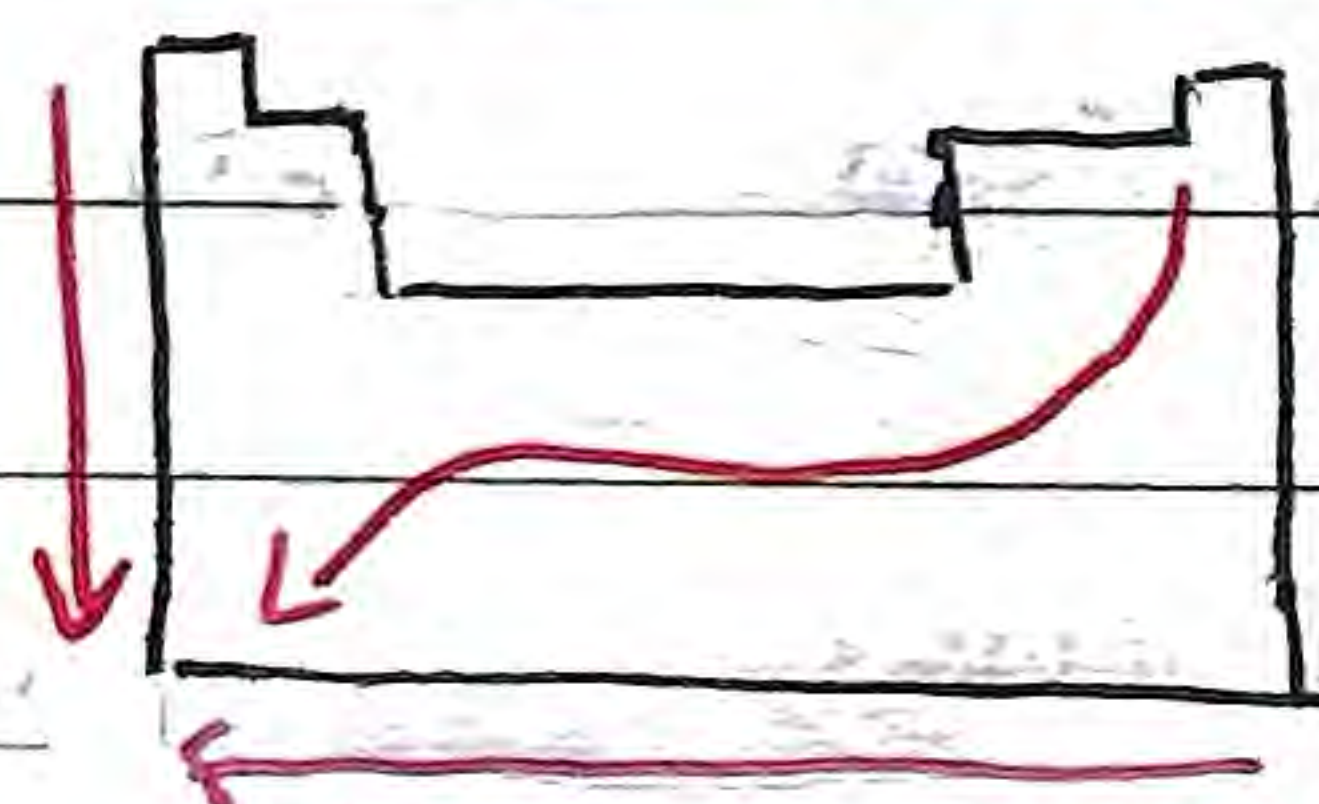
⇒ More protons, smaller radius,
greater electrostatic attraction

F is greatest, Fr is weakest
(no units)

Metallic Character

- The ability for an atom to lose electrons
- Non-metallic character – Ability to gain electrons

General Trend



increase
down
group

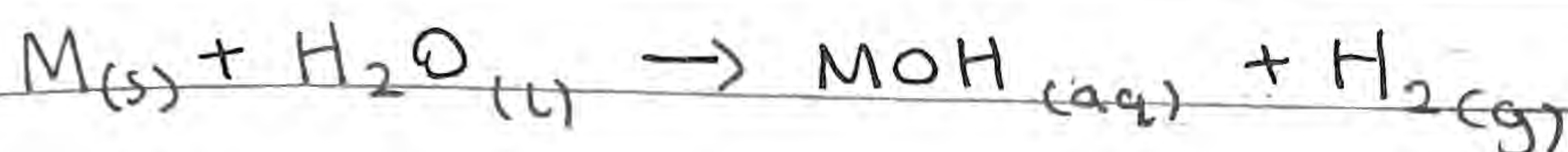
⇒ more e^- shells,
greater electron shielding,
easier to lose electrons

decreases across
period

⇒ greater protons,
nuclear charge $\uparrow$,
harder to lose electrons

Metallic character

Group 1 metals are hazardous because high ability to lose electrons.



M^+ produced in hydroxide

H_2 gas produced, resulting in fizzing and bubbling

Group 1 metal observations (with water)

Li

- ! Moves on surface of water,
- ! fizzing heard, effervescence / gas
- ! is produced (does not ignite)

Na

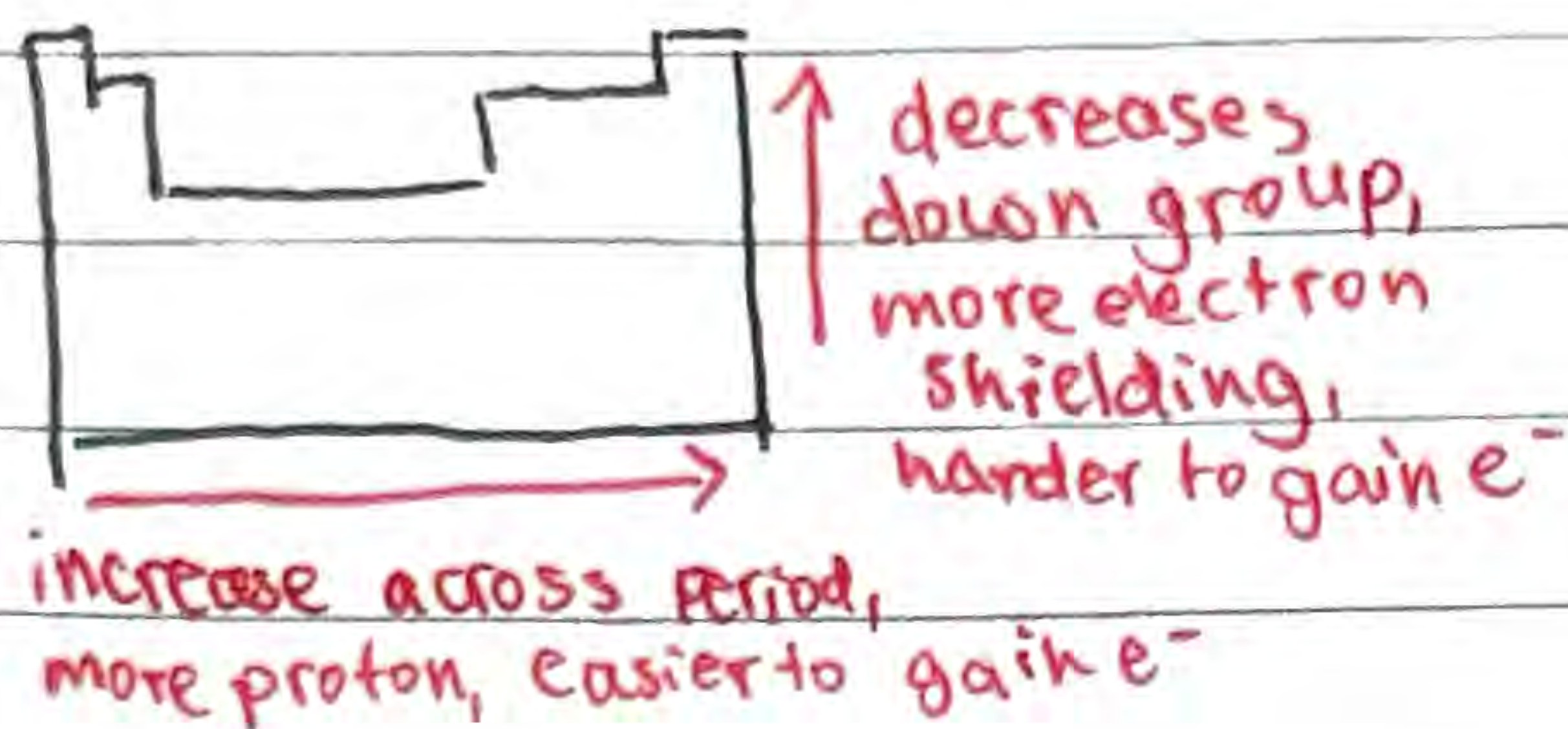
- ! Same as Li but also melts into a ball, may produce orange flame,
- ! may explode if H_2 is ignited

K

- ! Same as Na but pink flame

More reactive,
thus
more explosive

Nonmetallic character



Group 17: Halogens

Chlorine: Yellow / green gas

Bromine: Volatile orange liquid

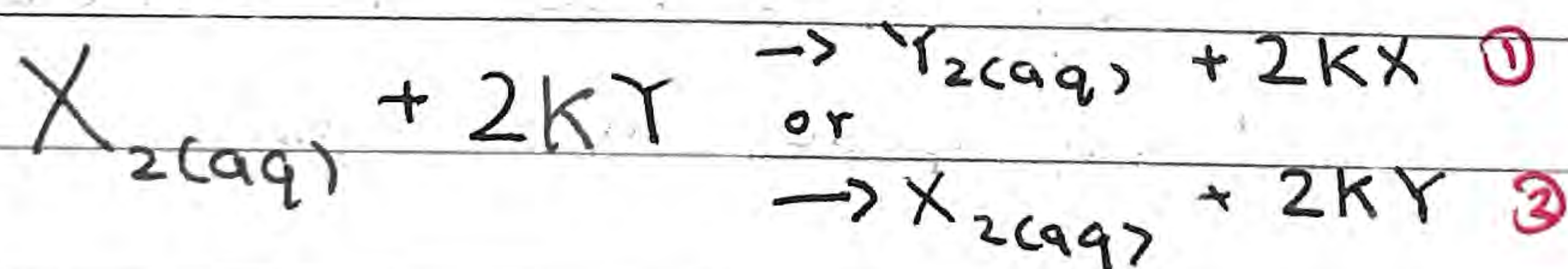
Iodine: Dark purple solid (sublimes upon heating)

Halogen - Elemental (diatomic), e.g. Fluorine

Halide - Ion, e.g. Fluoride

We can compare ability to gain electron

- More reactive, greater ability to gain electron, greater non metallic character



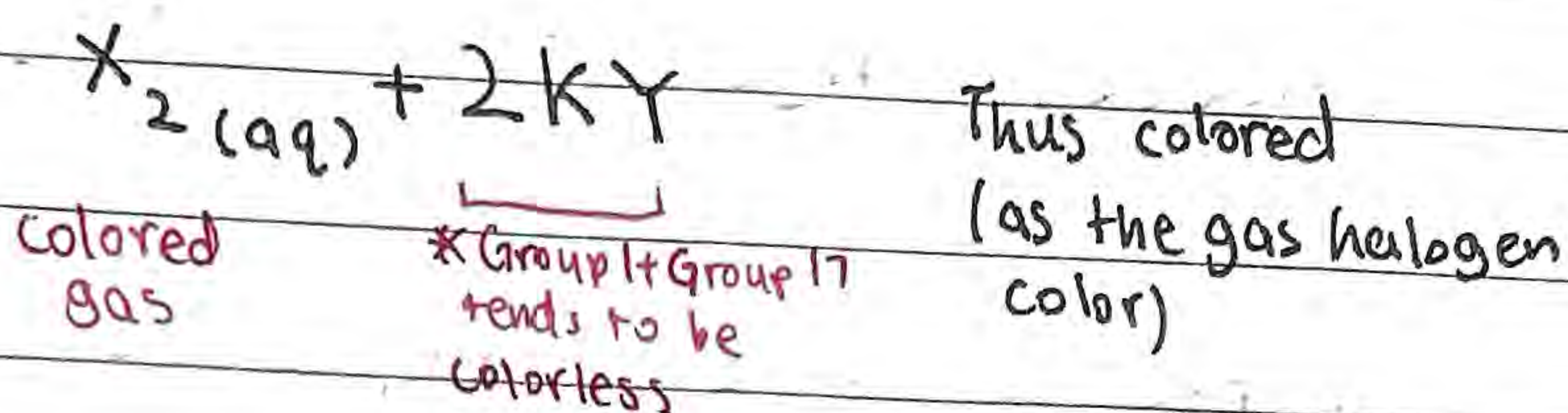
if $\textcircled{1}$, Reaction occurs, Y (initially in favorable position), displaced by X. Indicates X is more reactive

if $\textcircled{2}$, Reaction does not occur, Y (stays in favorable position), not displaced by X. Indicates Y is more reactive

Halogens Displacement reactions

Like mentioned before, these reactions indicate the nonmetallic character and reactivity of an element

- We can see whether a reaction occurs by viewing its color



if reaction occurs, then new color would be that of the displaced elemental gas

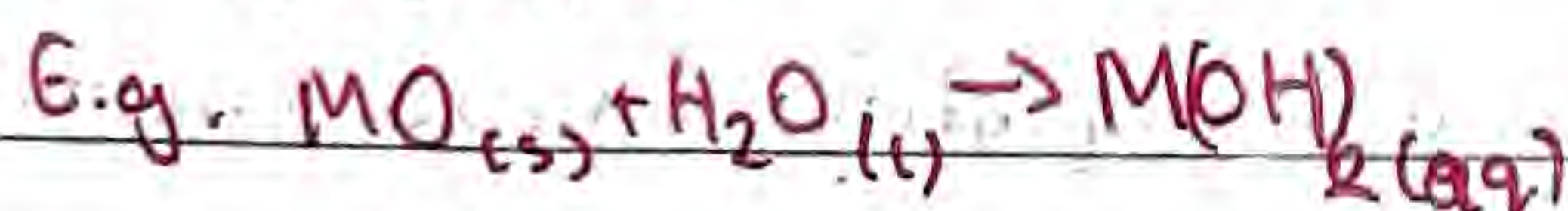
Acid-base character of metal and nonmetal oxides

	Group 1	Group 2	Group 13	Group 14	Group 15	Group 16	Group 17
Period 3 oxides (e.g.)	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_6 P_4O_{10}	SO_2 SO_3	Cl_2O Cl_2O_7

Acid Base	strong base	weak base	Amphoteric	Weak Acid	→ Strong acid		
			when in acid, acts as base, when in base, acts as acid				

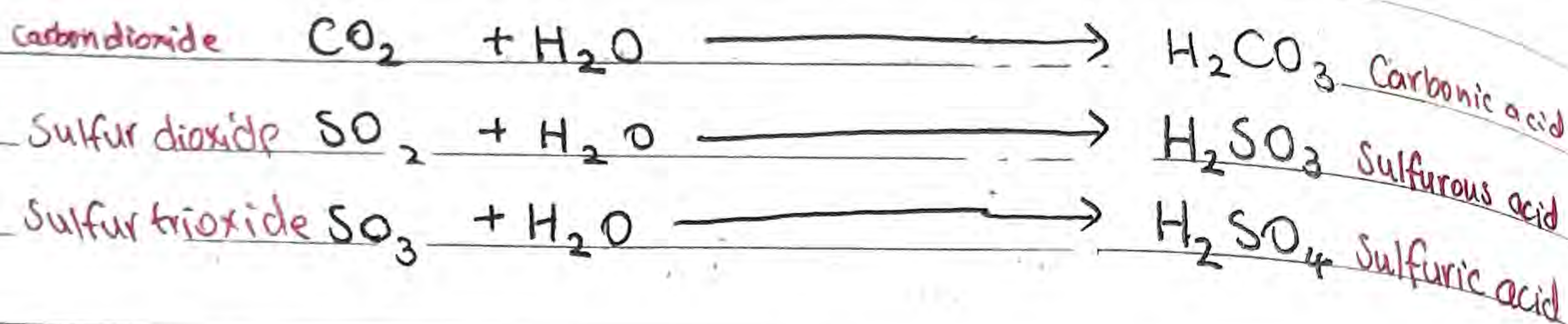
General trend	Metal oxide (basic)	→					Non metal oxides (acidic)
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produces OH^- in water



Acid-base of nonmetallic and metallic

Non metallic oxides - React with water to form acids



As mentioned, Al_2O_3 is amphoteric

- Will adjust its properties based on other reactant
 - If reactant is acid, Al_2O_3 behaves as base
 - If reactant is base, Al_2O_3 behaves as acid
- As such, will always produce salt when reacted with acid/base

Acid rain - Defined as $\text{pH} < 5.6$, but why not $\text{pH} < 7$?

- This is because of naturally existing CO_2
- Carbonic acid will naturally be produced
- It's only when it reacts with pollutants (e.g. produce nitric acid) does it become an issue

* Acidic gases are produced by combustion of fossil fuels.

- NO_x formed at high temps in combustion engines

SOME can be natural from volcano eruptions

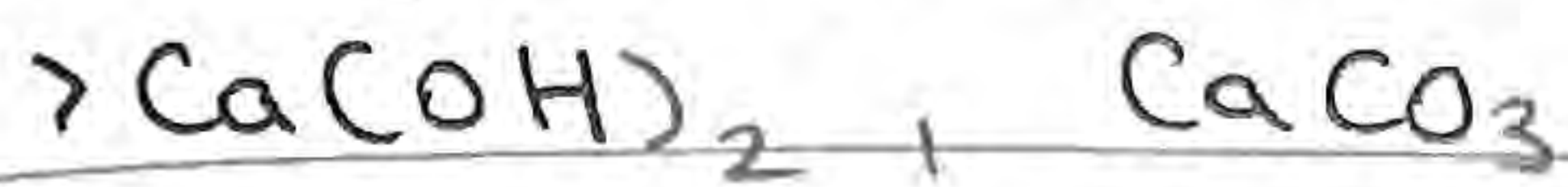
(O_2 , N_2)

- SO_x formed when sulfur impurities in fuel react with O_2 in the air

- Decreased pH is harmful for vegetation, etc.

Acidic Gases

- In industries, "scrubbers" are used to neutralize acidic gases produced



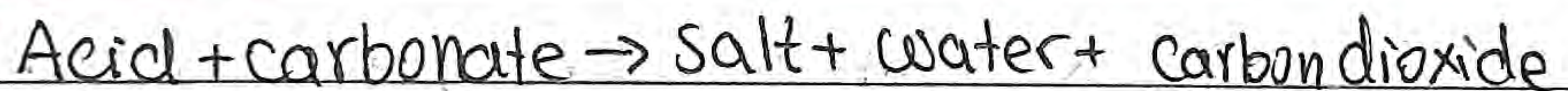
- In waterways, adding limestone can help raise the pH

Neutralization Reactions

Acid-Base



Acid-carbonate $\leftarrow$ tends to be safer (e.g. CaCO_3 is chalk)



Ocean acidification

- Increased CO_2 in atmosphere (burned fossil fuel)

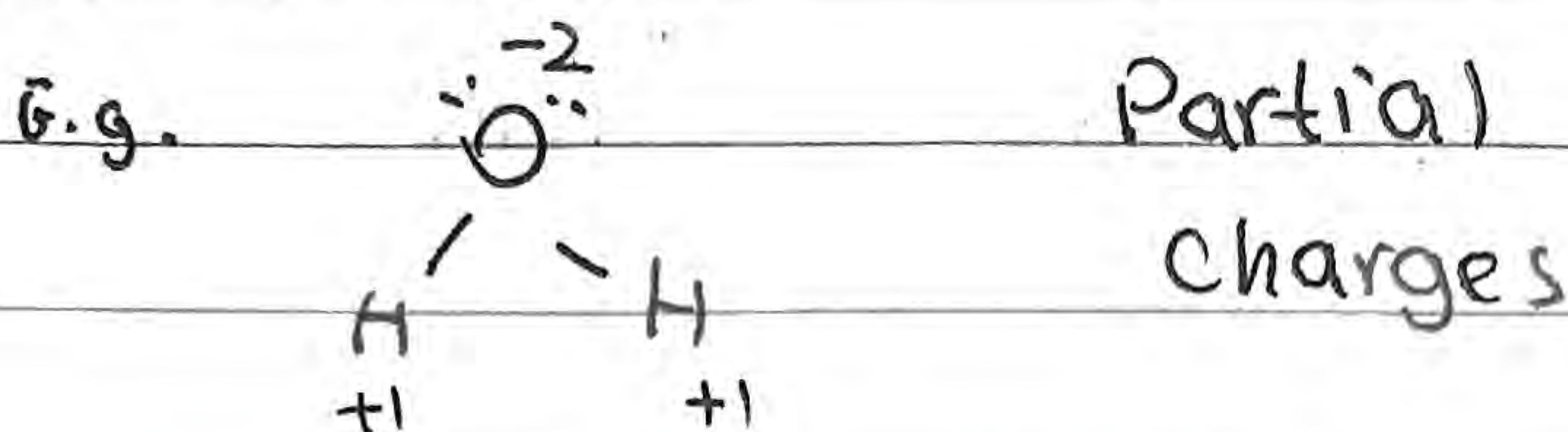
$\hookrightarrow$ Dissolves in ocean to form carbonic acid

$\hookrightarrow$ Acids harm marine creatures; formation of shell by

mollusks, coral reef (due to CaCO_3)

Oxidation State / number

Charge of an atom in a compound if it was composed of ions



* sign first for oxidation

Use these rules to assign oxidation states to atoms:

$X = 0$
or
 $X_n = 0$

Rule 1 - Oxidation state of any free element is zero (including atomic molecules, e.g. Cl_2)

neutral compound = 0

Rule 2 - The sum of the oxidation states of all atoms in a neutral compound is zero

polyatomic ion = charge

Rule 3 - The sum of the oxidation states of all atoms in a polyatomic ion is equal to the charge of the ion.

$F = -1$

Rule 4 - The oxidation state of fluorine is -1 in all compounds

$O = -2$

unless
 OF_2 , $O = +2$
peroxides, $O = -1$

Rule 5 - The oxidation state of oxygen is -2 except in OF_2 , where fluorine is more electronegative, so oxygen has an oxidation state of +2, and in peroxides

such as H_2O_2 , where it is -1.

Rule 6 - The oxidation state of the group 1 metals is always

Group 1 = +1

+1 and the oxidation state of the group 2

Group 2 = +2

metals is always +2.

Rule 7 - The oxidation state of hydrogen is +1 when it is

$H = +1$ or -1

with more elect. neg. with less elect. neg.

combined with more electronegative elements (most non-metals), and -1 when combined with less electronegative elements (metals). Hydrogen in a -1 oxidation state is called a hydride ion.

Oxidation states

→ typically asked to find transition metals or other tricky cases

You can find oxidation states through

calculations: ← summing the oxidation states of each individual charge/oxidation state

if $A_a B_b^x$, then $a \cdot (\text{oxid. of A}) + b \cdot (\text{oxid. of B}) = x$
can be $-x, 0, +x$

E.g. H_2O_2 , $2 \cdot (+1) + 2 \cdot (-1) = 0$

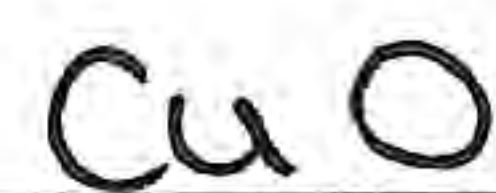
Transition metals

Transition metals have multiple different oxidation states as a result of their flexible subshells (d-block)

• Naming transition metal compounds involve putting the transition metal's oxidation state next to the metal when naming the compound (in roman numerals)



copper(I) oxide



copper(II) oxide

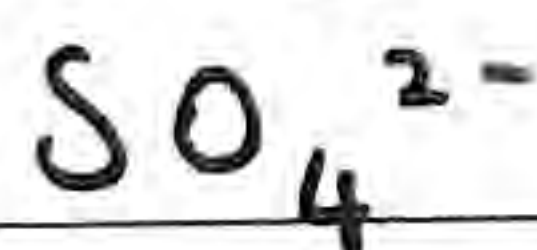
Oxyanions

Similar situation with oxyanions naming



Sulphite

Sulphate



the more common ion ends with -ate
↳ less oxygen ends with -ite

Alternatively, you can name with oxidation state

() ion, e.g. sulphate (IV) ion and sulphate (VI) ion
common name of ion oxidation state of non oxygen

Oxidation state

Oxidation states are assigned based on the assumption that a compound is composed of ions.

> We make this assumption because...

- it allows us to systematically track electrons during chemical process

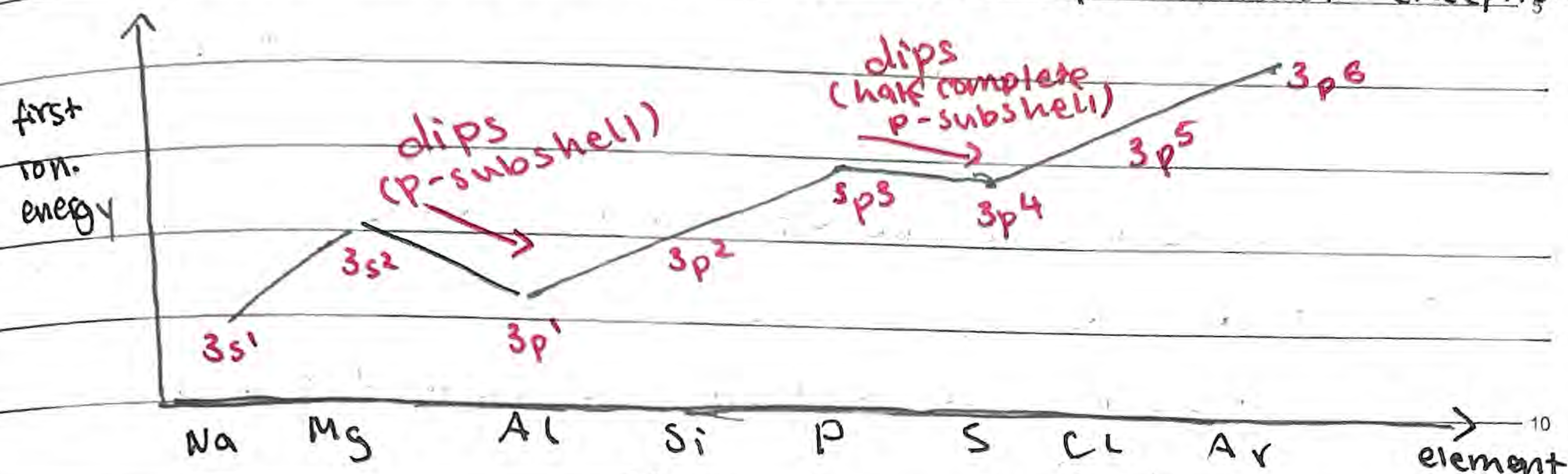
- also since even in covalent bonds, e^- not shared equally

- it is more accurate for ionic

- Simplified way to keep track of electrons, especially how they are transferred in redox.

Discontinuities in first ionization energies

It is expected that ionization energies increase across a period but there are a few notable exceptions



Between Mg, Al

Mg: $3s^2$, Al: $3p^1$; Al after losing $1e^-$ means empty subshell, thus lower energy
Mg not as energetically favorable

* Also, p subshell is further away from nucleus than s subshell (due to higher energy level)

⇒ Thus, lower ionization energy in Al than Mg.

Between P, S

P: $3p^3$, S: $3p^4$

$\uparrow \uparrow \uparrow$

$\uparrow \uparrow \uparrow$

exists greater e^-, e^- repulsion

(prefers half-full subshell, so

energetically unfavorable for S to

maintain $3p^4$)

Transition element

all transition elements must be d-block, but not all d-block elements are transition

- Recall not the same as d-block element
- It's transition when either the element forms an incomplete (non-empty) d-sublevel or forms stable ions with incomplete (non-empty) d-sublevel.

Magnetism (only d-block/transition elements)

only electrons in same orbital cancel, diff. orbitals are added

generates magnetic dipole

- Electrons have spins (paired electrons have spins cancelled out, thus only unpaired electrons contribute to magnetism)
- When these electrons are unpaired, they exhibit magnetism (are magnetic)

each pair of unpaired electrons means two different spins

↑ or ↓

so two different paths when affected by magnet

- The greater # of unpaired electrons, mostly d, but really any the greater the magnetism of the atom/ion is
- If no unpaired electrons (all paired/empty) then the atom/ion is not magnetic

- The unpaired electrons must be exhibited by the ion/atom present

E.g. Mn^{2+} : $[Ar]3d^5$, 5 unpaired so magnetic

Mn^{7+} : $[Ar]3d^0$, all paired so not magnetic

Properties

Variable oxidation states

- Since ns and (n-1)d sublevels have similar energy levels, transition elements have more places to lose electrons from, thus multiple oxidation states

Properties

High melting points

> For metallic bonds, electrostatic attraction between cations and delocalized electrons

• Since transition metals can lose more electrons than non transition due to variable oxidation states, they exhibit stronger bonds to overcome (higher melting points)

Used as Catalysts

> Transition elements can form more than one stable oxidation state (can lose/accept electrons easily)

↳ Can be oxidized/reduced repeatedly, allows for catalyst behavior in redox

• Heterogeneous catalyst (different physical state/phase from reactants)

• Homogeneous catalyst (same physical state/phase as reactants)

Forms colored compounds

• Compounds are often colored (due to ligands, see next page)

Transition element complexes and their color

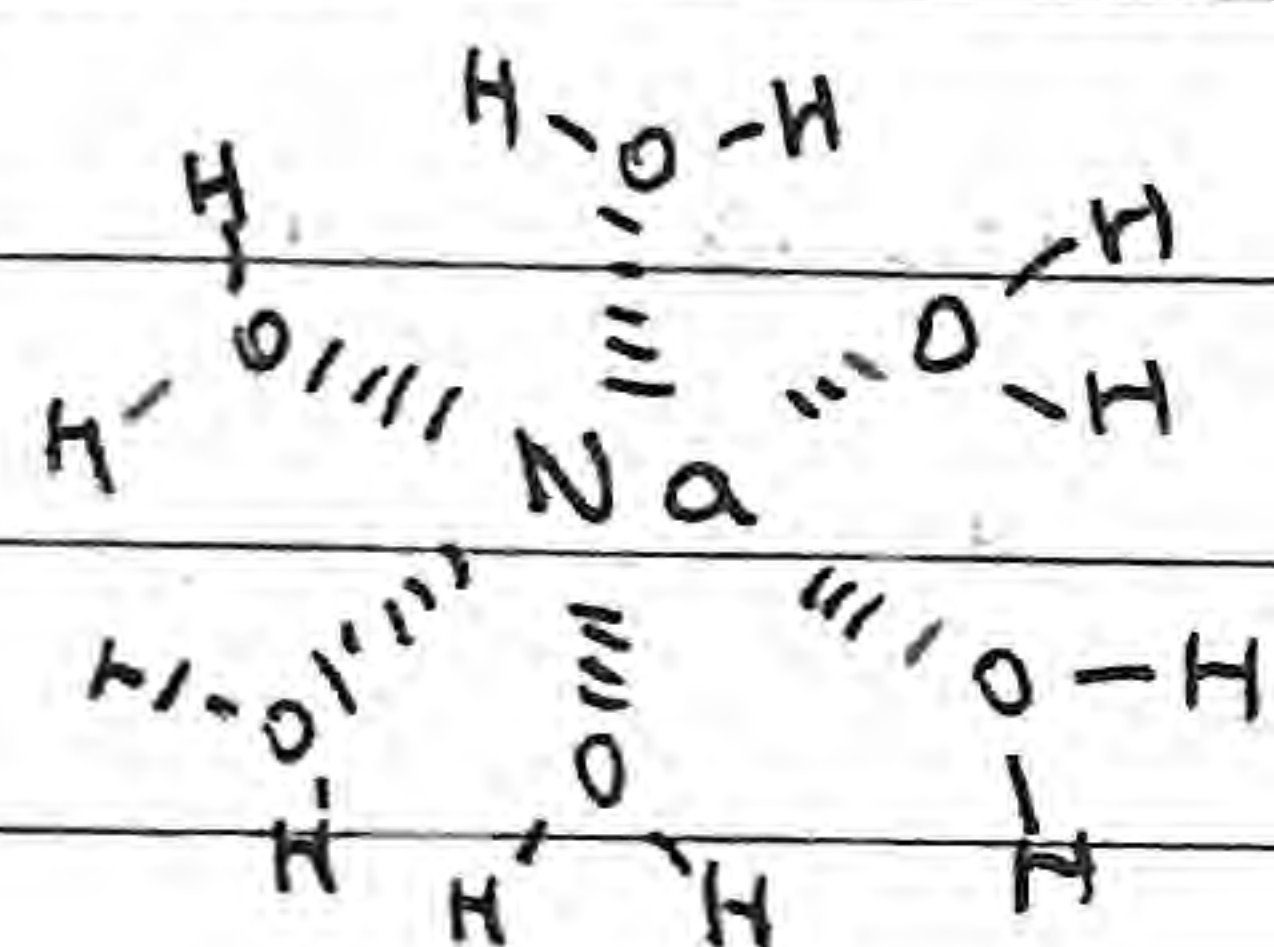
Has to do with ligands. (which allows them to form complex ions)

E.g. Anhydrous CuSO_4 is colorless but becomes blue when hydrated as $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 $\rightarrow$ Cu "bonds" with H_2O , SO_4 unbonded

Transition metals, with d-sublevel, allows for more room to form bonds (whereas normal ionic would only be IMF's)

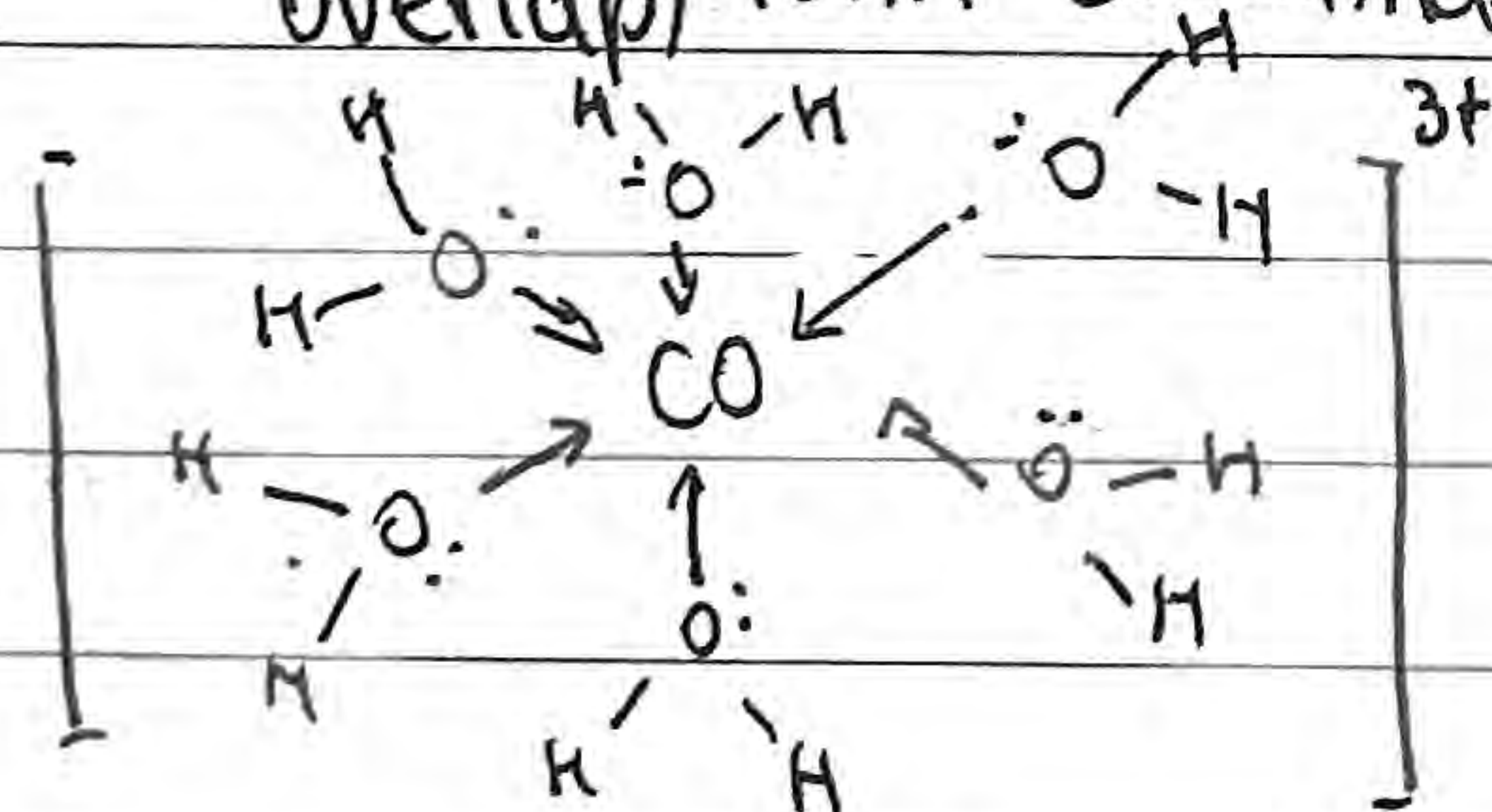
so e.g. Zinc not included

E.g. - When added to water, ionic compounds typically overcome their ionic bond, resulting in dipole-ion intermolecular forces



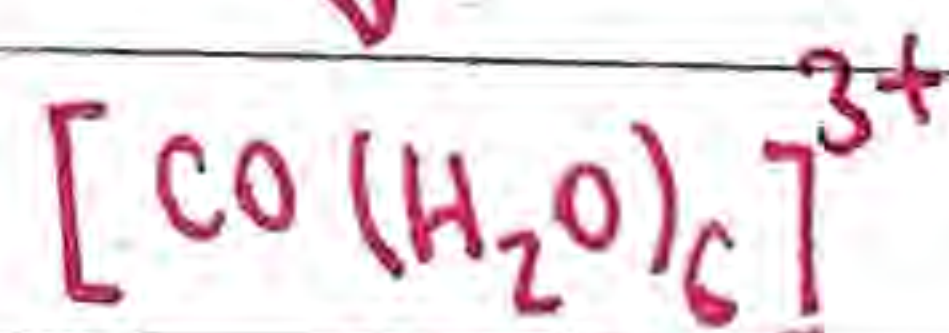
a "ring" of water around ion

• But transition metals (due to d-sublevel, more spaces for overlap) form coordinate bonds instead



water is neutral, Cobalt is 3+ so entire compound is 3+

chemical formula



this is a Complex ion

Complex ion - central metal ion surrounded by ligands
 The water here act as ligands

Ligand donates a pair of electrons to form a coordinate bond with a metal ion (they are a species)

molecule $\downarrow$ $\rightarrow$ ion

Ligands

Common examples of ligands include...



thus oxidation state of metal ion must be properly calculated

(some ligands are not neutral and will contribute to the net charge of the complex ion)

Co-ordinate bonds are typically 2, 4, 6 (coordinate numbers)

How do we know how many ligands?

• Metal ion size - larger metal ions accommodate more ligands

• Ligand size and shape - large, bulky ligands take up more space

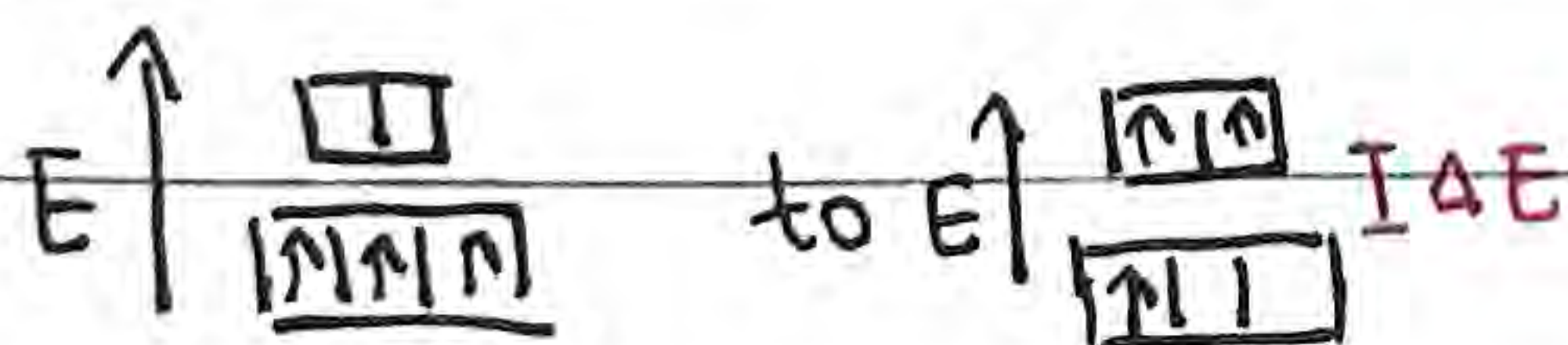
• Ligand charge - more negative ligands increase electron density so less can fit around

• 18 electron rule - Stable complex ions typically have 18 outer electrons (some exceptions)

Ligands lead to d-orbital splitting

• No need to know why but d-orbital splitting and exciting of electrons happen simultaneously

In other words, electrons are excited absorbing specific light waves



ΔE depends on ligand
(CN^- is highest, I^- is lowest)

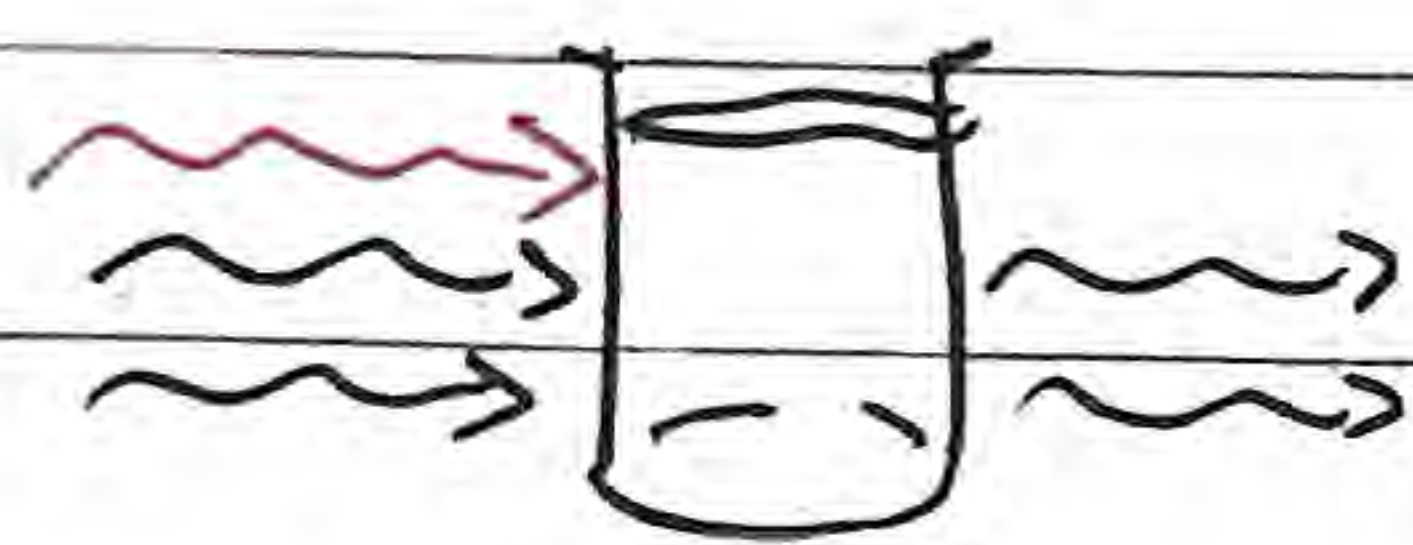
Ligands and d-orbital splitting

Some ligands result in greater d-orbital splitting due to different ΔE s

That means that exchanging ligands^{or adding ligands} can result in a change in color

Specifically it's dependent on the energy absorbed

$$E = h \cdot f \Rightarrow c = \lambda \cdot f$$



The actual wavelength is not the wavelength absorbed

Thus, λ_{out} is $\Leftarrow$ (it's the λ reflected)

Complementary of λ_{in} (use databooklet)

So when ligand is replaced such that

$E \uparrow$ where $E \propto \frac{1}{\lambda}$, then λ_{out} must decrease

Enthalpy - Energy Changes

Energy Transfer

- Most chemical reactions take place in open system
- Can exchange energy or matter with surroundings

↓
e.g. air escapes

- However, some occur in a closed system
- Exchanges energy but not matter with surroundings

↓
e.g. closed lid, no air escapes

Conservation energy

- Total energy must not change during the process
- Energy lost is gained by surroundings

Calorimeter can be used to determine whether a reaction is endothermic or exothermic

↳ A calorimeter is ANY apparatus used to measure the amount of heat being exchanged between system and surroundings that contains the solvent

Typically focused on ΔT of reaction solvent (e.g. water)

Reaction	Heat	Temperature	ΔH
<u>Exothermic</u>	<u>Release</u> heat	Becomes <u>hotter</u>	<u>(-)</u>
<u>Endothermic</u>	<u>Absorbs</u> heat	Becomes <u>cooler</u>	<u>(+)</u>

Energy Transfer

Heat - Energy transfer between 2 objects of
(a) different temperatures

Law of thermodynamics - Heat flows from HIGH to LOW temperature until thermal equilibrium is reached

Enthalpy - Heat content of a system
(H)

Kelvin - SI unit for temperature

$$K = ^\circ C + 273.15$$

Absolute zero - No movement of particles, 0 K
(0 K)

Thermochemistry - study of heat changes that occur during a chemical reaction

An equal heat in different masses will correspond to lower ΔT for larger masses than smaller masses

↳ Similarly, same ΔT , larger mass more heat than smaller mass

Enthalpy - Heat released or absorbed by a system (given constant pressure)

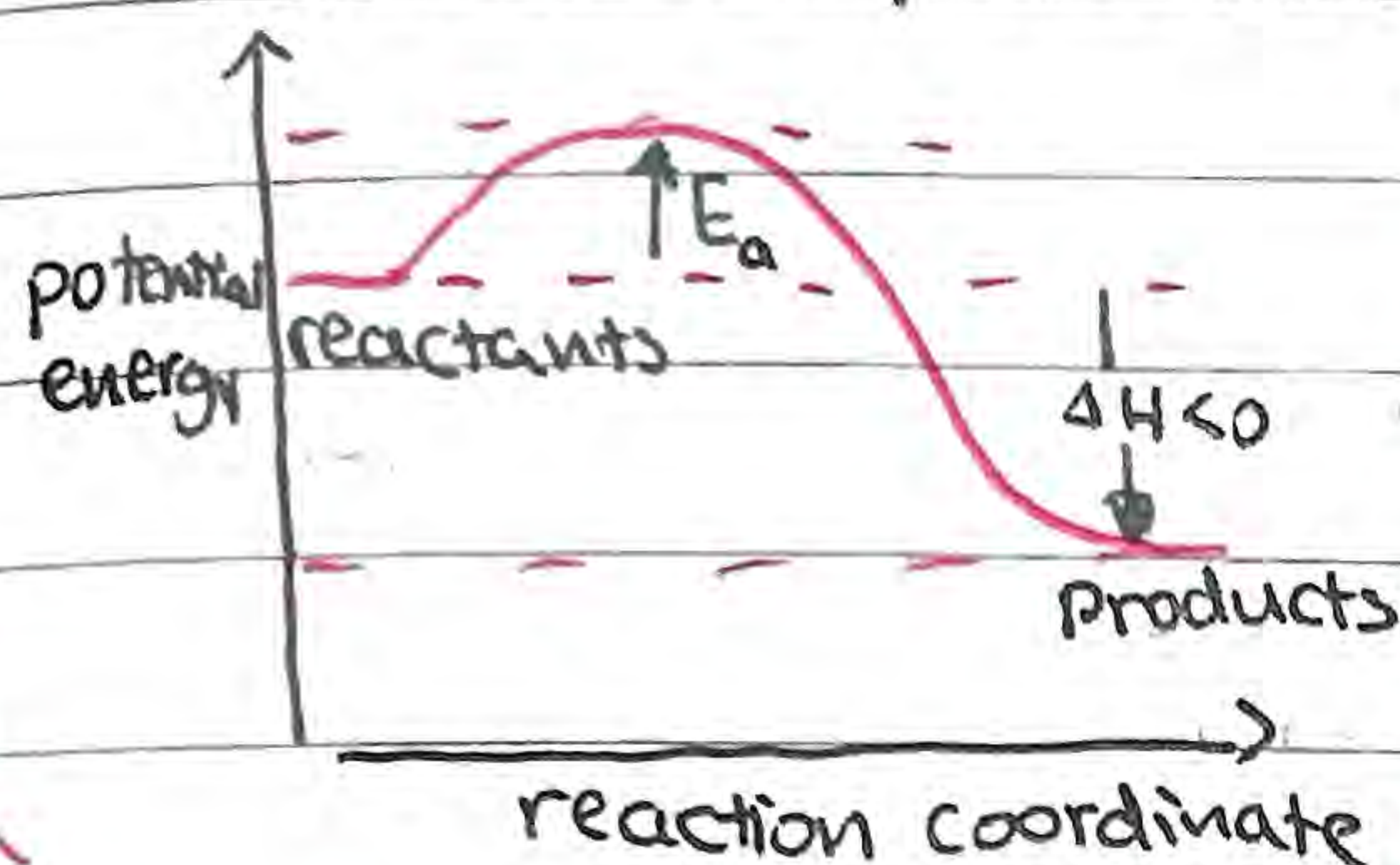
• Total enthalpy cannot be measured though change in enthalpy can

$\Delta H^\ominus$ ← under standard conditions

Energy Transfer

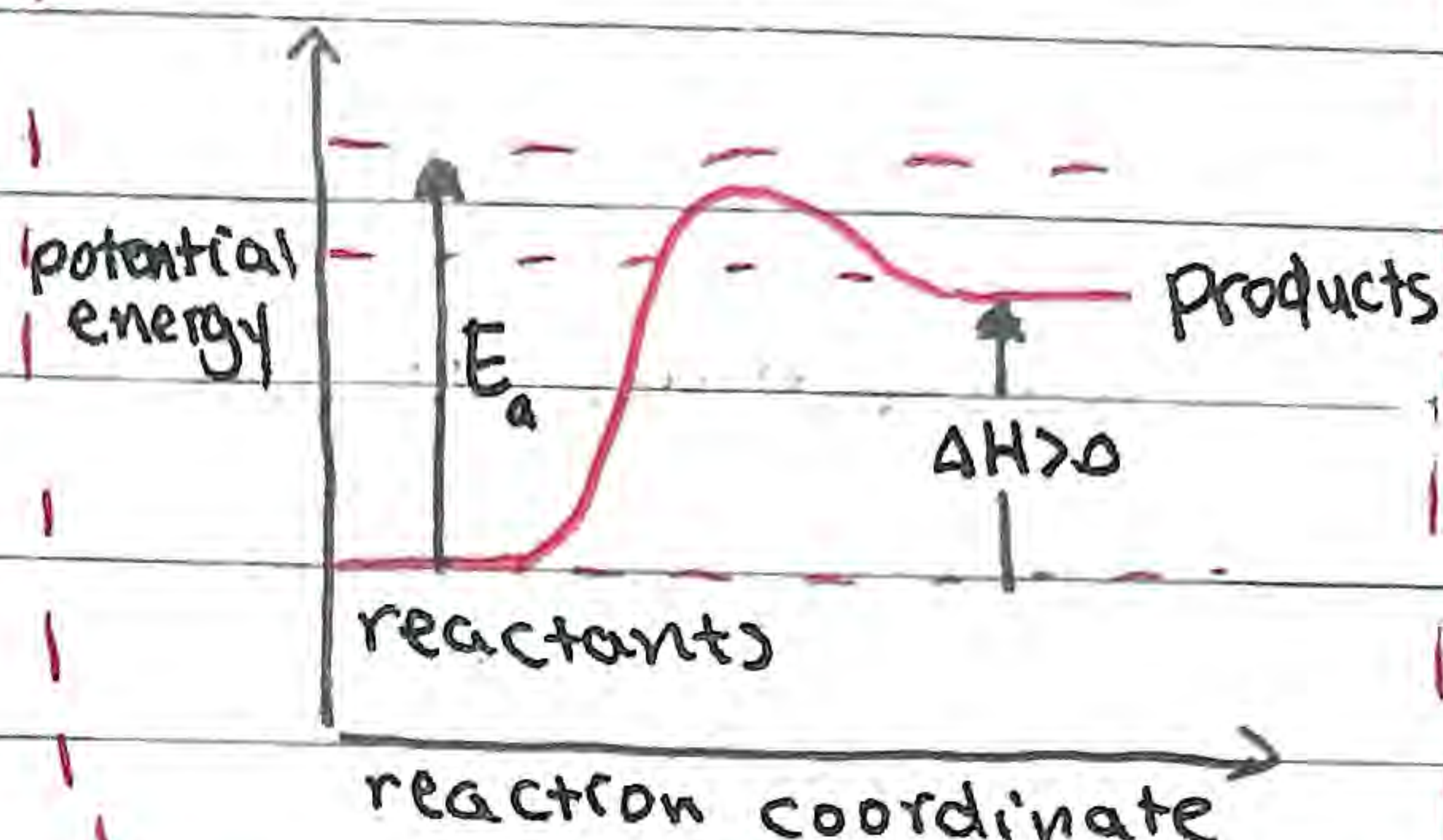
Exothermic

- Most chemical reactions are exothermic (energy favorable)
- Give out heat, makes bonds



Endothermic

- Absorbs heat, breaks bonds



Standard Enthalpy Change ($\Delta H^\ominus$) - Heat transferred (constant pressure) under standard conditions and states
 units = $\text{kJ} \cdot \text{mol}^{-1}$

- Determined from the change in temperature of a pure substance

Assumes: Pressure of 100 kPa, all solution concentrations = 1 mol dm^{-3}
 all substances in standard state

STP = 273.15 K and 100 kPa (denoted by $\ominus$)

SATP = 298.15 K and 100 kPa

Can be calculated with heat absorbed or release

- Specific heat capacity (c) required

Standard Enthalpy Change

specific heat capacity

• Amount of heat needed to raise the temperature of 1 kg of a substance by 1°C or 1 K

→ typically use that of the solvent, e.g. water

↳ Especially in aqueous, thus you have to assume "pure water" even if it's diluting something else

↳ water = $4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$, Ethanol = $2.44 \text{ kJ kg}^{-1} \text{ K}^{-1}$

Calculating heat

$$Q = m \cdot c \cdot \Delta T$$

mass $\rightarrow$ specific heat capacity $\rightarrow$ change temp. (+ or -)
 heat gained/lost $\rightarrow$ units of m, c, & must match (is it in kg, kJ etc.?)

Calculating enthalpy

$$\Delta H = -Q / n$$

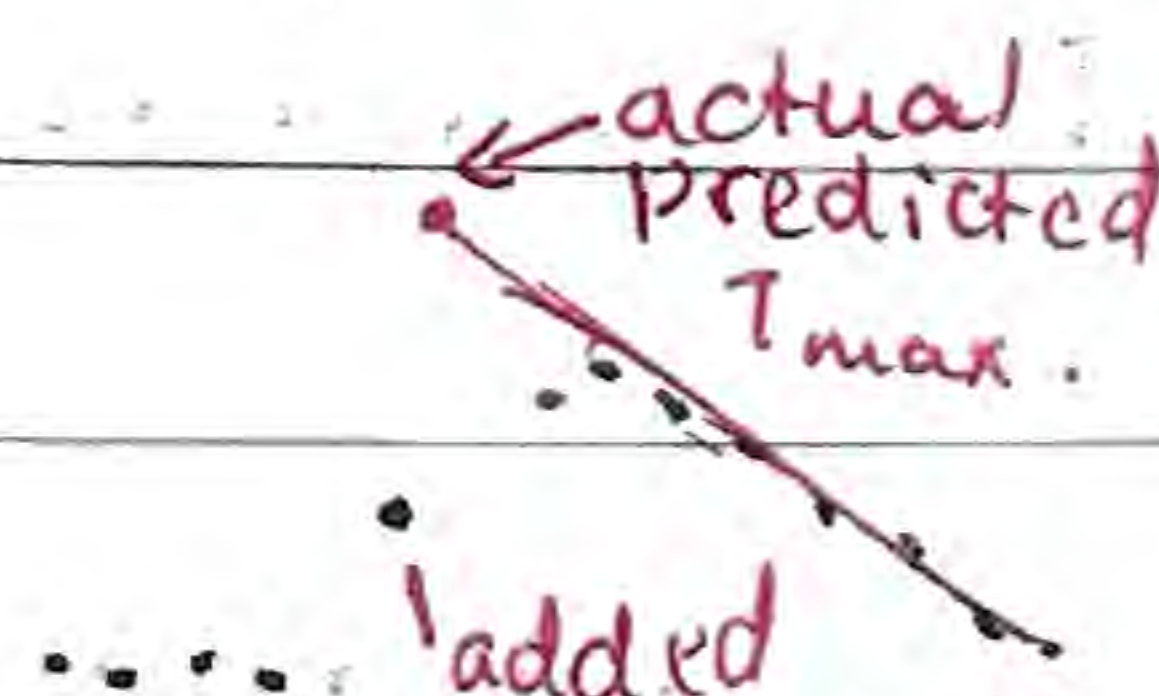
change in enthalpy $\rightarrow$ moles consumed (pay attention to limiting reactant)
 heat gained or lost (make sure kJ)

Calorimetry

• When recording max. temp for an exothermic reaction, it's necessary to extrapolate back when reaction started (regression)

• This is due to temp. lag

There was actual temp. change we couldn't observe



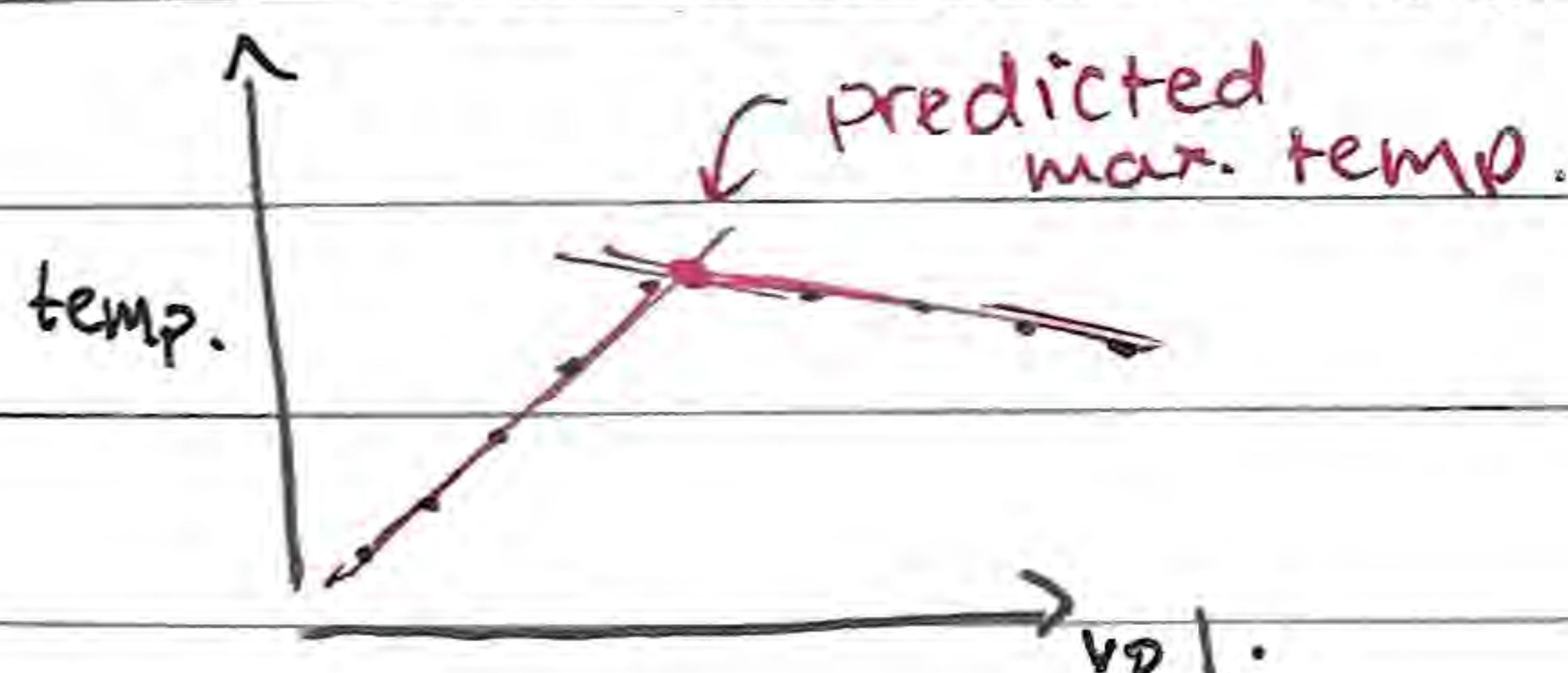
Calorimetry Assumptions

- Heat is completely transferred to the water
- Negligible heat capacity of container
- Reaction proceeds perfectly (extrapolation works)
- Maximum temperature reached is an accurate representation of heat evolved during the reaction
- Specific heat capacity of an aqueous solution is same as that of water

Errors: Mainly loss of heat to surroundings (max. temp. recorded lower than theoretical)

Thermometric Titration

- Neutralization reaction between acid/base = exothermic
- Determining the unknown concentration can be done when measuring Δ temp. since $T_{\max}$ occurs when acid and base are mixed in stoichiometric amounts



The intersection between the two lines of best fits is the

$T_{\max}$.

same thing as enthalpy, just for formation
 ↑ similarly applied to enthalpy of dissolution, etc.

Enthalpy of formation ($\Delta H_f^\ominus$)

e.g.
 $O_2 + C \rightarrow CO_2$
 is $CO_2 - 0 - 0$
 $= -393.5 kJ$

Enthalpy change when 1 mol of a compound is made from its elements in their standard states under standard conditions

* Elements have a $\Delta H_f^\ominus$ of zero: $\Delta H^\ominus = \sum \Delta H_f^\ominus (\text{products}) - \sum \Delta H_f^\ominus (\text{reactants})$

Forming 1 mol from its elements $\rightarrow$ change in enthalpy of reaction

> Since it's 1 mol restricted, when writing reaction equations

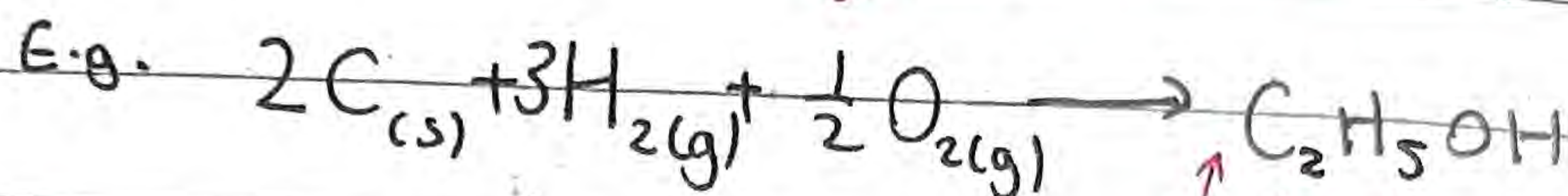
use fractional coefficients when balancing $\leftarrow$ to not change coefficient of compound

$\sum \Delta H_f^\ominus$ refers to weighted sum (coefficients)

Enthalpy of formation

The enthalpy of formation reaction equation is a synthesis reaction

fractional is allowed

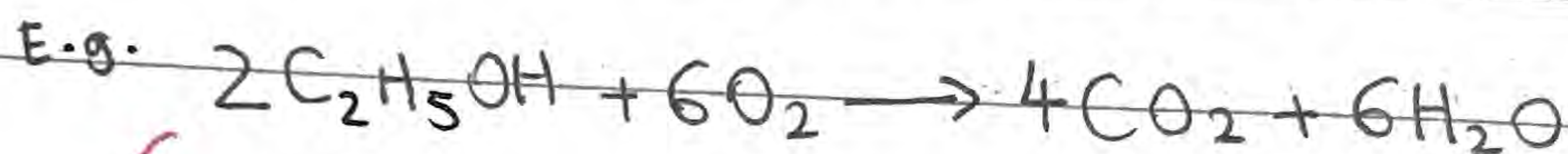


fixed at 1 mol

* Carbon standard state is $C_{(s)}$

Enthalpy of Combustion (ΔH_c°)

Enthalpy change when 1 mol of compound burns in excess oxygen under standard condition

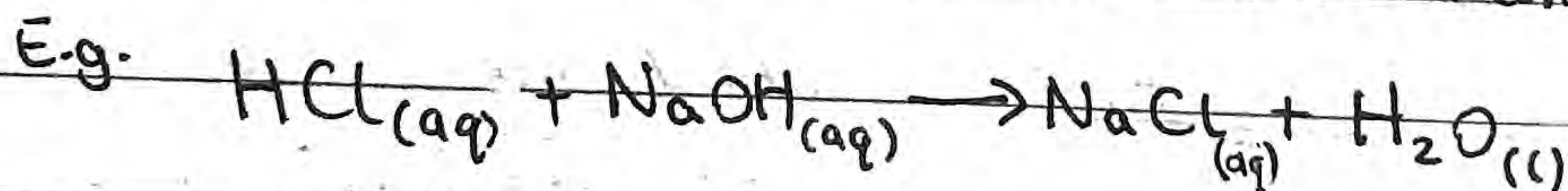


Assumes all heat from reaction is absorbed by the water

* Enthalpy must be done with respect to what is heated and specifics of question (heat transferred must be mass of ΔT and c of ΔT , must correspond)

Enthalpy of neutralization (ΔH_n°)

Enthalpy change when a strong acid and base are reacted together to form 1 mol of water under standard conditions



Just know that you can use heat to calculate enthalpy since combustion, neutralization, formation typically results in a temperature change.

Energy

Bond enthalpies

Chemical reaction is the rearrangement of atoms

Involves making or breaking bonds

Breaking bonds = endothermic Forming bonds = exothermic

Can be used interchangeably with enthalpy of formation to find enthalpy.

> Energy required to break 1 mol of bonds by homolytic fission in 1 mol of gas covalent molecules in standard conditions

↳ to overcome bonds rather than also IMFs in solids or liquids

Homolytic fission - Covalent bond breaks evenly, with each atom retaining one electron from the shared pair, forming two neutral free radicals

Bond enthalpy values are derived from experimental data taken from breaking the bond in different compounds

↳ Thus, they're average values and only an approximation

* particularly since the chemical environment changes as you remove a hydrogen each time

↳ surrounding bonds

not the most accurate for the specific compound

So how to calculate enthalpy

or $\sum$ BE of reactants

$\sum$ BE of products

(bonds)

$$\Delta H = \sum (\text{bond enthalpy, bond broken}) - \sum (\text{BE, bond formed})$$

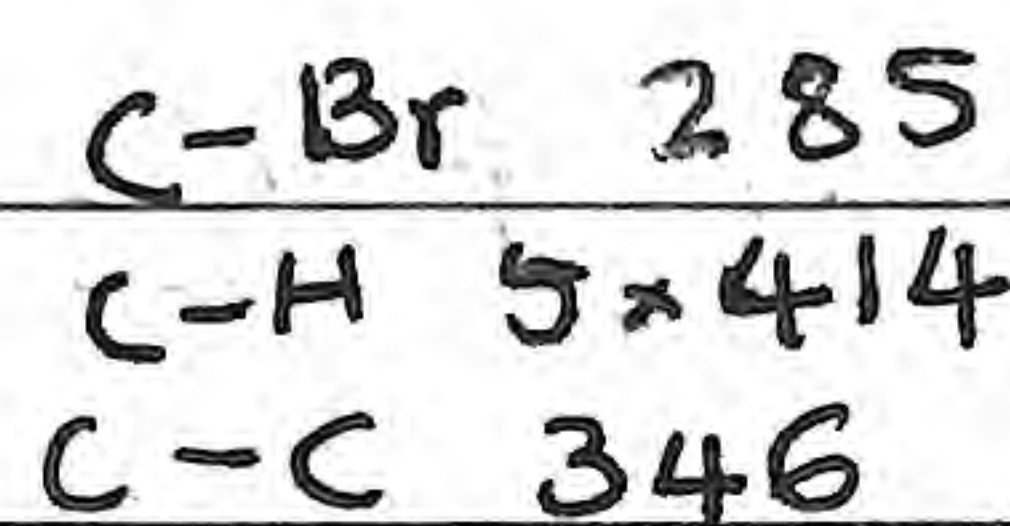
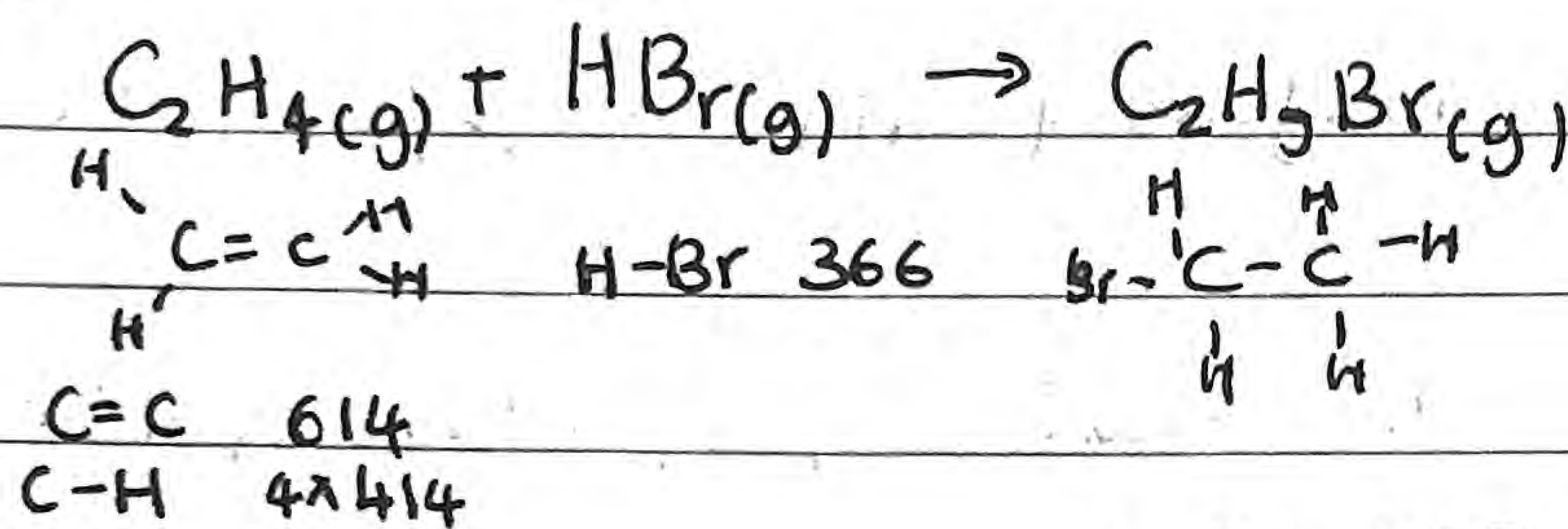
* answers with bond enthalpy data will differ from real data due to average nature of BE.

Bond Enthalpy

Enthalpy

· Always draw structure to visualize type of bonds and # of bonds

E.g.



Hess's Law

• Describes the different ways a reactant can yield a product

↳ A simple reaction like $A+B \rightarrow C$ may through $A+X \rightarrow B+Y$ etc.

• A chemical equation typically shows net equations

> Made up of other reactions $\rightarrow$ when added together gives overall reaction

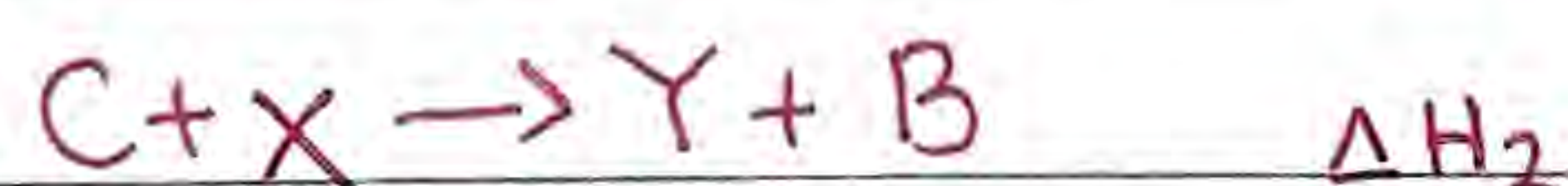
> Difficult in one step, so typically multiple

"Regardless of the route by which a chemical reaction proceeds, the enthalpy change will always be the same (given initial and final are same)"

↳ The enthalpy change of the net equation is the sum of the component reaction enthalpies:

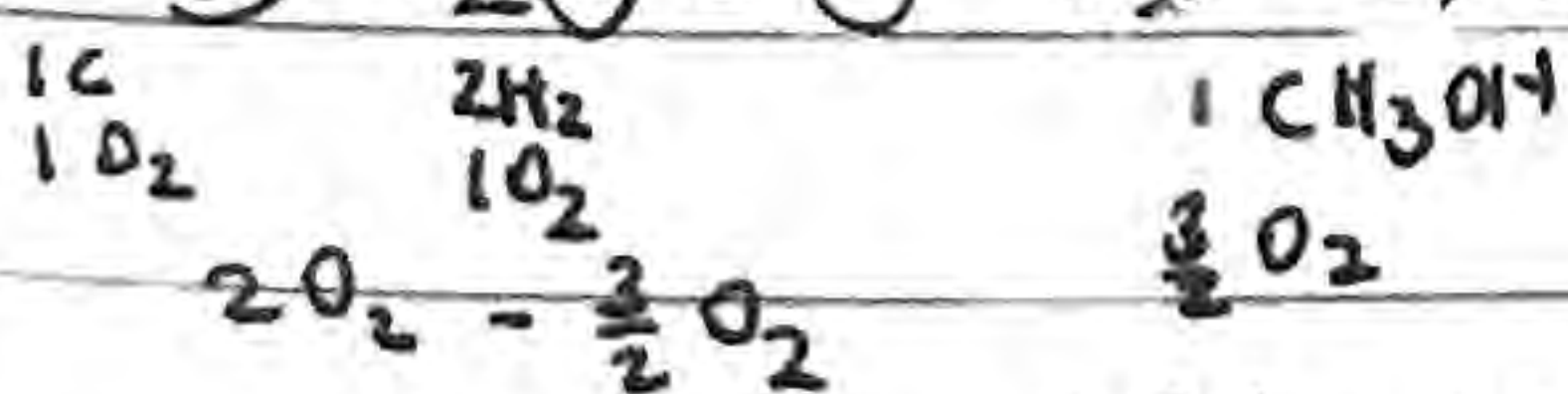
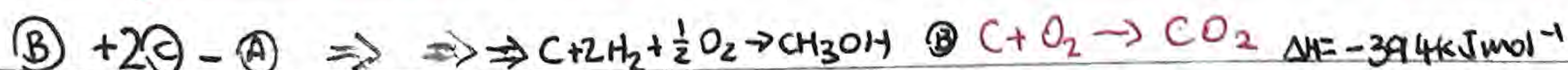
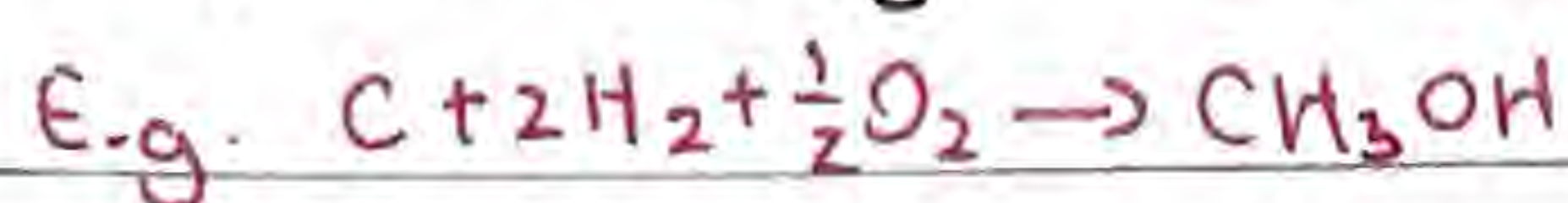


$\Delta H = \Delta H_1 + (-\Delta H_2)$



← Since the reaction required is the reverse of the given, the enthalpy must change signs

* The enthalpy of a reaction reversed is just the negative of the original reaction

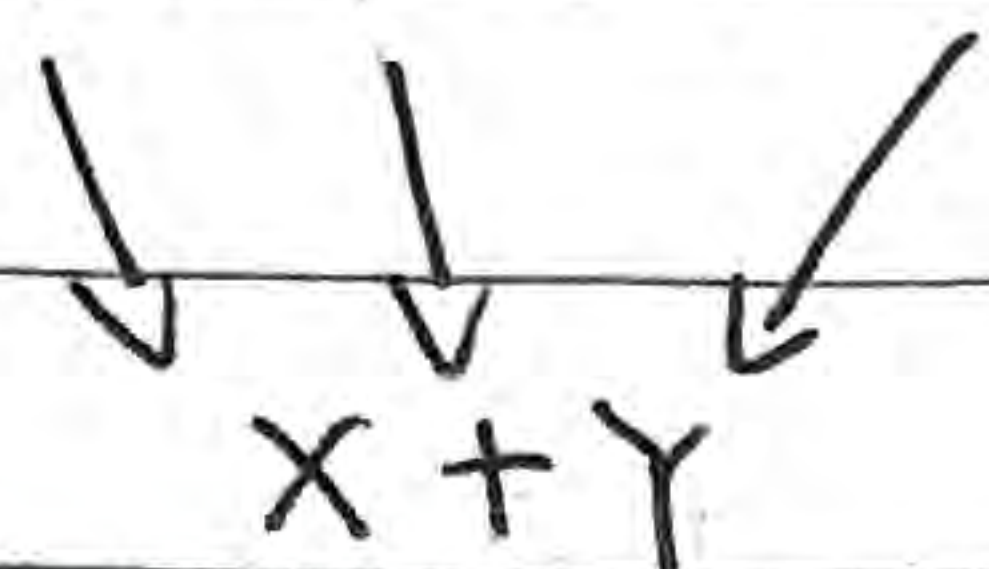
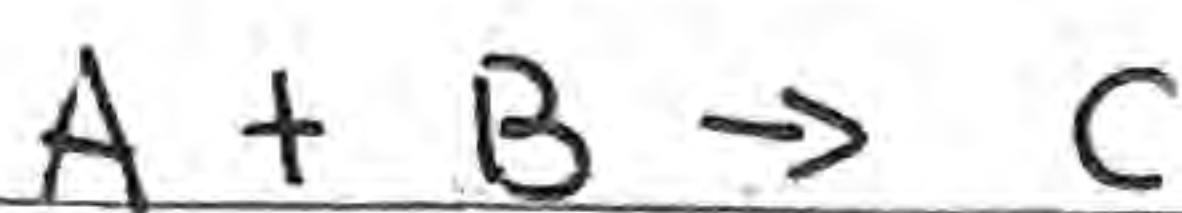


$\Delta H_{\text{net}} = -394 + 2(-286) - (-726 \text{ kJ mol}^{-1}) = -240 \text{ kJ mol}^{-1}$

Hess's Cycle

- Enthalpy cycle method

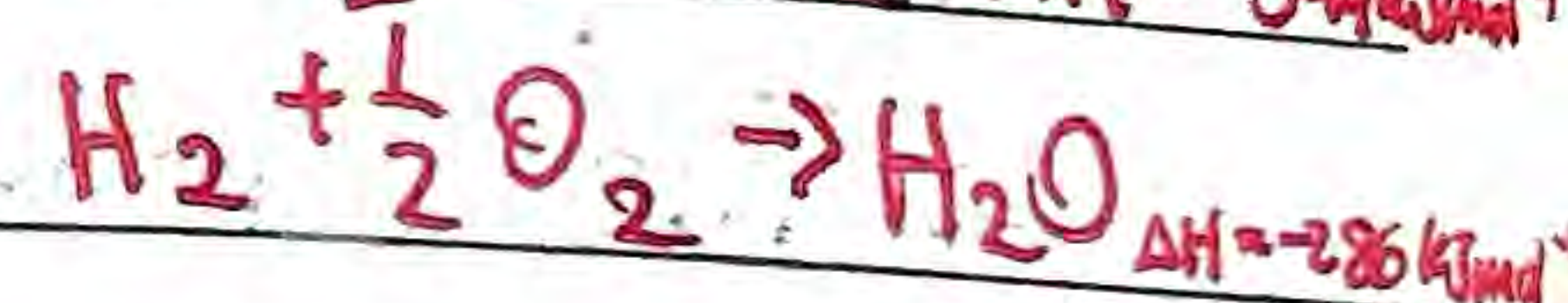
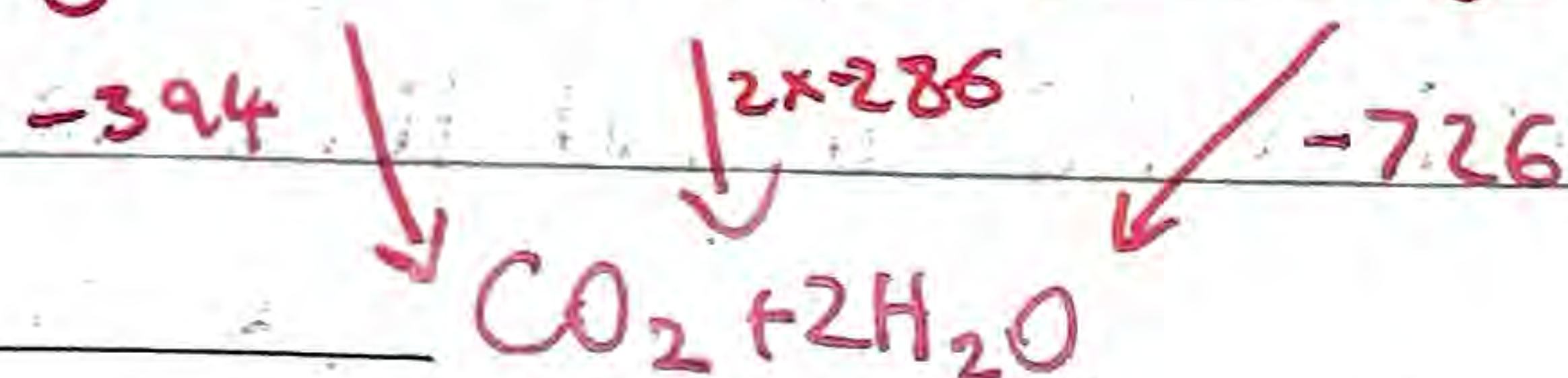
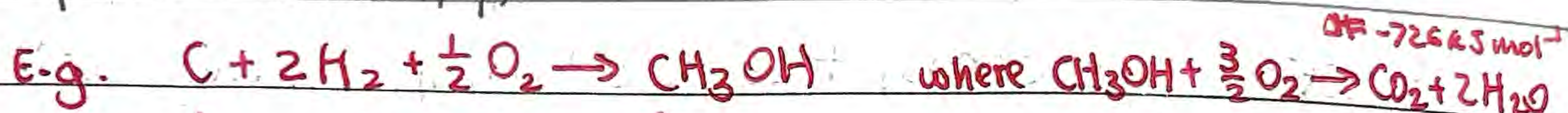
- This method deals with exploring intermediates



The goal is to go from A+B to C (arrow into C)

where

Final enthalpy is sum of enthalpies of individual reactions (when an arrow is inverted, take the negative of the enthalpy)



$$\Delta H_{\text{net}} = -394 + 2(-286) - (-726) = -240 \text{ kJ mol}^{-1}$$